ESE 4013 Complete Study Pack

Energy in Built Environment · KUET · past papers 2020, 2021, 2022, 2024 · answers from the course slides

24-hour priority list

You answer **3 of 4 questions in each part**, so each part has its own list. Study down both lists until the green **pass line**: about 6 h for Part A and 6 h for Part B, which leaves time for sleep and a last read of the one-shot notes. Alternate between parts (A block 1, B block 1, A block 2 …) so neither is left short. **Chance**: how often the topic came up, newer papers weigh more (2024 ×4, 2022 ×3, 2021 ×2, 2020 ×1). **Covered**: share of that part's past-paper marks you have studied once you finish the row.

Next if you have time: the two numericals, overhang sizing (A12) and the sol-air wall heat gain (B4), are short and pure formula marks: learn them with the calculators even if they sit below the pass line. In Part B, B6 (transient heat flow) and B8 (degree-days) are long but come every year, so learn their diagrams and the one key equation each.

Section A: Built environment, climate, sun, shading, daylighting

Block 1: very likely and easy · 1.7 h

Topic	Chance	Level	Covered
A1 Indoor environment control and IEQ	Very likely 100%	Easy	10%
A4 Reducing energy use: new and existing buildings, lighting, building categories	Very likely 100%	Easy	18%
A11 Sun shading devices: types, functions, selection	Very likely 100%	Easy	25%
A6 Noise and acoustic comfort	Very likely 90%	Easy	30%

Block 2: very likely, medium effort · 3.8 h

Topic	Chance	Level	Covered
A7 Sun-earth relationship, solar radiation and building orientation	Very likely 100%	Medium	42%
A10 Wind flow patterns and natural ventilation around buildings	Very likely 100%	Medium	48%
A13 Daylighting systems: passive, active (sun tracker), top and side lighting	Very likely 90%	Medium	60%
A8 Ambient temperature and sky temperature	Very likely 90%	Medium	64%
A14 Daylighting design principles and building orientation/form	Very likely 80%	Medium	71%

Block 3: very likely numericals · 1.0 h

Topic	Chance	Level	Covered
A12 Shadow angles (HSA, VSA) and overhang sizing numerical	Very likely 90%	Hard	76%

✔ Pass line: 76% of Section A marks covered, enough to choose 3 of the 4 questions (about 6.4 h)

Block 4: likely · 0.8 h

Topic	Chance	Level	Covered
A2 Indoor air contamination: sources and control	Likely 70%	Easy	82%
A3 Energy use in buildings: envelope, use pattern, performance factors	Likely 60%	Easy	89%

Only if time is left · 1.5 h

Topic	Chance	Level	Covered
A5 Thermal and visual comfort in building design	Possible 50%	Medium	96%
A9 Urban heat island and urban wind island effects	Possible 40%	Medium	100%

Section B: Thermal performance, loads, annual energy, NZEB

Block 1: very likely and easy · 1.7 h

Topic	Chance	Level	Covered
B11 Net-zero energy and green buildings, LEED	Very likely 100%	Easy	14%
B3 Smart windows (thermochromic, photochromic, electrochromic, LC)	Very likely 100%	Easy	19%
B10 Zoning, building shape and one-zone heat balance	Very likely 90%	Easy	22%
B5 Internal heat gains: people, lights, equipment; sensible vs latent	Very likely 80%	Easy	31%

Block 2: very likely, medium effort · 2.2 h

Topic	Chance	Level	Covered
B7 Infiltration, ventilation and the stack effect	Very likely 100%	Medium	42%
B1 Heat transfer through windows: U-factor, double pane, low-e, glazing selection	Very likely 100%	Medium	52%
B12 Evaporative cooling and solar-assisted desiccant dehumidification	Very likely 90%	Medium	60%

Block 3: very likely numericals · 2.0 h

Topic	Chance	Level	Covered

B8 Annual energy: balance point, degree-day and bin methods	Very likely 100%	Hard	70%
B6 Transient heat flow: diurnal wall response, CTF, finite difference	Very likely 100%	Hard	80%

✔ Pass line: 80% of Section B marks covered, enough to choose 3 of the 4 questions (about 5.9 h)

Block 4: likely · 1.8 h

Topic	Chance	Level	Covered
B2 Solar heat gain coefficient (SHGC)	Likely 70%	Medium	83%
B4 Sol-air temperature, wall colour, and the sun-lit wall numerical	Likely 70%	Hard	93%

Only if time is left · 1.2 h

Topic	Chance	Level	Covered
B13 Energy audit and building energy management economics	Possible 50%	Easy	96%
B9 Below-grade heat loss: basement walls, floors, slab on grade	Possible 50%	Medium	100%

CONTENTS

1. One-shot notes
2. **Section A: Built environment, climate, sun, shading, daylighting**: A1 Indoor environment control and IEQ, A4 Reducing energy use: new and existing buildings, lighting, building categories, A11 Sun shading devices: types, functions, selection, A6 Noise and acoustic comfort, A7 Sun-earth relationship, solar radiation and building orientation, A10 Wind flow patterns and natural ventilation around buildings, A13 Daylighting systems: passive, active (sun tracker), top and side lighting, A8 Ambient temperature and sky temperature, A14 Daylighting design principles and building orientation/form, A12 Shadow angles (HSA, VSA) and overhang sizing numerical, A2 Indoor air contamination: sources and control, A3 Energy use in buildings: envelope, use pattern, performance factors, A5 Thermal and visual comfort in building design, A9 Urban heat island and urban wind island effects
3. **Section B: Thermal performance, loads, annual energy, NZEB**: B11 Net-zero energy and green buildings, LEED, B3 Smart windows (thermochromic, photochromic, electrochromic, LC), B10 Zoning, building shape and one-zone heat balance, B5 Internal heat gains: people, lights, equipment; sensible vs latent, B7 Infiltration, ventilation and the stack effect, B1 Heat transfer through windows: U-factor, double pane, low-e, glazing selection, B12 Evaporative cooling and solar-assisted desiccant dehumidification, B8 Annual energy: balance point, degree-day and bin methods, B6 Transient heat flow: diurnal wall response, CTF, finite difference, B2 Solar heat gain coefficient (SHGC), B4 Sol-air temperature, wall colour, and the sun-lit wall numerical, B13 Energy audit and building energy management economics, B9 Below-grade heat loss: basement walls, floors, slab on grade

★ One-shot notes

Section A: Built environment, climate, sun, wind, shading, daylight

Building and energy

Envelope = 5 elements: foundation, walls, roof, windows, doors (separates in/out). **Fabric** = the materials/components it is made of.

Buildings: ~40 % of world energy, ~39 % of CO_2; people indoors 80-90 % of life.

Life-stage energy: **embodied, operational, maintenance, demolition**. End use: HVAC 35 %, lighting 11 %, appliances 18 %, other 36 %.

Six factors: **climate, envelope, systems, O&M, IEQ, occupants** (+ social).

IEQ and IAQ

IEQ = **IAQ + thermal + visual + acoustic**. Poor IEQ: eye/nose/throat irritation, eczema, headache, fatigue, low concentration, sick leave.

Contaminant sources: **outside, equipment (HVAC/non-HVAC), human activities, building components/furnishings, other**. Control: source → ventilate (7.5 L/s/person, CO_2 <1000 ppm) → filter → measure.

Comfort

Thermal comfort (ASHRAE): state of mind expressing satisfaction with the thermal environment. 4 environmental (T, MRT, air speed, RH) + 2 personal (met = 58.2 W/m², clo = 0.155 m²K/W).

Visual comfort: subjective visual well-being; task lighting, local control, no over-illumination, daylight.

Noise: sound that disturbs silence/intended sound, annoys or harms. Acoustic comfort by **insulation** (mass, airtight) + **absorption** (porous finishes); reverberation time, dB(A), STC, NRC.

Saving energy (7 areas)

Heating, cooling, ventilation, lighting, hot water, equipment, good housekeeping. Lighting: daylight, efficient luminaires, controls, sun-pipes; personal control saves 10-20 %.

New building: orient, form, envelope, window, shade, daylight, ventilate, efficient systems, renewables.

Sun-earth

Rotation 24 h → day/night. **Revolution** 365¼ d + 23½° tilt → seasons.

Solstice: axis farthest toward/away (21 Jun over Tropic of Cancer; 21 Dec over Capricorn). **Equinox**: sun over equator, day = night (21 Mar, 23 Sep).

Weather = short-term local; climate = long-term average.

```
1367 W/m2 outside atmosphere: 26 %
reflected, 18 % absorbed, 56 % reaches
ground
G = B + D + R   (diffuse 10-20 % clear,
up to 100 % overcast)
```

Bangladesh: long axis **E-W**, glazing N/S, shade south, avoid west glass.

Sky temperature

```
Tsky = Ta - 6 ;  Swinbank Tsky = 0.0552
Ta^1.5 (K)
Lclear = 5.31e-13 Ta^6 ; Lcloud =
ec*sigma*(Ta-5)^4
Lsky = (1-c)Lclear + c*Lcloud ; c = (Kd-
0.165)/0.835 ; Tsky = (Lsky/sigma)^0.25
```

Clear night: big radiative loss (roof colder than air); cloudy: small.

Urban effects

UHI: city hotter than country: waste heat, dark low-albedo surfaces, less green/water, canyons trap radiation, low wind.

Urban wind island: city wind obstructed, deflected, sometimes faster (around sunrise/sunset); roughness, down-wash, Venturi, canyon vortex.

Wind patterns (increase)

Wind shadow 3-4× building size. Bar 1.4×; Venturi 1.3× (30 m), 1.6× (50 m); gap 1.2× (7 storeys), 1.5× (60); corner 1.2× (15 m), 1.5× (>35 m), 2.2× between 100 m towers; wake 1.4-2.2×; gap in windbreak 100-120 %.

Ventilation: face the wind, inlet low/outlet high, outlet ≥ inlet, wing walls, thin plan, avoid wind shadows.

Shading

Internal (glare), external (best, up to 80 % cut), interpane. **South** = horizontal; **E/W** = vertical/movable; hot climate = egg-crate; north little; deciduous trees.

```
HSA = wall azimuth - solar azimuth
(fins)
tan VSA = tan(altitude)/cos(HSA)
(overhang)
Overhang depth D = H/tan VSA ; side
extension = D tan HSA
```

Critical: Sept noon (S), Sept 10 am (E), Sept 3 pm (W). Fulbarigate Sept noon: alt 67°, az −8° → VSA 67.2°, D = 0.9/2.379 = 0.38 m, width 0.61 m.

Daylighting

Controlled admission of sunlight + skylight to cut electric light. **Passive** (fixed) vs **active** (tracking mirrors, GPS, light well, diffuser). Top: skylight, monitor, saw-tooth (faces north), clerestory, tubular. Side: windows, light shelf.

Roof light = 3× window per m², ≤ 12 % of floor. Glazing ratio 25-50 %. Savings 15-40 %. Higher window head → deeper light. Office sill 1-1.2 m; house 0.7-0.9 m.

Glare: excessive brightness contrast; discomfort and disability.

Six principles: orientation, form, glazing ratio, window height, overhead, redirection.

Section B: Thermal performance, loads, annual energy, NZEB

Windows

```
Uw = (Ucg Acg + Ueg Aeg + Uf Af)/Aw
Single: R = 1/hi + L/k + 1/ho =
0.121+0.003+0.029 = 0.153 (glass 2 %)
Double: 1/U = 1/hi + 1/hspace + 1/ho ;
e_eff = 1/(1/e1 + 1/e2 - 1)
```

Air k = 0.025 (1/30 of glass). Gap heat: half radiation (low-e, ε 0.1 halves it), half conduction (optimum ~13 mm, then convection). Argon/krypton, triple.

Cold: high solar τ, low-e for far IR, low U. Hot: visible in, IR out (absorbing film outside, low-e inside), −60 % solar gain, −15-30 % cooling load.

Selective low-e also reflects near-IR (low SHGC + high VT).

SHGC

```
tau + rho + alpha = 1
SHGC = solar gain / incident solar
Q/A = U(Ti-To) + SHGC*IT
single: SHGC = tau + alpha*U/ho ,  U =
hi*ho/(hi+ho)
double: SHGC = tau + ao*U/ho + ai*U(1/ho
+ 1/hs)
```

Low SHGC not always best: cold climates and south sun want gain; daylight; orientation.

Smart windows

Thermochromic (temperature), photochromic (light), liquid crystal (voltage, privacy), electrochromic (voltage, controllable). Bangladesh: **thermochromic** (follows heat, not light).

Sol-air

```
Tsa = Tamb + as*q/ho - e*sigma*(Tamb^4 -
Tsurr^4)/ho
Q = U A (Tsa - Ti) ; dTsolar = as*q/ho
solar fraction = (Tsa - Tamb)/(Tsa - Ti)
fraction of incident solar inside =
U*as/ho
as/ho: light 0.026, dark 0.052 m2K/W (ho
= 17)
```

Radiation term ~0 walls, ~4 °C roofs. 2021: 1.8×20×26 = **936 W**; fraction 13/26 = **50 %**.

Internal gains

```
Lights: Q = PL * FUL * FSA   (FSA 1.2
fluorescent)  or  LPD * area
Equipment: Q = (PM/etaM) * FUM * FLM
Infiltration: Qs = rho*cp*V*(Ti-To) ; Ql
= rho*hfg*V*(Wi-Wo), rho*hfg = 3010
W/(L/s)
```

Person 100 W rest to 500 W heavy; latent ≥ 30 W (1 L water/day). Latent = instant; sensible delayed by mass.

Transient

```
dT/dt = alpha d2T/dx2 , alpha = k/(rho
cp)
Interior: Tm' = Fo(Tm-1 + Tm+1) + (1-
2Fo)Tm , Fo = alpha dt/dx^2 ⩽ 1/2
Surface: T0' = 2Fo(T1 + Bi*Tinf) + (1 -
2Fo - 2Bi*Fo)T0 , Bi = h dx/k
h = hc + hr , hr = e*sigma*(T0+Ts)
(T0^2+Ts^2)
CTF: Qt = -sum dn Q(t-n) + A[sum bn
Tos(t-n) - Trc sum cn]
```

4 reasons: envelope mass, stored solar, internal schedules, controls. Heavy wall: lag 6-12 h, smaller peak.

Infiltration

```
ACH = V(m3/h) / Volume ; min 0.35 ACH,
7.5 L/s per person
Crack: V = C dp^n (n ~0.65) ; orifice V =
CD A sqrt(2dp/rho)
dp = dp_wind + dp_stack + dp_vent ;
dp_wind = Cp rho v^2/2
dp_stack = -rho_i g dh (Ti-To)/To  (x Cd)
LBL: V = Aleak sqrt(as dT + aw v^2) ;
tracer: N = ln(C0/C)/t
```

Stack (winter): in at bottom, out at top, neutral plane mid-height. Prevent: caulk fixed joints, weather-strip moving joints.

Annual energy

```
Tbal = Ti - Qgain/Ktot ; Ktot = sum UA +
rho cp V = Qdesign/(Ti-To)des
HDD = 1 day * sum(Tbal - To)+ ; Qyr =
Ktot * HDD * 24 / eta
CDD = sum(To - Tbal)+ ; HDD - CDD = 365
(Tbal - To_avg)
Tmax = Ti - Qgain/Kmax ; Qc =
Ktot[CDD(Tmax) + (Tmax-Tbal)Nmax]/eta
Bin: Q = sum Nbin * Ktot/eta * (Tbal -
To)+
```

Peak load = worst weather, sizes equipment; annual = average weather + efficiency.

Ground and zoning

```
Tsoil(0,t) = Tavg + dT sin(2 pi t/8760 -
phi - pi/2)
Basement wall: Q = Uavg,bw * Pb * hbw *
(Ti - Tsoil)  (radial paths)
Slab: Q = Fp * P * (Ti - To)
Tunheated = [(UA)c Ti + (UA)u To]/[(UA)c
+ (UA)u]
```

Zone = one thermostat. Long narrow plan > square in heat loss (more perimeter).

NZEB and green

US DOE: produces its own annual energy by renewables. EU: **nearly** ZEB, very high performance, very low need mostly renewable on/near site. Types: **site, source, cost, emission**.

Design order: **envelope → efficiency → renewables** (+ commissioning 5-10 %). LEED: register, prerequisites + credits, submit, GBCI review; Certified 40, Silver 50, Gold 60, Platinum 80.

Desiccant and evaporative

Direct evaporative: cooler + more humid (dry climates). Indirect: cools at constant W. Desiccant (silica gel, alumina, LiCl, TEG), regenerate 60-90 °C by evacuated-tube solar. Sequence: desiccant wheel → thermal wheel → evaporative cooler; exhaust: evap → thermal wheel → solar heater → desiccant → out. Hot-humid NZEB: desiccant DOAS for latent + high-temperature radiant/chilled beam for sensible + PV.

Night cooling offsets 20-30 W/m² (natural), 50 W/m² (mechanical), night < 20 °C.

Abbreviations used in this pack

Every short form, written out. The answers also spell each one out the first time it appears.

Short form	Full meaning
ACH	air changes per hour
ALT	solar altitude angle
ASHRAE	American Society of Heating, Refrigerating and Air-Conditioning Engineers
BEMS	building energy management system
CDD	cooling degree-days
COP	coefficient of performance
CTF	conduction transfer function
dB	decibel (sound level unit)
DF	daylight factor
EUI	energy use intensity
HDD	heating degree-days
HSA	horizontal shadow angle
HVAC	heating, ventilation and air-conditioning
IAQ	indoor air quality
IEQ	indoor environmental quality
LC	liquid crystal
LED	light-emitting diode
LEED	Leadership in Energy and Environmental Design (green building rating)
NRC	noise reduction coefficient
NZEB	net-zero energy building
PMV	predicted mean vote (comfort index)
PPD	predicted percentage dissatisfied
PV	photovoltaic (solar cells)
SC	shading coefficient
SF	solar factor
SHGC	solar heat gain coefficient
STC	sound transmission class
UHI	urban heat island
VOC	volatile organic compound
VOCs	volatile organic compounds
VSA	vertical shadow angle
VT	visible transmittance

Section A: Built environment, climate, sun, shading, daylighting

A1 Indoor environment control and IEQ Must study Very likely · 100% Easy

PAST-PAPER QUESTIONS

- **2024 Q1(b)** [11] What is meant by IEQ (indoor environmental quality)? What are the adverse effects of poor IEQ? How good housekeeping can provide good IEQ?
- **2022 Q1(a)** [8] What is meant by 'Indoor Environment control'? To control indoor environment, provide some technical suggestions with examples.
- **2021 Q1(c)** [11] What is indoor environment control (IEQ) of a building? What are the internal and external factors upon which energy consumption of a building depend? What are the adverse effects of IEQ?
- **2020 Q1(a)** [10] What is meant by Indoor Environment Control of a building? What are the internal and external factors upon which energy use of a building depend?
- **2020 Q1(b)** [10] What are the four major aspects of indoor environment quality? How can ventilation contribute to maintain good indoor air quality?

★ REMEMBER IT

- IEQ has four parts: **T-A-V-A** = Thermal, Air quality, Visual, Acoustic ("tava" is a frying pan: a bad building fries you).
- Energy use depends on **external** factors (climate, sun, wind, site, orientation) and **internal** factors (people, lights, equipment, schedules, set-points, envelope).

Before you start: what the built environment is about

People spend about 90 % of their time indoors. A building should keep them **comfortable and healthy** (right temperature, humidity, fresh air, light, low noise) while using as little energy as possible. All of these together are called indoor environmental quality (IEQ).

Indoor environment control / IEQ (2020 1a, 2021 1c, 2022 1a, 2024 1b)

Indoor environmental quality (IEQ) is the overall quality of the conditions inside a building as they affect the health, comfort and performance of the occupants. It results from the interaction of the site, climate, building systems, construction, contaminant sources and the occupants. **Indoor environment control** means keeping these conditions inside acceptable limits by design (envelope, windows, shading) and by active systems (HVAC, lighting, controls).

Four major aspects of IEQ (2020 1b): (1) indoor air quality (IAQ), (2) thermal conditions (thermal comfort), (3) illumination (visual comfort), (4) acoustics (acoustic comfort).

Taken from: doi:10.3390/en12081414

Table 1. Indoor environment quality factors and their parameters.

	Thermal Comfort	Visual Comfort	Acoustic Comfort	Indoor Air Quality
Note		Requirements for lighting and daylighting		Close connection to thermal environment
Parameters	PMV/PPD or operative temperature; Humidity; Air velocity; Draught; Vertical air temperature differences; Radiant temperature asymmetry; Surface temperature of the floor.	Daylight provision; View out; Exposure to sunlight, Illuminance; Luminance; Light uniformity; Glare; Color (color rendering, light source color);	Sound level difference; Sound insulation; Sound absorption; Noise level(s) Frequency; Reverberation time.	Indoor sources of pollution; Outdoor sources of pollution; Ventilation parameters; Airflow patterns and pressure relationships; Air filtration system.
Influence factors	Clothing; Activity	Light source, Visual task; Use of room		Sources of heat gains; Outdoor conditions and outdoor air ventilation rate

IEQ factors and their parameters: thermal, visual, acoustic comfort and IAQ (slide table)

Factors on which building energy use depends

External (physical, hard to change)	Internal (human-influenced)
Climate: temperature, humidity, solar radiation, wind of the site	Indoor environmental quality demanded (set-points, light level, fresh air)
Building envelope: insulation, windows, shading, orientation	Occupant activities and behaviour (schedules, habits)
Building energy systems: HVAC (heating, ventilation and air-conditioning), lighting, equipment and their efficiency	Operation and maintenance of the systems
Site exposure, surroundings, building age and type of construction	Social factor: energy price, attitudes, ability to pay

The six factors (slides): climate, envelope, energy systems, operation and maintenance, IEQ provided, occupant behaviour; plus the separate social factor.

Adverse effects of poor IEQ

- Health complaints: eye, nose and throat irritation, eczema, headache, fatigue, lack of concentration (sick building syndrome)
- Lower work and study performance: about 2 % drop per °C above 25 °C
- More sick leave and absenteeism; comfort complaints; damage to building fabric (mould, condensation)

Technical suggestions to control the indoor environment (2022 1a)

- Thermal: insulation, shading, HVAC with zone thermostats, avoid radiant asymmetry (e.g. low-e glass near seats)
- Air: remove pollutant sources first, then ventilate (CAV/VAV with minimum outdoor air), filtration, exhaust in kitchens/toilets
- Light: daylight with shading and light shelves, LED (light-emitting diode) task lighting, occupancy and daylight sensors
- Sound: sound insulation of facade and partitions, absorptive ceilings, inlets away from traffic

How ventilation keeps good IAQ (2020 1b)

After eliminating sources, ventilation dilutes and removes the unavoidable pollutants (CO_2, odours, VOCs (volatile organic compounds), moisture) with outdoor air: about 7.5 L/s per person keeps CO_2 below 1000 ppm. Good "ventilation efficiency" (supply air mixing well with room air), filtration, and pressure control (exhaust from kitchens/toilets) complete it.

Good housekeeping for good IEQ (2024 1b)

- Regular cleaning with low-emission cleaning materials; vacuum with HEPA so dust is not re-circulated
- Store supplies, solvents and trash away from occupied rooms and air intakes
- Maintain HVAC: change filters, clean coils, drain pans, ducts and cooling towers (stops microbial growth)
- Fix leaks fast, avoid standing water; ban smoking indoors; run exhaust fans while cooking

Remember: IEQ = **Air, Heat, Light, Sound**. Energy factors: **C-E-S-O-I-O** (Climate, Envelope, Systems, O&M, IEQ, Occupants) + social.

A4 Reducing energy use: new and existing buildings, lighting, building categories Must study Very likely · 100% Easy

PAST-PAPER QUESTIONS

- **2024 Q1(d)** [8] When designing a new building, what key factors should be considered to minimize dependence on artificial energy consumption?
- **2022 Q1(d)** [10] Based on indoor activities, categories the building and mention the energy saving options for each category.
- **2022 Q2(a)** [7] For an existing building, discuss the technical issues to reduce energy consumption. Explain the strategies to ensure good acoustic environment in building.
- **2021 Q1(d)** [7] When developing a new building, what are the factors should be considered to reduce artificial energy consumption?
- **2020 Q2(c)** [8] How can energy use for lighting be reduced?

Before you start: saving energy in buildings

Three steps, in order: (1) reduce the need (insulation, shading, daylight, airtightness); (2) use efficient equipment (LED lights, efficient chillers, controls); (3) supply the rest from renewables (solar PV (photovoltaic (solar cells))).

Key factors for a new building to cut artificial energy (2021 1d, 2024 1d)

1. **Site and orientation:** long axis east-west, main glazing north/south, use prevailing wind
2. **Form:** compact shape (low surface/volume) in cold climates; thin plan or courtyard for daylight and cross ventilation in hot-humid climates
3. **Envelope:** insulation of walls and roof, light-coloured (low absorptivity) surfaces, thermal mass where diurnal swing is large
4. **Windows:** right window-to-wall ratio (25-50 %), low-e / low-SHGC glass, avoid large east/west glazing
5. **Shading:** overhangs on south, fins on east/west, trees
6. **Daylighting:** high windows, light shelves, skylights/light tubes to cut electric lighting
7. **Natural ventilation and passive cooling:** cross and stack ventilation, night cooling, evaporative cooling
8. **Efficient systems and controls:** right-sized HVAC (heating, ventilation and air-conditioning), LED (light-emitting diode), sensors, BMS
9. **Renewables:** rooftop PV, solar water heating

Approaches to reduce energy in an existing building (2022 2a, slides)

Area	Measures
Heating demand	Limit exposed surface area, improve insulation, reduce ventilation losses, efficient heating with controls
Cooling demand	Control solar gain through glazing, reduce internal gains, effective natural ventilation, reduce lighting loads
Ventilation	Maximise natural ventilation, effective window design, efficient fans
Lighting (2020 2c)	Use daylight while avoiding solar gain; efficient luminaires (LED, high lm/W); controls that stop lights being left on (timers, occupancy and daylight sensors); sun-pipes for windowless areas; individual control saves 10-20 %
Hot water	Time controls, correct thermostat setting, switch off immersion heaters and pumps when not needed, insulate pipes and cylinders
Office equipment	Switch off / power-down mode, upgrade to efficient models, match equipment to the task
Good housekeeping	Energy policy, energy champion, involve staff, set targets, notices, walk-rounds, meter readings

Strategies for a good acoustic environment (2022 2a second part): see A6: insulate against outdoor noise (facade, windows), isolate equipment noise, absorptive finishes, sound masking.

Building categories by indoor activity and saving options (2022 1d)

ICC group (activity)	Typical energy-saving options
A Assembly (theatres, restaurants, stadiums)	Demand-controlled ventilation by CO_2, VAV, scheduled HVAC pre-cooling, LED stage/hall lighting, kitchen heat recovery
B Business (offices, banks)	Daylight + dimming, occupancy sensors, efficient office equipment, VAV, night set-back, BMS
E Educational (schools)	Daylighting, natural ventilation, ceiling fans, timers following class schedule
F Factory / industrial	Efficient motors and VSDs, waste-heat recovery, high-bay LED, compressed-air leak control
H High hazard	Safe ventilation with heat recovery, sealed efficient process equipment
I Institutional (hospitals, prisons)	24 h loads: heat recovery, CHP, efficient chillers, solar hot water
M Mercantile (shops, markets)	LED display lighting, efficient refrigerated cases with doors, air curtains
R Residential (houses, hotels, dorms)	Insulation, shading, efficient AC/fans, solar water heating, occupant habits
S Storage, U Utility	Minimal conditioning, motion-sensor lighting, roof PV

Remember: New building: **Orient, Form, Envelope, Window, Shade, Daylight, Ventilate, Efficient systems, Renewables**.

A11 Sun shading devices: types, functions, selection Must study

Very likely · 100% Easy

PAST-PAPER QUESTIONS

- **2024 Q3(b)** [12] Select appropriate shading devices for different sides of a building and mention the reasons behind their selection.
- **2022 Q3(d)** [7] Discuss about the factors that should be considered to design and select a sun-shading device.
- **2021 Q4(c)** [12] What are the advantages and disadvantages of sun shading devices? Explain the horizontal and vertical shadow angle to determine the dimension of shading device.
- **2020 Q3(b)** [5] What are the functions of sun shading devices? Describe how to select a sun shading device for a building.

★ REMEMBER IT

- Which device on which side: **South = overhang (horizontal), East/West = fins or egg-crate (vertical or both), North = nothing or small fins**. Sun high → horizontal blocks it; sun low and sideways → vertical blocks it.

Before you start: shading

The best way to stop solar heat is to stop sunlight before it reaches the glass. **Horizontal** devices (overhangs) block the high sun (south facades); **vertical** fins block the low sun from the side (east/west); **egg-crate** combines both.

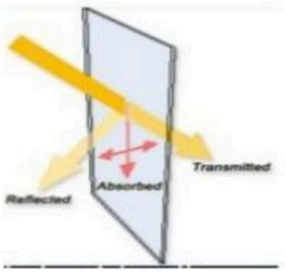
Sun shading devices: functions (2020 3b)

Sun shading devices are fixed or movable elements (mechanical equipment, textiles, vegetation) placed inside, outside or between glazing to block, allow or redirect the solar radiation falling on a building, giving cool interiors in summer and warm in winter.

- Block direct sun to cut solar heat gain and cooling load (external shading can cut window gain by up to 80 %)
- Admit winter sun when designed for it
- Control glare and sky glare; increase useful daylight
- Allow a view out; protect from rain; give privacy

Solar shading

When sunlight hits a pane of glass, it splits into three components



Reflected: no effect on heating.

Absorbed: glass heats up which would transfer heat by conduction and also emit heat

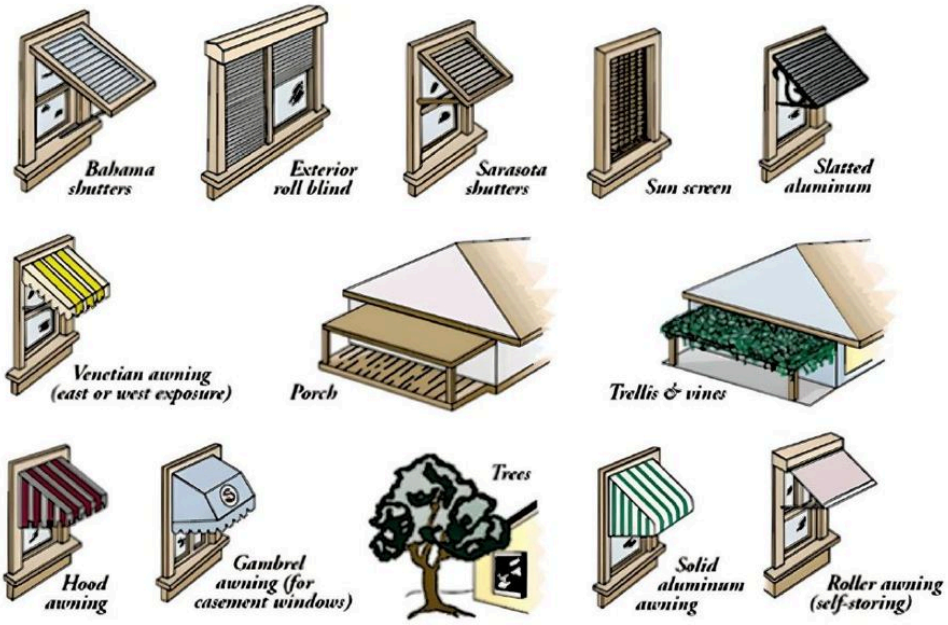
Transmitted: Heat up surface behind it.

The proportion between the three components is determined by the 'angle of incidence' and by the type of glazing. For most types of glazing, the transmitted component is very small if the angle of incidence is larger than 45° from the normal to the glazing. If the angle is more than 60°, most of the radiation is reflected.

Sunlight on glass splits into reflected, absorbed and transmitted parts; beyond 45° incidence little is transmitted

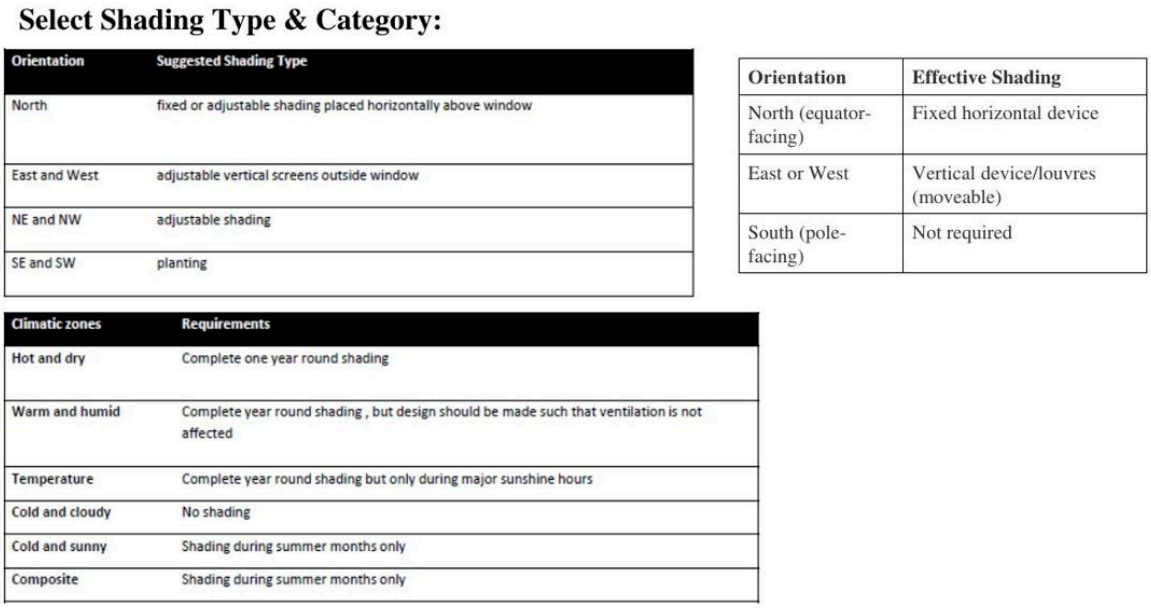
Types

By position: internal (curtains, Venetian blinds, vertical louvre blinds, roller, pleated, blackout blinds: mainly glare control, adjustable); external (overhangs, fins, egg-crate, awnings, louvres, vegetation, other buildings: most thermally effective since the heat is stopped before the glass); interpane (blinds between panes).



External shading devices: shutters, awnings, louvres, porches, trellis, trees

Device	Works best on	Why
Horizontal overhang / louvres	South (equator-facing) side	High summer sun at noon is cut by a small depth; low winter sun gets under it
Vertical fins (slanted, movable)	East and west	Sun is low and comes from the side; fins cut it at an angle; movable louvres best
Egg-crate (horizontal + vertical)	Hot climates, SE/SW, walls with sun from many angles	Very high shading efficiency; horizontal part also cuts ground glare
Deciduous trees	East, west, south	Shade in summer, let sun through in winter after leaf fall
North side	Pole-facing (north in Bangladesh)	Mostly diffuse light; small fins only for early morning/late evening summer sun



Select Shading Type & Category:

Orientation	Suggested Shading Type
North	fixed or adjustable shading placed horizontally above window
East and West	adjustable vertical screens outside window
NE and NW	adjustable shading
SE and SW	planting

Orientation	Effective Shading
North (equator-facing)	Fixed horizontal device
East or West	Vertical device/louvres (moveable)
South (pole-facing)	Not required

Climatic zones	Requirements
Hot and dry	Complete one year round shading
Warm and humid	Complete year round shading , but design should be made such that ventilation is not affected
Temperature	Complete year round shading but only during major sunshine hours
Cold and cloudy	No shading
Cold and sunny	Shading during summer months only
Composite	Shading during summer months only

Selecting shading type by orientation and by climate zone (slide tables)

Shading devices for different sides of a building in Bangladesh (2024 3b)

Bangladesh is at about 23° N, so the equator-facing side is **south** (the slide table is written for the southern hemisphere where north is equator-facing).

- **South:** fixed horizontal overhang or horizontal louvres/light shelf: the noon sun is high, so a shallow projection shades the window from March to September but lets the low winter sun in
- **East:** vertical fins or movable vertical louvres; morning sun is low and cooler, so partial shading or adjustable blinds is acceptable
- **West:** the most critical: low, hot afternoon sun with glare. Use movable vertical louvres, egg-crate, or deep vertical fins plus trees; keep glazing small
- **North:** little direct sun; small vertical fins against early-morning and late-afternoon summer sun, otherwise leave open for daylight
- **Roof:** double roof, pergolas, green roof

Factors in designing/selecting a sun-shading device (2022 3d)

1. Orientation of the facade and the sun path of the location (altitude, azimuth for the months to be shaded)

2. Type: horizontal, vertical or egg-crate; internal or external
3. Dimensions from horizontal and vertical shadow angles (A12)
4. Category: fixed, adjustable, movable, manual or automatic; durability of moving parts
5. Climate and the season to be shaded (cut-off date), daylight and view needs, ventilation and rain protection
6. Material and colour (light, low thermal mass, not re-radiating to the glass), cost, maintenance, appearance

Advantages and disadvantages (2021 4c)

Advantages	Disadvantages
Better thermal comfort inside	Internal blinds and curtains are hard to handle
Lower HVAC (heating, ventilation and air-conditioning) loads and energy cost (block summer gain, admit winter gain)	Light shelves can let in low winter sun and cause glare
Prevent glare	May clash with the building style; always blocks part of the view
More useful daylight	Fixed devices are orientation-specific; leak sun if not facing exactly south
Privacy and beauty (curtains, trees give oxygen)	Can reduce air flow in non-AC buildings; cost and maintenance

Remember: South = horizontal, East/West = vertical (movable), Hot = egg-crate, North = little. External beats internal.

A6 Noise and acoustic comfort Must study Very likely · 90% Easy

PAST-PAPER QUESTIONS

- **2024 Q1(c)** [8] What is noise? Briefly describe various methods of maintaining acoustic comfort in built environment.
- **2022 Q2(b)** [8] What is meant by acoustic comfort? Discuss about different building materials that can be used to maintain good acoustic comfort.
- **2021 Q2(c)** [8] What is meant by acoustic comfort? Explain the parameters that should be maintained for good acoustic condition in building.

Before you start: sound basics

Sound level is measured in decibels (dB), a logarithmic scale: +10 dB (decibel (sound level unit)) sounds about twice as loud. Noise enters a room through air (gaps, windows) and through the structure (footsteps, machines). Control: block (mass, airtight), absorb (soft materials) and isolate (separate the source).

Noise and acoustic comfort (2021 2c, 2022 2b, 2024 1c)

Noise is "the sound occurring within the frequency range of human hearing which disturbs silence or an intended sound perception and results in annoyance or risks the health". **Acoustic comfort** is a state of contentment with the acoustic conditions: an acceptable noise level, isolation of unwanted sound and presence of pleasant sound, so that speech is intelligible and concentration, rest and sleep are possible. Poor acoustics is the top complaint of occupants, even in green buildings.

Methods of maintaining acoustic comfort

1. **Sound insulation** of facade, roof and windows against outdoor noise; of floors, partitions and doors against indoor sources (neighbours, AC units)
2. **Sound absorption** inside the room so reflections do not build up (control reverberation): absorptive ceilings, panels, carpets, drapes, banners
3. Limit glass to what daylighting needs; double-glazed, well-sealed windows
4. Place natural-ventilation inlets away from traffic and other noise sources
5. Sound-masking systems in very quiet open offices for speech privacy
6. Isolate and silence equipment (vibration mounts, duct silencers, low-velocity ducts); zone noisy and quiet spaces apart

Parameters to maintain (2021 2c)

Parameter	Meaning / good practice
Sound pressure level (dB(A))	Background noise below about 35-45 dB(A) in offices/classrooms, 30 dB(A) in bedrooms
Reverberation time	Time for sound to decay 60 dB; short (0.4-0.8 s) for speech rooms
Sound insulation (STC / R_W)	High transmission loss of walls, floors, windows, doors
Sound absorption (NRC, α)	Enough absorptive area for the room volume and shape
Frequency spectrum, speech intelligibility, vibration	Control low-frequency HVAC (heating, ventilation and air-conditioning) rumble; clear speech; no structure-borne vibration

Building materials for good acoustics (2022 2b)

- **Absorbers:** mineral wool, glass wool, acoustic ceiling tiles, perforated gypsum/wood panels with backing, acoustic foam, carpets, curtains, upholstered furniture, recycled-fibre (green) panels
- **Insulators (mass):** dense concrete, brick, double-leaf walls with cavity insulation, laminated or double glazing, solid-core doors with seals
- **Vibration isolation:** resilient channels, floating floors, rubber mounts

Remember: Two tools: **INSULATE** (keep noise out, needs mass and airtightness) and **ABSORB** (stop echo inside, needs porous material).

A7 Sun-earth relationship, solar radiation and building orientation Must study Very likely · 100% Medium

PAST-PAPER QUESTIONS

- **2024 Q2(a)** [10] Write short notes on (i) Solstice, and (ii) Equinox. Describe briefly about various factors that influence the variation of radiation at ground level.
- **2022 Q3(a)** [10] On the basis of energy consumption view point, signify the sun-earth relationship for a building.
- **2021 Q3(a)** [8] Distinguish between weather and climate. Explain the effects of building orientation on the solar heat gain of a building.
- **2021 Q3(b)** [11] What does mean by sun-earth relationship? With the help of necessary sketch, explain the significance of earth relation and revolution on building heat gain.
- **2020 Q2(a)** [10] What are meant by earth rotation and earth revolution? Explain with necessary figure.
- **2020 Q2(b)** [12] How is built environmental influenced by the orientation of a building? Explain.

LEARN IT FROM ZERO

Why the sun matters to a building, from zero

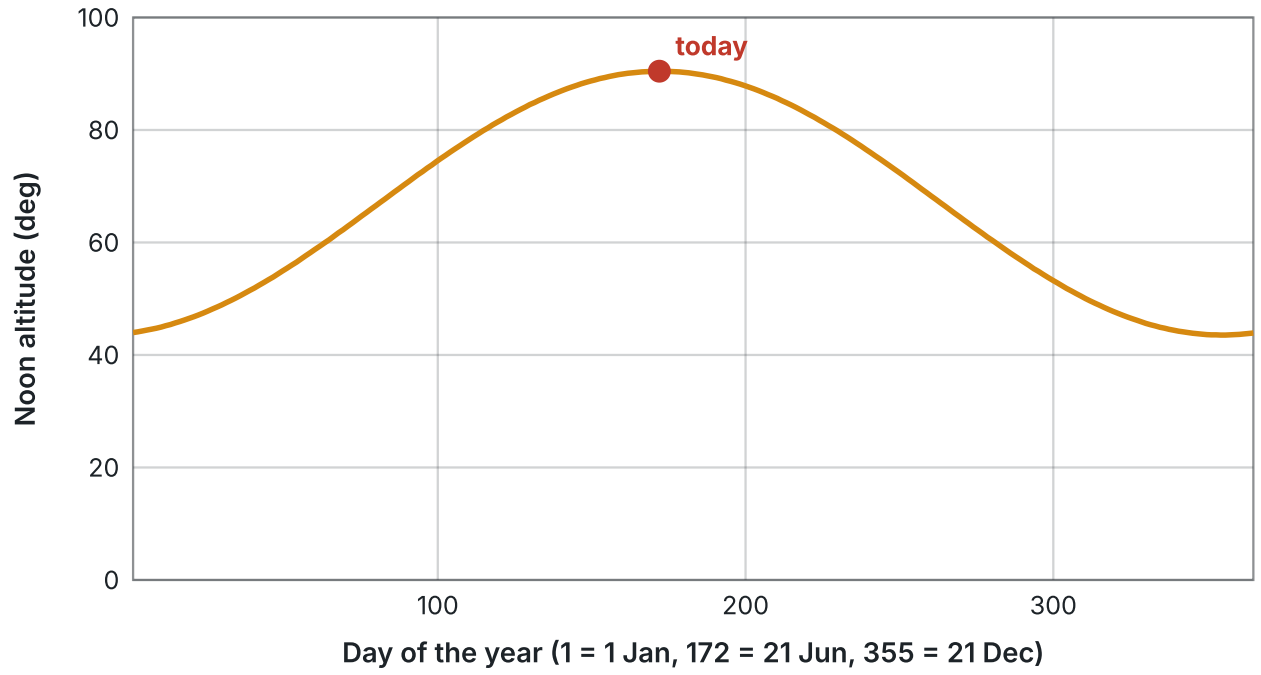
The earth does two things. It **spins** once a day (rotation), so the sun rises in the east, crosses the sky and sets in the west: the east wall is hit in the morning, the west wall in the afternoon. It also **goes round the sun** once a year (revolution) with its axis **tilted 23.5°**. Because of the tilt, the noon sun is high in June and low in December. That is the seasons.

For a building in Khulna (latitude about 23° N) this means: the noon sun is almost overhead in summer (easy to block with a small overhang on the south side), but low in winter (it comes deep into south rooms, which is welcome). East and west sun is always low and hard to block, so those walls get the worst afternoon heat and glare.

Two numbers do all the work: the **declination** δ (how far north or south of the equator the sun is overhead today, from +23.45° in June to −23.45° in December) and your **latitude** φ. Noon sun altitude = $90° - \varphi + \delta$. Play with the graph below.

INTERACTIVE **Noon sun altitude through the year**

Move the sliders and watch the curve change.



Declination δ: **23.4 °** · Noon altitude: **90.0 °**
Khulna is 22.9° N. Equinox (day 80 or 266): altitude = 90 − φ = 67°. Summer solstice: 90.5° (sun just north of overhead). Winter solstice: 43.6°.

★ **REMEMBER IT**

- **Solstice** = sun "stands" at its farthest north (21 June) or south (22 Dec). **Equinox** = "equal night": day = night, sun over the equator (21 March, 23 September).
- Weather = today (mood); climate = 30-year average (personality).

Before you start: how the sun moves

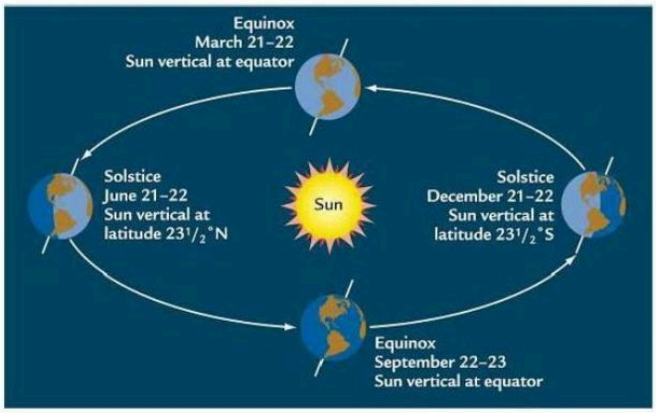
The Earth's axis is tilted 23.45°, so the sun is high in summer and low in winter, and rises north-east in summer, south-east in winter (northern hemisphere). **Solar altitude** = angle of the sun above the horizon; **azimuth** = its compass direction. Khulna is at about 22.9° N, so the noon sun at the equinox is 90 − 22.9 = 67.1° high, in the south.

Earth rotation and revolution (2020 2a, 2021 3b)

Rotation: the earth spins west to east on its own axis once every 24 h, giving day and night and the daily rise and fall of solar radiation on each facade (east wall in the morning, west in the afternoon).

Revolution: the earth travels round the sun in an elliptical orbit in 365¼ days with its axis tilted 23½° from the perpendicular to the orbit plane. The tilt makes the sun's noon altitude and day length change through the year, which creates the seasons.

Earth Revolution: While Earth is rotating on its axis, it is also traveling in an orbit around the sun—this revolution takes the Earth 365 ¼ days to complete.



Revolution of the earth: solstices (June 21-22, December 21-22) and equinoxes (March 21-22, September 22-23)

Solstice and equinox (2024 2a)

Solstice: the moment when the earth's axis points farthest toward or away from the sun. Summer solstice, 21 June: sun directly over the Tropic of Cancer (23½° N), longest day in the northern hemisphere. Winter solstice, 21/22 December: sun over the Tropic of Capricorn (23½° S), shortest day.

Equinox: the axis is tilted sideways, neither toward nor away from the sun; the sun is directly over the equator and day equals night everywhere. Spring (vernal) equinox about 21 March, autumnal equinox about 22-23 September.

Significance for building heat gain (2021 3b, 2022 3a)

- Seasonal sun altitude: high in summer, low in winter. A south overhang sized on this blocks the summer sun and admits the winter sun.
- Daily path: east and west facades get low-angle sun (hard to shade, glare, high gain in the afternoon); roofs get the most in summer.
- Solar data (altitude, azimuth for each month and hour) fix cooling load peaks, shading design, PV (photovoltaic (solar cells)) tilt and daylighting.
- From the energy view point: correct orientation and shading using the sun-earth geometry can cut cooling energy a lot, while winter sun can heat for free.

Weather vs climate (2021 3a)

Weather	Climate
Short-term conditions of the atmosphere at a place: what it is like outside here and now	Long-term average (30 years or more) of weather over a large area
Changes hour to hour, day to day	Changes over decades
Example: rain today in Khulna	Example: tropical monsoon climate of Bangladesh

Radiation at ground level and what changes it (2024 2a)

Of the extraterrestrial radiation (about 1367 W/m²), about 26 % is reflected/scattered back by clouds and particles, 18 % absorbed (ozone absorbs UV, CO_2 and water vapour absorb IR) and about 56 % reaches the surface. At ground level: **G = B + D + R** (beam + diffuse + reflected). Diffuse is 10-20 % on clear days and up to 100 % on overcast days.

Solar Radiation at the Ground Level

The solar radiation that is able to pass through the atmosphere and reaches the ground level has two main components according to the direction where these come from, as schematically presented in Fig: the *direct (beam)* and the *diffuse* solar radiation.

A third component, *the reflected/Albedo* solar radiation may also occur under certain conditions.

Global (total) solar radiation can be calculated using

$G = B + D + R$

where

– G is the global solar irradiance.

– B is the direct solar irradiance.

– D is the diffuse solar irradiance.

– R is the reflected solar irradiance.

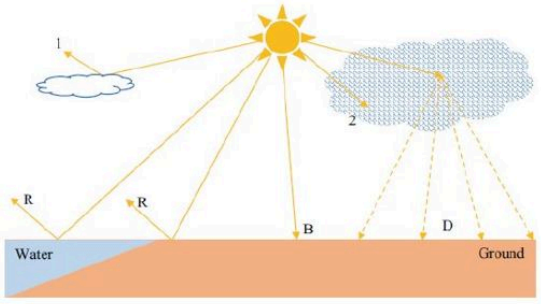


Fig. 2.4 Three components (*B, D, R*) of the solar radiation at ground level as it passes through the atmosphere where part of it is: reflected by the clouds (1) or absorbed by the clouds (2)

Components of solar radiation at the ground: direct (beam), diffuse and reflected (albedo)

Factors: latitude and longitude (sun angle, day length); season and time of day; topography (hills block, water/snow reflect); climate: cloud cover, humidity, dust and pollutants that scatter; altitude (thinner atmosphere); and in a built environment: orientation and tilt of the surface, shading by neighbours, albedo of surroundings, urban heat island and street canyons.

Orientation and the built environment (2020 2b, 2021 3a)

- Long axis east-west so the large faces look north and south: south face gets high summer sun that a small overhang blocks, north face gets diffuse light, little gain
- Minimise east and west glazing: low morning/afternoon sun gives large gain and glare and is hard to shade
- Orient openings to the prevailing summer wind (south/south-east in Bangladesh) for cross ventilation
- Surface tilt: a collector or roof facing south at about the latitude angle (about 23° in Khulna) gets the most yearly energy
- Wrong orientation can double the cooling load; the right one improves daylight, comfort and PV yield

Remember: Rotation = **day/night, 24 h**. Revolution + 23½° tilt = **seasons**. G = **B + D + R**. Bangladesh: **long axis E-W, shade south, avoid E/W glass**.

A10 Wind flow patterns and natural ventilation around buildings Must study

Very likely · 100% Medium

PAST-PAPER QUESTIONS

- **2024 Q3(d)** [7] Mention some design modifications that can influence the natural ventilation of a building.
- **2022 Q4(d)** [7] How to increase natural wind flow around building? Discuss with schematically.
- **2021 Q3(d)** [10] Draw five wind flow patterns around a building which will increase the natural wind flow and mention the amount of increment.
- **2020 Q3(a)** [6] Why is control of wind motion necessary for a building? Draw four wind flow patterns around a building which can help to increase air flow.

Before you start: wind around buildings

When wind hits a building it creates **positive pressure** on the windward face and **negative pressure (suction)** on the leeward face and roof. Openings on opposite sides let air flow through: that is **cross ventilation**.

Why control wind (2020 3a)

Wind is air in motion caused by uneven heating of the earth's surface and pressure differences; it is specified by speed and direction. For a building it changes from a resource to a nuisance with the seasons: in summer (hot-humid Bangladesh) we want more breeze for natural ventilation and comfort; in winter or storms we want shelter, less infiltration and safe pedestrian areas. Wind also controls infiltration, structural loads and pollution dispersal. So we design to increase summer wind and shelter from unwanted wind.

Basic behaviour

Before an obstacle the wind slows, builds positive pressure and turns up or sideways (windward = luv). As it passes, it speeds up and a low pressure forms at the sides and behind (leeward = lee). Behind the building is a **wind shadow** of quiet air extending 3-4 times the building size, with reverse eddies. Wind speed increases with height (height effect).

Luv, Lee and Wind Shadow

Wind Shadow: Due to wind being blocked by an obstruction such as a land-form, building or trees, a windless area of quiet air or low-winds is created on leeward side. This wind shadow area extends on downwind-side of the building for a length of 3-4 times its size, depending on its roof profile and dimensions (Figure 2a, 2b, 2c). Due to negative pressure zone formed, this zone may experience reverse downward currents or breeze with turbulence.

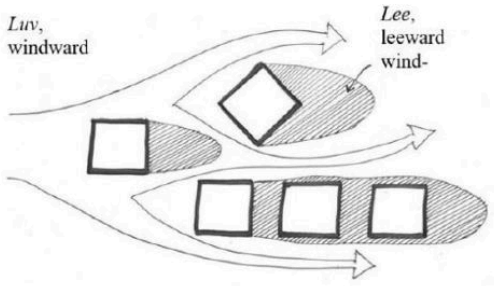


Fig.1: Luv, Lee, Wind Shadow

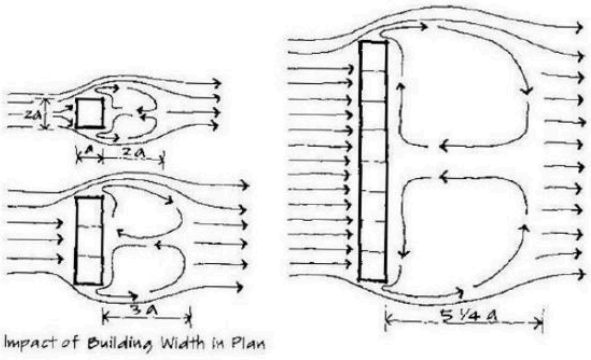


Fig.2a: Building width and Wind Shadow zone (Brown, DeKay, 2001)

Luv, lee and wind shadow behind buildings of different width and arrangement

Five wind patterns that increase wind flow (2021 3d, 2020 3a, 2022 4d)

Following Figures, show wind behavior to be expected from typical building arrangements.

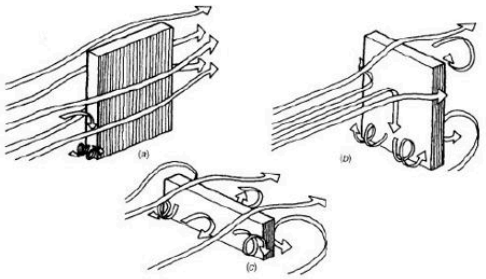


Fig. 1: *Wind patterns around single buildings. (a) Tall, slender buildings: height greater than 2.5 times the width. (b) Tall, rather wide buildings; height between 2.5 and 0.6 times the width. (c) Long buildings; height less than 0.6 times the width.*

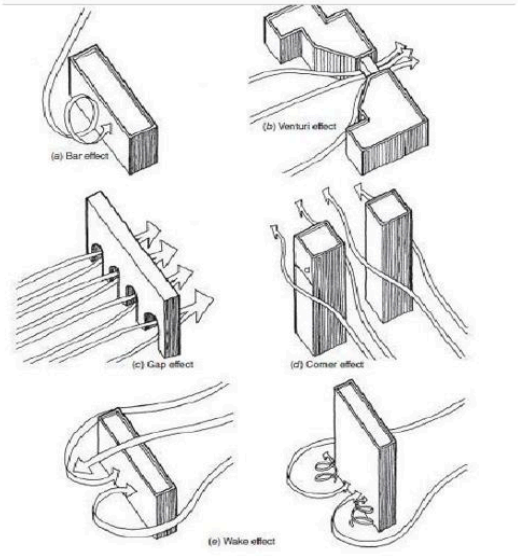
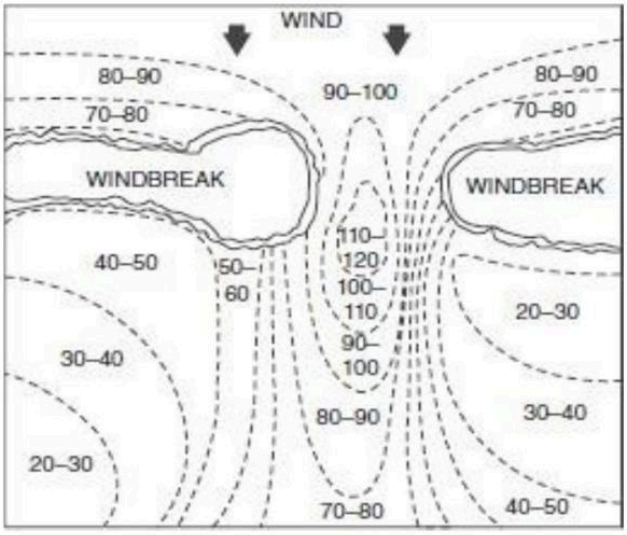


Fig. 2: *Wind patterns among building clusters*

Wind patterns around single buildings and building clusters (bar, Venturi, gap, corner, wake)

Pattern	Increase over ambient wind
(a) Bar effect: down-spinning wind behind a long building	up to 1.4×
(b) Venturi effect: wind squeezed through the narrow neck between buildings	1.3× up to 30 m height, 1.6× at about 50 m
(c) Gap effect: gap between buildings of 5+ storeys	1.2× at 7 storeys, 1.5× at 60 storeys
(d) Corner effect: higher buildings	1.2× at 15 m, 1.5× above 35 m; 2.2× between two 100 m towers
(e) Wake effect: turbulence behind 16-30-storey towers	1.4 to 2.2×
Gap in a windbreak (trees)	100-120 % in the gap (vs 20-60 % behind the solid part)

Increased wind speed can be produced by a gap in a windbreak. Figure below shows this effect.



Although a gap is a threat to protection from unwanted winter winds, it is an opportunity for enhancement of desirable summer winds.

Fig. *Wind speeds accelerate through a gap in a windbreak. Numbers indicate the percentage of the incoming (unaffected) wind speed.*

Wind accelerates through a gap in a windbreak; numbers are % of incoming wind speed

Design modifications for natural ventilation (2024 3d, 2022 4d)

- Orient the building and its openings to face the prevailing summer wind (within 45°)
- Cross ventilation: inlets on the windward side, outlets on the leeward side; outlet area equal or larger than inlet (raises speed inside)
- Inlets low (at body level), outlets high to add the stack effect; roof vents, ridge vents, solar chimneys, atrium
- Wing walls and vertical fins beside windows turn the wind into the room; casement windows that open outward as scoops
- Thin plan (depth less than 5 times ceiling height for single side, 15 m for cross ventilation), open plan, courtyards
- Space buildings so they are not in each other's wind shadow; stagger them; use gaps and Venturi channels; raise on stilts
- Landscaping: trees with clear trunks to channel breeze, hedges to deflect wind into windows, avoid dense windbreaks upwind

Remember: Wind speeds up at **gaps, corners, necks**; it stops in the **shadow (3-4× size)**. Ventilation: **face wind, inlet low/outlet high, outlet ≥ inlet, wing walls**.

A13 Daylighting systems: passive, active (sun tracker), top and side lighting

Must study · Very likely · 90% · Medium

PAST-PAPER QUESTIONS

- **2024 Q4(a)** [10] Between active and passive day lighting systems, which one is preferable and why? Write down working principle of sun-tracker system with schematic.
- **2024 Q4(b)** [9] Among different passive day lighting system, which one is more suitable for building in Bangladesh? Explain with diagram and examples.
- **2024 Q4(c)** [6] What is glare? Explain its causes and effects on visual comfort.
- **2022 Q4(b)** [10] With the help of neat sketch, describe the active daylighting system.
- **2021 Q4(a)** [8] What are the different options of natural lighting in building? Among top and side daylighting, which one is preferable? Explain.
- **2021 Q4(b)** [6] What is daylighting? What are the benefits and drawbacks of daylighting?
- **2021 Q4(d)** [9] Write short notes on: (i) Saw tooth lighting, (ii) Tubular daylighting, (iii) Daylight reduction [redirection].

★ REMEMBER IT

- **Passive** = no moving parts (light shelf, skylight, light pipe, saw-tooth roof). **Active** = moves to follow the sun (heliostat / sun tracker with mirrors or fibre optics). Passive suits Bangladesh: cheap, no maintenance, plenty of sun.
- Glare = too much brightness contrast in the field of view: causes discomfort and fatigue.

Before you start: daylighting

Daylighting = using sunlight/skylight to light rooms so electric lights can stay off. **Passive** systems are fixed (windows, light shelves, skylights, light pipes). **Active** systems move (sun trackers, mirrors) to follow the sun.

Daylighting (2021 4b)

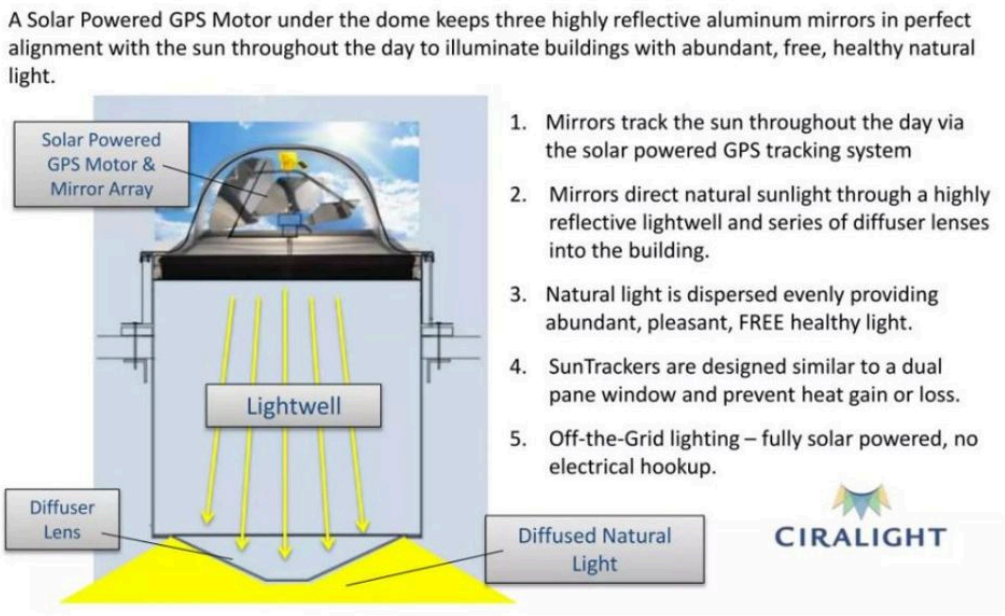
Daylighting is the controlled admission of natural light (direct sunlight and diffuse skylight) into a building to reduce electric lighting and save energy. Daylight enters by direct sun, clear sky, clouds (diffuse) and reflection from the ground and nearby objects.

Passive vs active (2024 4a, 2022 4b)

Passive daylighting	Active daylighting
Collects sunlight with stationary, non-tracking elements: windows, glass doors, most skylights, light tubes, light shelves	Uses a mechanical device (motorised mirrors, sun tracker) that follows the sun to collect more light
No moving parts, cheap, almost no maintenance, no energy input	Higher, steadier light output through the day; can light deep or windowless spaces
Output falls in early morning/late afternoon	Costly, needs power (often a small PV (photovoltaic (solar cells)) cell), maintenance of moving parts

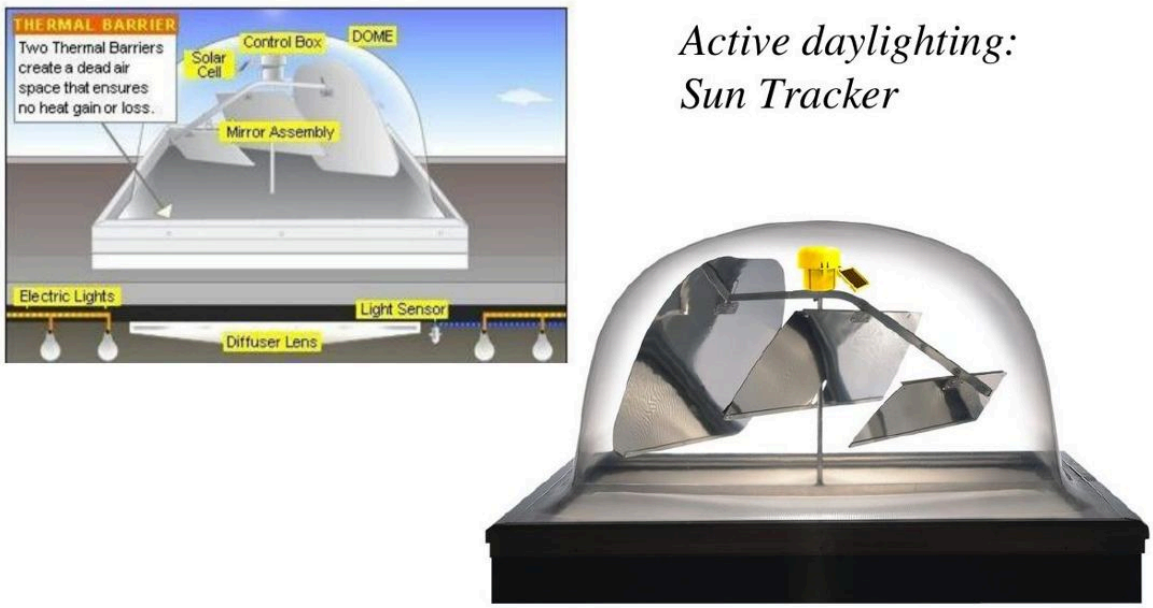
Which is preferable? For most buildings, and in Bangladesh with plenty of sunshine and limited budgets, **passive** daylighting is preferable: it is cheaper, reliable and maintenance-free and gives most of the 15-40 % lighting savings. Active systems are justified for deep-plan or windowless spaces (warehouses, gymnasiums, factories) where passive light cannot reach.

Sun tracker working principle (2024 4a, 2022 4b)



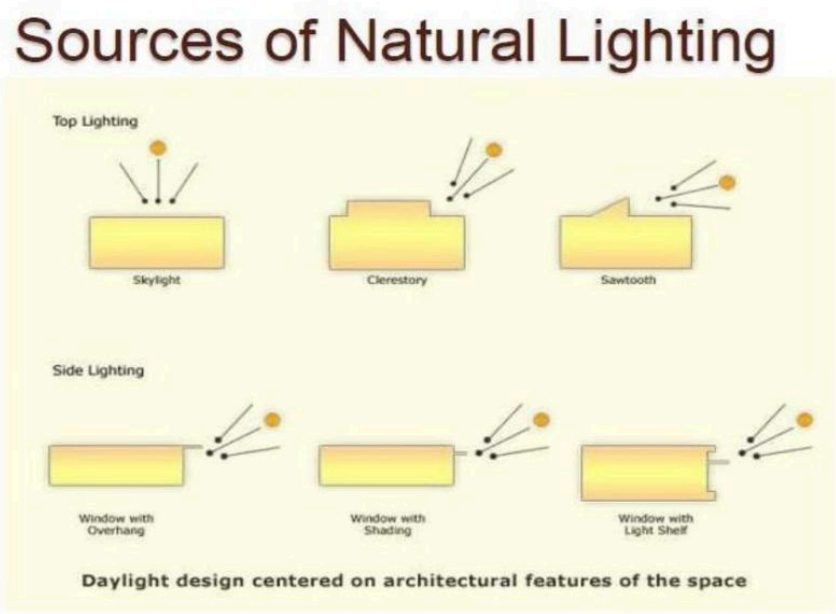
Active daylighting with a sun tracker: GPS-controlled mirror array under a dome sends sunlight down a reflective light well to a diffuser

1. A clear dome on the roof protects a mirror assembly
2. A small solar-powered GPS/motor controller turns three highly reflective aluminium mirrors to follow the sun all day
3. The mirrors reflect sunlight straight down a highly reflective lightwell (shaft)
4. A diffuser lens at the ceiling spreads the light evenly in the room; double-pane barriers stop heat gain/loss
5. Off-grid: the motor runs on its own PV cell, so no electrical hook-up



Sun tracker: dome, mirror assembly, solar cell, light well and diffuser

Options of natural lighting: top vs side (2021 4a)



Sources of natural lighting: top lighting (skylight, clerestory, saw-tooth) and side lighting (windows with overhang, shading, light shelf)

Top lighting (roof): skylights, monitors, saw-tooth roofs, clerestories, tubular devices. **Side lighting** (walls): windows, light shelves, clerestory windows high in walls. **Which is preferable?** A horizontal roof light admits about three times more daylight per m^2 than a vertical window and lights deep spaces evenly, so top lighting is better for single-storey, deep-plan spaces (factories, halls); it is limited to about 12 % of floor area because of heat gain. Side lighting gives view and ventilation and works for multi-storey buildings, but only to a depth of about 2-2.5 times the window head height. In practice a combination is best.

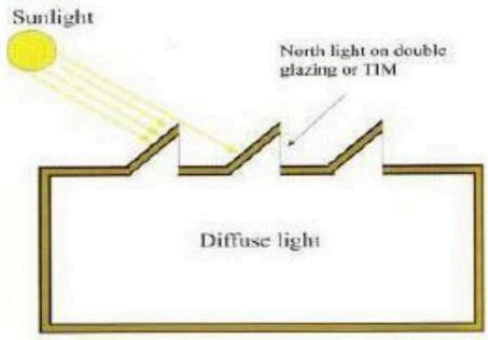
Short notes (2021 4d)

Saw-tooth lighting: a roof of repeated sloping sections with vertical or steep glazing facing the pole (north in Bangladesh). It admits diffuse north light evenly and avoids direct sun and heat gain; used in factories and workshops.

Saw Tooth Lighting

Heat gains can be reduced by tilting roof lights towards the North in order to utilize diffuse north lighting.



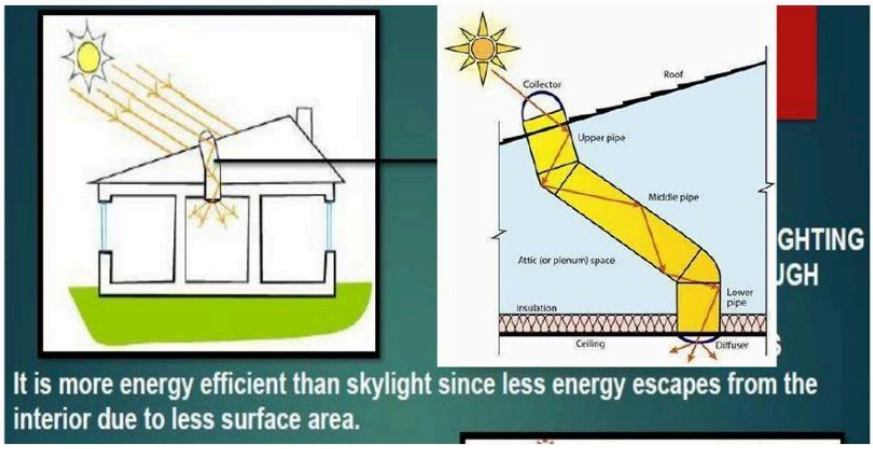


Saw-tooth roof lighting: glazing tilted toward the north gives diffuse light

Tubular daylighting (solar light tubes): a small clear dome on the roof captures sunlight; a tube lined with a highly reflective film carries it down by repeated reflections to a diffuser in the ceiling. Much smaller than a skylight, so less heat escapes; cheap, easy to install, good for corridors and windowless rooms.

Tubular daylight devices (Solar Light Tubes)

Tubular daylight devices are another type of top lighting device. Solar Light Tubes are tubes that capture sunlight from the rooftops and pipe them to the indoors of the buildings.

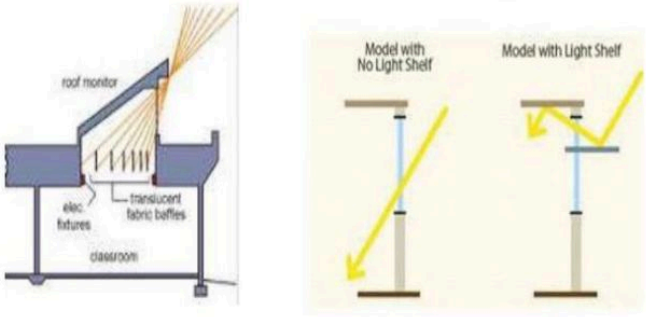


Tubular daylight device (solar light tube)

Daylight redirection (light shelf): devices that take direct beam sunlight and redirect it, usually onto the ceiling. Two functions: glare control (sun kept away from occupants' eyes) and deeper daylight penetration. Forms: a large horizontal element (light shelf) or louvre systems.

Daylight Redirection

□ Daylight redirection devices take incoming direct beam sunlight and redirect it, generally onto the ceiling of a space. These devices serve two functions: glare control, where direct sun is redirected away from the eyes of occupants, and daylight penetration, where sunlight is distributed deeper into a space that would not be allowed otherwise. Daylight redirection devices generally take one of two forms: a large horizontal element, or louvered systems. Horizontal daylight redirection devices are often called light shelves.



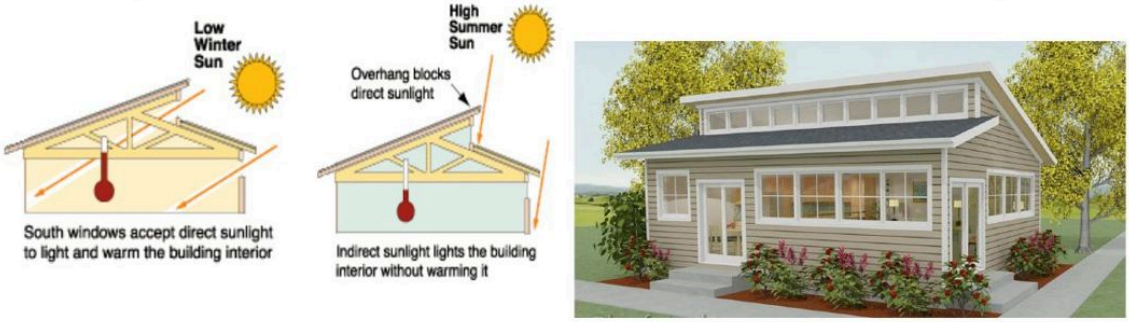
Daylight redirection: light shelf and louvres bounce sunlight onto the ceiling

Passive system for Bangladesh (2024 4b)

Bangladesh has strong sun, high cooling load and hot-humid climate, so the best passive daylighting gives **diffuse** light without heat: **side windows on north and south walls with overhangs and light shelves**, plus **clerestory windows** (which also vent hot air) and **tubular daylight devices** for deep or windowless rooms. Large unshaded skylights are not suitable because they bring heat. Examples: the National Parliament House (Louis Kahn) uses large light wells and openings that bring in diffuse, reflected light; Bait-ur-Rouf Mosque (Marina Tabassum) uses a skylight and perforated brick walls for soft daylight; factories in EPZs use saw-tooth/monitor roofs.

Clerestory Windows

- Clerestory windows are usually situated at a high level (near the ceiling of the room) - always above eye level.
- They provide an effective source of natural light and ventilation whilst reducing glare.



Clerestory windows: high-level glazing gives light and ventilation, overhang blocks high summer sun

Benefits and drawbacks (2021 4b)

Benefits	Drawbacks
15-40 % energy savings in lighting, lower HVAC (heating, ventilation and air-conditioning) load (less heat from lamps)	More heat gain if unshaded
Better light quality and colour rendition	High initial cost of high-performance systems
Occupant satisfaction, health, connection to nature; about 15 % higher productivity	Glare on work surfaces and screens
Less GHG emission	Roof shape, building angles may stop daylight reaching spaces

Glare (2024 4c)

Glare is excessive brightness or brightness contrast within the field of view that causes discomfort or reduced vision. **Causes:** direct sun or bright sky seen through windows, reflections from glossy surfaces or screens (reflected glare), bare bright lamps, large luminance contrast between a bright window and a dark wall, low sun on east/west facades. **Effects:** discomfort glare (annoyance, eye strain, headache, fatigue) and disability glare (reduced contrast and visibility, errors, accidents); occupants close blinds and switch lights on, wasting energy. Control with shading, light shelves, blinds, matte finishes, diffusers and correct screen orientation.

Remember: Passive = **still**, Active = **tracks the sun**. Top light = **3×** a window but ≤ **12 %** of floor. Savings **15-40 %**.

A8 Ambient temperature and sky temperature

Must study · Very likely · 90% · Medium

PAST-PAPER QUESTIONS

- **2024 Q2(d)** [5] Define ambient temperature and sky temperature. Explain how the difference between these two affects heat loss from building surfaces during nighttime.
- **2022 Q3(c)** [8] How to calculate sky temperature? How sky temperature influences the analysis of solar energy utilization in building?
- **2021 Q3(c)** [6] What is the relationship between ambient temperature and sky temperature? How sky temperature can be calculated?

UNDERSTAND **Read this to understand; you do not need to write it**

The sky is not empty: water vapour and CO_2 in the air radiate heat down to us. A clear dry sky radiates like a black body that is **colder than the air**. That imaginary temperature is the **sky temperature**.

A roof at night sends heat up as long-wave radiation, $q = \varepsilon\sigma(T_s^4 - T_{sky}^4)$. The colder the sky, the more it loses: roofs can drop below air temperature on clear nights (dew, frost, or free cooling).

CALCULATOR **Sky temperature and night radiation loss**

Change any number; every step is recalculated and shown with the numbers put in.

1. Air temperature in kelvin: **30** + 273 = **303.0** K
2. Swinbank sky temperature: $T_{sky} = 0.0552\ T_a^{1.5} = 0.0552 \times \text{pow}(\mathbf{303},1.5) = \mathbf{291.1}$ K
3. Sky temperature in °C: **291.1** - 273 = **18.1** °C
4. Surface in kelvin: **28** + 273 = **301.0** K
5. Net radiation loss: $q = \varepsilon\sigma(T_s^4 - T_{sky}^4) = \mathbf{0.9} \times 5.67\times10^{-8} \times (\text{pow}(\mathbf{301},4) - \text{pow}(\mathbf{291.1},4)) =$ **52.2** W/m²

Quick rule from the slides: $T_{sky} \approx T_a - 6$ °C. Clouds push T_{sky} up towards T_a, so the loss falls.

Before you start: sky temperature

A surface facing the sky loses heat by radiation to the sky. The clear night sky behaves like a very cold surface (often 10-20 °C colder than the air), which is why roofs can cool below air temperature at night.

Definitions (2021 3c, 2024 2d)

Ambient temperature T_a: the temperature of air measured in a shaded but well-ventilated enclosure (Stevenson screen). It depends on wind and local factors (shade, water bodies, sunshine).

Sky temperature T_{sky}: the equivalent temperature of the sky hemisphere treated as a black body, with which a surface on earth exchanges long-wave radiation. Because the clear atmosphere emits little, T_{sky} is lower than T_a.

How to calculate it (2022 3c)

Simplest: $T_{sky} = T_a - 6$ °C

Simplified Swinbank: $T_{sky} = 0.0552\ T_a^{1.5}$ (both in K)

Clear sky flux: $L_{clear} = 5.31\times10^{-13}\ T_a^6$; cloudy: $L_{cloud} = \varepsilon_c\sigma(T_a - 5)^4$

Real sky: $L_{sky} = (1 - c)L_{clear} + cL_{cloud}$, $c = (K_d - 0.165)/0.835$

$T_{sky} = (L_{sky}/\sigma)^{1/4}$

The actual sky condition lies between a clear sky and a totally overcast one. Sky radiation intensity for the sky close to real conditions is approximated with the relation:

$$L_{sky}(t) = (1 - c(t))\cdot L_{clear}(t) + c(t)\cdot L_{cloud}(t) \qquad c(t) = (K_d(t) - 0.165)/0.835$$

After transformations using Eqns, the sky temperature may be found as:

$$T_{sky}(t) = \sqrt[4]{\frac{\left(1 - \frac{K_d(t)-0.165}{0.835}\right)\left(5.31\times10^{-13}\cdot T_a(t)^6\right) + \frac{K_d(t)-0.165}{0.835}\left(\varepsilon_c\sigma(T_a(t) - 5)^4\right)}{\sigma}}$$

Sometimes a simplified Swinbanck formula (T_{sky} in a function of T_a (K)) is expressed as: $T_{sky}(t) = 0.0552\cdot(T_a(t))^{1.5}$

Detailed analyses of solar energy utilization in the building sector should use the above Eqn to determine the sky temperature, as for certain elements of a building envelope at certain weather conditions, heat exchange with the sky (hemisphere) significantly influences their thermal conditions.

Sky temperature model for a real sky between clear and overcast (course slide)

Example: T_a = 30 °C = 303 K → Swinbank $T_{sky} = 0.0552 \times 303^{1.5}$ = 291 K = 18 °C, i.e. about 12 °C below ambient on a clear night.

Effect on night-time heat loss (2024 2d)

Net long-wave loss from a surface is $q = \varepsilon\sigma(T_s^4 - T_{sky}^4)$. On a clear, dry night the sky is much colder than the air, so roofs and walls facing the sky radiate strongly and cool below the air temperature (radiative cooling): roofs can be several degrees colder than the air, causing dew or frost, extra heating load in cold climates, and a useful free cooling in hot climates (roof ponds, night sky radiators). Clouds and humidity raise T_{sky} close to T_a, so the loss falls. Horizontal surfaces lose most (they see the whole sky); vertical walls see only half.

Influence on solar energy analysis (2022 3c)

The sky temperature sets the radiation loss term of every sky-facing element: the sol-air temperature correction (about 4 °C for roofs, near 0 for walls), the top heat loss and efficiency of solar collectors and PV (photovoltaic (solar cells)), night cooling of roofs and the frost/condensation risk. Detailed analyses must use the full equation because for some envelope elements the exchange with the sky dominates their thermal condition.

Remember: $T_{sky} < T_a$; quick: $\mathbf{T_a - 6}$; Swinbank $\mathbf{0.0552\ T_a^{1.5}}$. Clear night = big loss; cloudy night = small loss.

A14 Daylighting design principles and building orientation/form Must study

Very likely · 80% Medium

PAST-PAPER QUESTIONS

- **2024 Q4(d)** [10] What should be the orientations and forms of the building according to the location of Bangladesh? Describe with schematic diagram.
- **2022 Q4(c)** [8] What factors should be considered while designing a natural daylighting system? Explain with proper example.
- **2020 Q3(c)** [16] Classify the daylighting system. In contrast of daylighting, write short note on (i) Glazing specification, (ii) Window height and location, (iii) Tubular daylight devices.

Before you start: designing for daylight

Light reaches about 1.5-2.5 times the window head height into a room. So narrow plans, high windows, light-coloured surfaces and the right orientation (north/south light is easiest to control) give good daylight with little glare and heat.

Classification of daylighting systems (2020 3c)

(1) By mechanism: passive (stationary) and active (tracking). (2) By position: top lighting (skylight, monitor, saw-tooth, clerestory, tubular devices) and side lighting (windows, clerestory windows, light shelves). (3) Redirection devices: light shelves and louvres. See A13 for each.

Principles / factors in designing natural daylighting (2022 4c)

1. **Orientation:** the equator-facing side is easy to shade with overhangs, light shelves or louvres; the pole-facing side gets steady diffuse light with no glare; east and west give inconsistent light and glare. So buildings longer on the east-west axis are better for daylight and visual comfort.
2. **Form:** thin linear, L, U or courtyard forms put every room near a facade; courtyards and thin buildings distribute light and heat to both sides.
3. **Glazing ratio and specification:** glazing ratio = glazing area / external wall area; 25-50 % is usually recommended. More glass gives more light but also summer gain and winter loss.
4. **Window height and location:** high, widely distributed windows; the higher the head, the deeper the daylight penetrates.
5. **Overhead daylighting:** roof lights give about 3 times the light of a vertical window per m^2, limited to 12 % of floor area.
6. **Daylight redirection:** light shelves and louvres to control glare and push light deeper.

Example: a classroom 7 m deep in Khulna: windows on the north and south walls, head height 3 m (daylight reaches about 2×3 = 6-7 m), glazing ratio about 30 %, south windows with an overhang and an internal light shelf, light-coloured ceiling; electric lights on daylight sensors.

Short notes (2020 3c)

(i) Glazing specification: the type of glazing controls both daylight and heat. Single glazing passes about 88 % of daylight and 83 % of solar heat; double 77-80 % and 65-70 %; tinted double only 29 % and 39 %; triple 70 % and 40-60 %. Triple glazing keeps the inner surface close to room temperature (better comfort), but tinted or reflective glass cuts daylight. Choose high visible transmittance with low solar heat gain (spectrally selective glass) for hot climates.

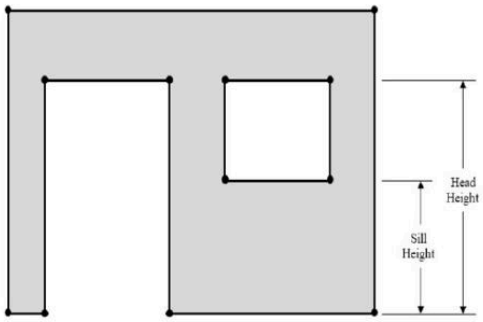
Window specifications	Daylight transmission	Solar transmission = direct heat from the sun
Single glazing	88%	83%
Double glazing	77-80%	65-70%
Double glazing - tinted	29%	39%
Triple glazing	70%	40-60%

Daylight and solar transmission of glazing types

(ii) Window height and location: windows should be high on the wall, widely distributed and of optimum area. A higher window head lets daylight penetrate deeper. Building code: offices, window opening 1.2 m with sill at 1.0-1.2 m above floor; residences, windows 1-1.1 m with sill at 0.7-0.9 m.

4. Window Height and location

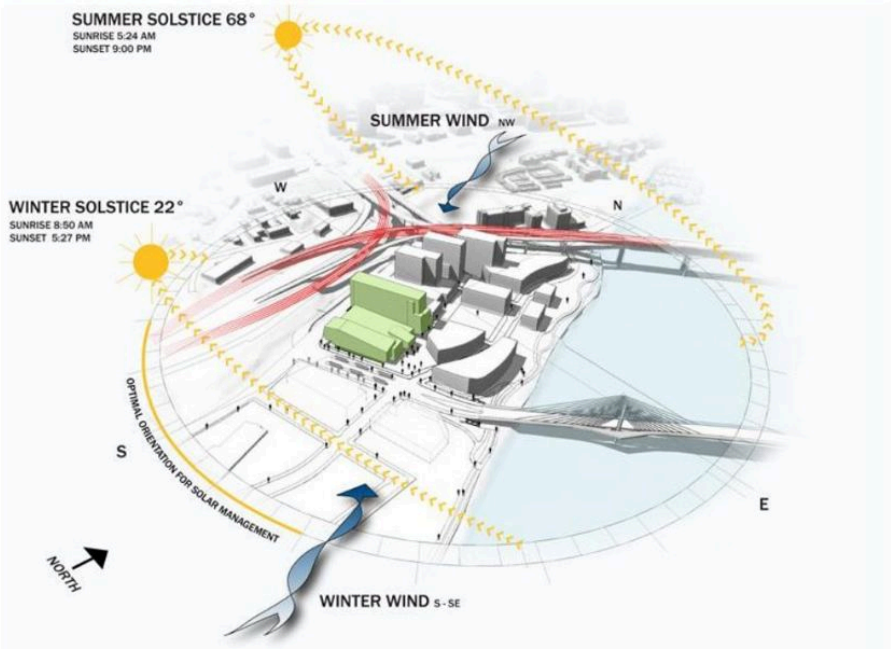
- Windows should be high on the wall, widely distributed and of an optimum area to achieve adequate daylighting.
- As a general rule, the higher the window head height, the deeper into the space the daylight can penetrate.
- Building code says-
- In office buildings, windows of 1.2m opening with sill height 1m-1.2m above floor and in residences, windows of 1- 1.1m and sill height to be 0.9-0.7m above floor.



Window head height and sill height

(iii) Tubular daylight devices: see A13: roof dome, reflective tube, ceiling diffuser; small, cheap, low heat loss.

Orientation and form for Bangladesh (2024 4d)



Orientation with summer and winter sun paths and winds (slide)

- **Orientation:** long axis east-west (rotated up to 15-20° toward south-east is acceptable to catch the south/south-east monsoon breeze). Main windows on north and south; minimum openings on east and especially west.
- **Form:** elongated rectangle or thin linear block (one or two rooms deep) for cross ventilation and daylight; L, U or courtyard forms for larger buildings; avoid deep compact plans that need artificial light and air.
- **Roof and openings:** wide overhangs and verandahs (sun + monsoon rain), high ceilings and clerestories to vent hot air, raised plinth for flooding.
- **Spacing:** buildings spaced so they are not in each other's wind shadow; trees on east/west.

> **Schematic (draw in the exam):** a long rectangle with its long sides facing N and S; arrows of summer wind from S/SE entering south windows and leaving north windows; overhang on the south face; few small windows on E and W with fins; trees on the west.

Remember: Six principles: **Orientation, Form, Glazing ratio (25-50 %), Window height, Overhead (≤12 %), Redirection**.

A12 Shadow angles (HSA, VSA) and overhang sizing numerical Must study

Very likely · 90% Hard

PAST-PAPER QUESTIONS

- **2024 Q3(c)** [8] Suppose a horizontal overhang is located at the south window of a building. The building is located at Fulbarigate facing due south. The height of the window from the sill is 0.9 m and the window is 0.5 m wide. Calculate the depth and width of the overhang considering the critical month and time.
- **2022 Q3(b)** [10] What are meant by horizontal shadow angle and vertical shadow angle? How to calculate them and mention the significance of them.
- **2022 Q4(a)** [10] How to identify the size of shading devices? Discuss with the help of sketchmatic diagram.
- **2021 Q4(c)** [0] (second half, marks counted in A11) Explain the horizontal and vertical shadow angle to determine the dimension of shading device.

LEARN IT FROM ZERO

Shadow angles, from zero

A shading device is just a sticking-out plate. To size it you need the angle of the sun **as seen from the window**, but split into two flat views, like an architect's drawings:

- **Plan view (from above):** how far the sun is to the side of the window's straight-out line. That is the **horizontal shadow angle, HSA** (horizontal shadow angle). It sizes **vertical fins**.
- **Section view (cut through the wall):** how high the sun looks when you stand facing the wall. That is the **vertical shadow angle, VSA** (vertical shadow angle). It sizes the **horizontal overhang**.

If the sun is straight in front of the window (HSA = 0) the VSA is just the sun's altitude. When the sun moves to the side, it "looks" higher in the section, so tan VSA = tan(altitude) / cos(HSA).

Then it is one right-angled triangle: the overhang depth D and window height H make tan VSA = H / D, so **D = H / tan VSA**.

Symbols used here

Symbol	Name and meaning	Unit
β (ALT)	Solar altitude: angle of the sun above the horizon	deg
γ_s	Solar azimuth: compass direction of the sun (0 = south here)	deg
γ_w	Wall azimuth: direction the window faces (0 = south)	deg
HSA	Horizontal shadow angle = $\gamma_w - \gamma_s$	deg
VSA	Vertical shadow angle, tan VSA = tan β / cos HSA	deg
H	Height of window to be shaded (sill to overhang)	m
D	Overhang depth (how far it sticks out)	m
W	Window width	m

CALCULATOR **Overhang sizing calculator (any window, any sun position)**

Change any number; every step is recalculated and shown with the numbers put in.

1. Horizontal shadow angle: HSA = $\gamma_w - \gamma_s$ = **0** - **(-8)** = **8.0** °
2. Vertical shadow angle: VSA = $\tan^{-1}$(tan β / cos HSA) = $\tan^{-1}$(tan(**67**) / cos(**8**)) = **67.2** °
3. Overhang depth: D = H / tan VSA = **0.9** / tan(**67.2**) = **0.378** m
4. Side extension each side: e = D tan HSA = **0.3783** × abs(tan(**8**)) = **0.053** m
5. Overhang width: W_o = W + 2e = **0.5** + 2 × **0.05317** = **0.606** m

Fulbarigate (22.9° N), September noon: altitude = 90 − 22.9 = 67°. Try 10 am (east window) or 3 pm (west) with your sun-path chart values.

Before you start: the two shadow angles

HSA (horizontal shadow angle) is seen from above (plan): how far round the sun is from straight in front of the window. **VSA** (vertical shadow angle) is seen from the side (section): how high the sun appears when you look straight out of the window. An overhang is sized with VSA; a fin with HSA.

Two formulas only: **tan VSA = tan(altitude) / cos(HSA)** and **depth = window height / tan VSA**.

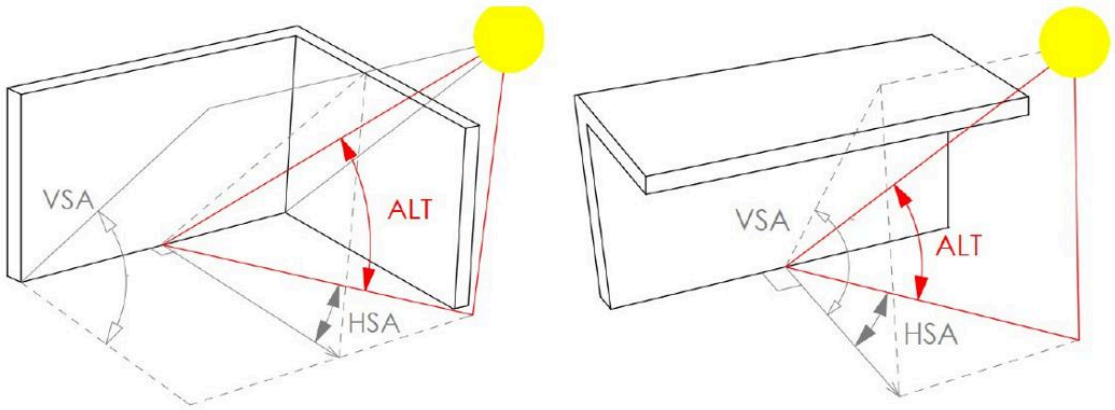
Horizontal and vertical shadow angles (2022 3b, 2021 4c)

Horizontal shadow angle (HSA): the angle, measured in plan, between the normal to the window and the solar azimuth. It is used to design **vertical** devices (fins).

```
HSA = wall azimuth - solar azimuth   (if |HSA| > 90° the sun is behind the facade)
```

Vertical shadow angle (VSA): the angle, measured in section on a plane perpendicular to the wall, between the horizontal and the line to the sun (the plane through the window's bottom edge and the sun). It is used to design **horizontal** devices (overhangs).

```
tan VSA = tan(solar altitude) / cos(HSA)
```



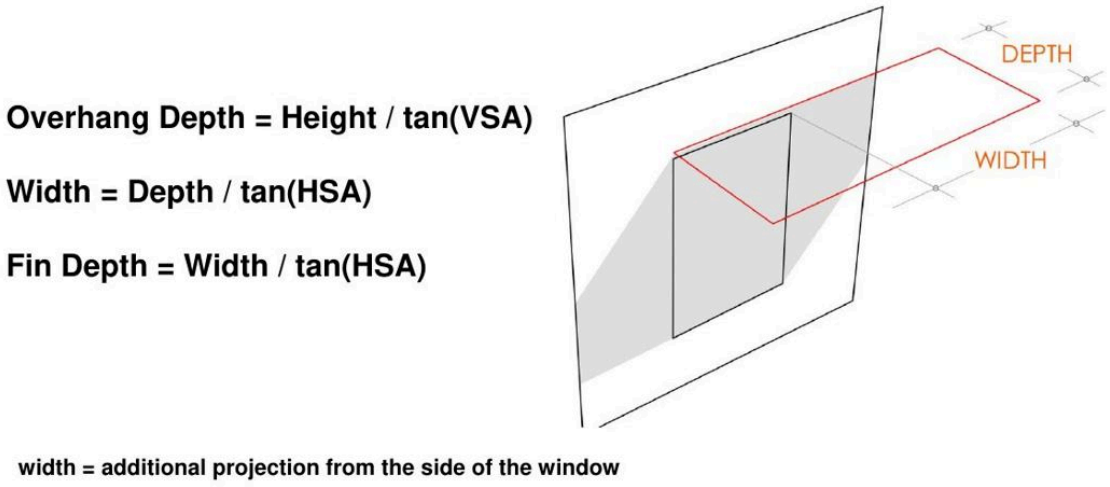
VSA and HSA for an overhang: altitude ALT (solar altitude angle), horizontal angle HSA, and the resulting vertical shadow angle VSA

Significance: they describe the performance of a shading device rather than just the sun position. A device with a given VSA shades the window completely whenever the sun's VSA is larger; a fin with a given HSA works whenever the sun's HSA is larger. They convert the 3-D sun path into the 2-D section and plan used for sizing.

Sizing a shading device (2022 4a)

1. Determine the cut-off date (before it, full shading; after, partial)
2. Fix start and end times of full shading (near sunrise/sunset the shade grows very large)
3. Read solar altitude and azimuth for each time on the cut-off date from a sun-path diagram or tables
4. Compute HSA and VSA
5. Compute depth and width

```
Overhang depth D = H / tan VSA      (H = window height from sill)
Fin depth = W / tan HSA      (W = window width to be covered)
Side extension of an overhang = D × tan HSA (each side)
```

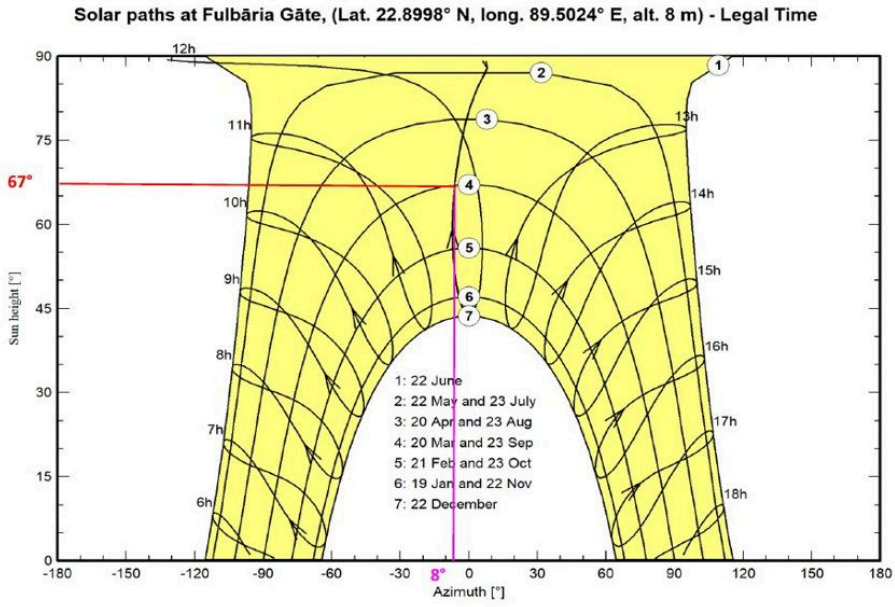


Overhang depth and width from VSA and HSA (slide)

The slide writes "Width = Depth / tan(HSA)"; geometrically the sideways shift of the shadow is D·tan(HSA), which is what we use below. Mention both if asked to quote the slide.

Critical months/times (slides): September noon for south windows, September 10 am for east, September 3 pm for west.

Numerical: overhang at Fulbarigate (2024 3c)



Sun path at Fulbarigate (22.9° N): at the September equinox, solar noon (legal time) the sun is at altitude 67°, azimuth ≈ −8°

Given: south-facing window (wall azimuth 0°), height H = 0.9 m, width = 0.5 m, location Fulbarigate, Khulna (φ = 22.9° N).
Critical time: September (equinox, 23 Sep), solar noon. From the sun-path chart: altitude β = 67°, azimuth ≈ −8° (check: 90° − 22.9° = 67.1°).
HSA = wall azimuth − solar azimuth = 0 − (−8°) = 8°
tan VSA = tan 67° / cos 8° = 2.356 / 0.990 = 2.379 ⇒ VSA = 67.2°
Depth D = H / tan VSA = 0.9 / 2.379 = **0.38 m**
Side extension = D × tan HSA = 0.38 × tan 8° = 0.38 × 0.141 = 0.053 m on each side
Overhang width = 0.5 + 2 × 0.053 ≈ **0.61 m** (round up to 0.65 m in practice)

If the examiner wants the slide form W = D/tan(HSA) = 0.38/0.141 = 2.7 m, that is the projection needed for a side fin, not the overhang width. Write the geometric answer and state the assumption (September noon, sun-path chart values).

Remember: HSA → fins (plan). **VSA** → overhang (section). **tan VSA = tan ALT / cos HSA**, **D = H / tan VSA**.

Same answer, step by step with reasons

Step (write this)	Why / what it means
`Wall azimuth = 0° (facing south); window H = 0.9 m, W = 0.5 m; φ = 22.9° N`	Write what you know.
`Design time: September equinox, solar noon: altitude β = 90 - 22.9 = 67.1° (chart 67°), azimuth ≈ -8°`	At the equinox the noon sun altitude = 90° − latitude.
`HSA = 0 - (-8°) = 8°`	How far round the sun is from straight in front of the window.
`tan VSA = tan 67° / cos 8° = 2.356/0.990 = 2.379 ⇒ VSA = 67.2°`	Converts real sun height into the height seen in the window's section.
`Depth D = H / tan VSA = 0.9/2.379 = 0.38 m`	Right triangle in section: the overhang must reach out far enough that the sun ray just grazes the sill.
`Side extension = D tan HSA = 0.38 × 0.141 = 0.053 m each side; width = 0.5 + 0.106 ≈ 0.61 m`	Because the sun is a little to one side, the shadow shifts sideways; extend the overhang to cover it.

Memory trick: "V for Verandah (overhang), H for Hand-fin": VSA sizes overhangs, HSA sizes fins.

WRITE IN EXAM **Write this in the exam (this much is enough for full marks)**

2024 3(c) Given: south window ($\gamma_w = 0°$), H = 0.9 m, W = 0.5 m, Fulbarigate $\varphi = 22.9°$ N

Critical time for a south window: September equinox, solar noon. From the sun-path chart: $\beta = 67°$, $\gamma_s \approx -8°$ (check: $90° - 22.9° = 67.1°$)

HSA: $\gamma_w - \gamma_s$

$= 0 - (-8°)$

$= \boxed{8°}$

VSA: $\tan^{-1}(\tan \beta / \cos \text{HSA})$

$= \tan^{-1}(2.356 / 0.990) = \tan^{-1}(2.379)$

$=$ **67.2°**

Depth: $D = H / \tan \text{VSA}$

$= 0.9 / 2.379$

$=$ **0.38 m**

Side extension: $e = D \tan \text{HSA}$

$= 0.38 \times 0.141$

$=$ **0.053 m each side**

Overhang width: $W + 2e$

$= 0.5 + 2 \times 0.053$

$=$ **0.61 m (use 0.65 m)**

Assumption: sun position from the Fulbarigate sun-path chart, September noon. A small sketch of the section (window, overhang, sun ray at VSA) gets an extra mark.

A2 Indoor air contamination: sources and control Must study Likely · 70% Easy

PAST-PAPER QUESTIONS

- **2024 Q3(a)** [8] What are the ways by which built environment can be contaminated by HVAC (heating, ventilation and air-conditioning) system? How to prevent that?
- **2021 Q2(b)** [12] What are the major sources of indoor air contaminations? Describe the personal responsibility to reduce the indoor air contaminations.
- **2020 Q1(c)** [10] What are the main sources of indoor air contamination?

Before you start: indoor air pollution

Indoor air can be dirtier than outdoor air: people breathe out CO_2, cooking and smoking give particles and CO, paints, furniture and cleaning products give off volatile organic compounds (VOCs), damp walls grow mould. Control = remove the source, ventilate (dilute with fresh air) or filter/clean the air.

Sources of indoor air contamination (2020 1c, 2021 2b)

Source group	Examples
1. Outside the building	Polluted outdoor air (dust, spores, industrial gases, vehicle exhaust); nearby sources: parking, loading docks, dumpsters; re-entrained exhaust; debris near the outdoor-air intake
2. Equipment	HVAC: dirt in ducts, microbes in humidifiers/coils/drain pans, wrong cleaning chemicals, poor venting of combustion products, refrigerant leaks. Non-HVAC: office machines (VOC, ozone), toners, solvents, labs, lift motors
3. Human activities	Personal: smoking, cooking, body and cosmetic odours. Housekeeping: cleaning agents, stored trash, dust raised by sweeping. Maintenance: cooling-tower mist, paints, adhesives, pesticides
4. Building components and furnishings	Carpets, curtains (dust, fibres), damaged asbestos, water-damaged furnishings, condensation, clogged drains, dry traps, VOCs (volatile organic compounds) from new materials
5. Other	Spills, floods, fire damage, smoking lounges, print shops, beauty salons, kitchens, renovation dust and fumes

Personal responsibility to reduce contamination (2021 2b)

- Do not smoke indoors; use the kitchen hood while cooking
- Keep the workplace clean, do not store food waste, empty bins daily
- Use low-VOC and fragrance-free products; close solvent and toner containers
- Do not block supply or return grilles; report leaks, odours and water damage
- Keep windows/doors closed near traffic when the HVAC is running so filters work

How the HVAC system contaminates a building and prevention (2024 3a)

Way of contamination	Prevention
Outdoor-air intake near exhaust, parking or dumpsters pulls in pollutants	Locate intakes away from (and upwind of) exhausts and roads; keep the area clean
Dirty filters, dust in ducts	Correct filter grade, scheduled filter change, duct cleaning
Microbial growth in wet coils, drain pans, humidifiers, cooling towers (Legionella)	Sloped drain pans, UV lamps, water treatment, regular cleaning
Poor ventilation rate or poor air distribution	Supply the design outdoor air (7.5 L/s per person), balance the system, check ventilation efficiency
Combustion products and refrigerant leaks	Proper flues, leak detection, regular maintenance
Wrong pressure relationships spread odours	Negative pressure in toilets, kitchens and labs with dedicated exhaust

Control order: eliminate the source → ventilate (dilute) → filter/clean the air → measure and verify.

Remember: Five sources: **Out-Equip-Human-Build-Other**. Control: **Source, Ventilate, Filter, Measure**.

A3 Energy use in buildings: envelope, use pattern, performance factors

Must study | Likely · 60% | Easy

PAST-PAPER QUESTIONS

- **2024 Q1(a)** [8] Define building envelop and building fabric? Why building energy is a concern of study in current times?
- **2021 Q1(a)** [8] How should building energy performance contribute to reduce the energy consumption?
- **2021 Q1(b)** [9] Explain the energy use in building. What are the activities influence the energy use in buildings?
- **2021 Q2(d)** [7] Explain the energy used pattern in a building.

Before you start: where a building's energy goes

A building uses energy mainly for cooling/heating (HVAC), lighting, hot water and equipment. The **envelope** (walls, roof, windows) decides how much heat leaks in or out; the **use pattern** (hours, occupants) decides how much internal heat and lighting is needed.

Building envelope and fabric; why building energy matters (2024 1a)

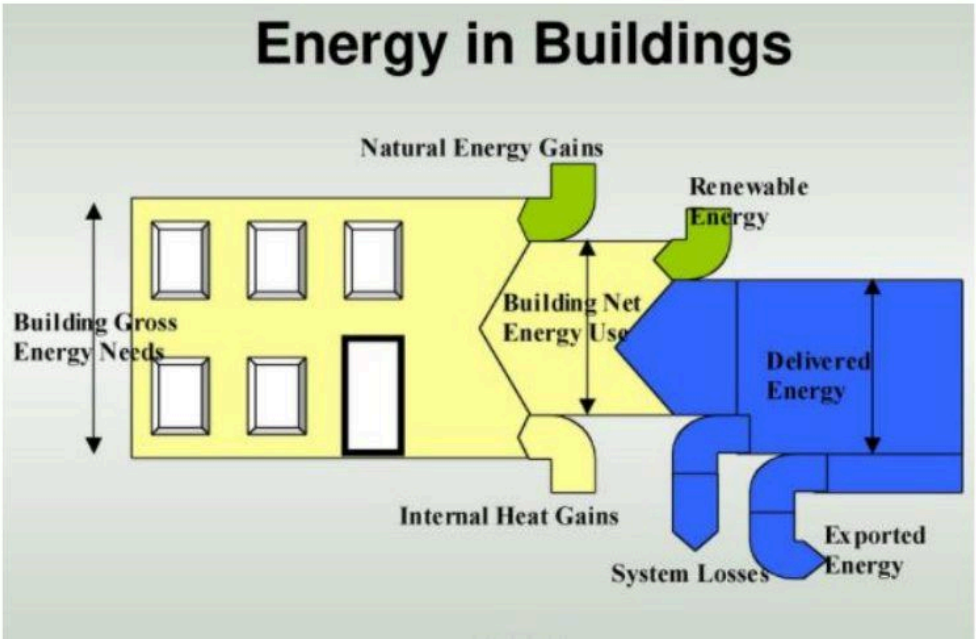
A **building** is a structure of roof, walls, floors and openings for human occupancy or activity. The **building envelope** is the boundary that separates the interior from the outside environment; it has five elements: foundation, exterior walls, roof, windows and doors. The **building fabric** is the components and materials the building itself is made of (walls, floors, roof, windows, doors and their layers).

Why it is a concern now: people spend 80-90 % of their lives indoors; buildings use about 40 % of world energy (36 % of final energy in 2018, up from 33.7 % in 1980) and cause about 39 % of energy-related CO_2 (about 19 % of all GHG). Comfort demand keeps rising, so cutting building energy is the cheapest way to fight climate change and energy shortage.

Energy use in a building (2021 1b, 2d)

By life stage: initial embodied energy (materials, transport, construction) → operational energy (heating, cooling, ventilation, lighting, hot water, cooking, appliances) → maintenance energy (replacement, refurbishment) → demolition energy.

Energy use pattern: energy enters as fuels and electricity plus "free" energy (sun, people). All of it ends up as heat leaving by transmission through the fabric, infiltration and ventilation, flue and cooling-tower losses. Typical end-use split: HVAC (heating, ventilation and air-conditioning) 35 %, lighting 11 %, major appliances (water heating, refrigerators, dryers) 18 %, others/electronics 36 %.



Energy flow of a building: gross need is met by natural gains, internal gains and delivered energy; part is lost in systems or exported

Terms: natural energy gains (passive solar, daylight, natural ventilation); internal gains (people, lights, appliances); delivered energy (bought to meet net demand, can be supplemented by on-site renewables); exported energy; system losses (conversion and distribution).

Activities and factors that influence energy use (2021 1b)

- Purpose of the building (office, lab, hospital), its age and type of construction, insulation, window type and shading
- Outside air temperature (climate), occupancy and HVAC schedules, sophistication of temperature control, plug loads and special equipment
- Indoor activities: metabolic rate of occupants (1 met = 58.2 W/m²), cooking, hot water, office equipment, lighting hours

How building energy performance reduces consumption (2021 1a)

Energy performance is how little energy the building needs to deliver the required indoor climate. It depends on (1) the properties of the building (envelope, form, orientation), (2) how efficiently delivered energy meets the net demand (system efficiency), (3) how efficiently people use energy, (4) the share supplied by renewables. Rating and benchmarking it (kWh/m²·yr, energy labels) shows where losses are, sets targets, and drives the measures of A4: better envelope, efficient HVAC/lighting, controls and occupant awareness. "If you cannot measure it, you cannot improve it" (Kelvin).

Remember: Envelope = **5 elements** (foundation, walls, roof, windows, doors). Life stages: **Embodied, Operational, Maintenance, Demolition**. Split **35-11-18-36**.

A5 Thermal and visual comfort in building design

Must study · Possible · 50% · Medium

PAST-PAPER QUESTIONS

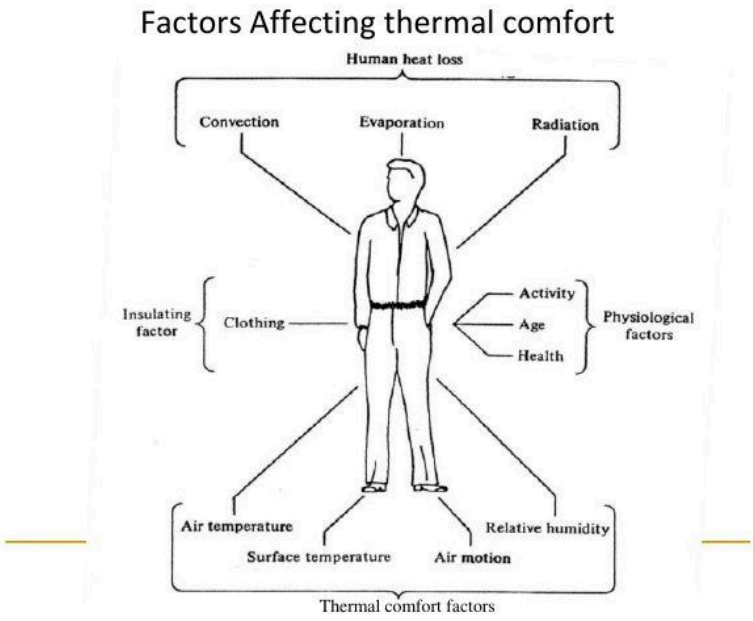
- **2022 Q1(b)** [8] What is meant by thermal comfort? Explain the impact of thermal control on building design.
- **2022 Q1(c)** [9] Discuss the aspects of buildings design on the followings: (i) thermal comfort, (ii) visual comfort, (iii) acoustic comfort.
- **2022 Q2(d)** [12] Discuss the design considerations of a building wall and roof for different climate conditions. Also discuss the effect of wall and roof materials on visual comfort.
- **2021 Q2(a)** [8] What is visual comfort? Explain the design consideration for visual comfort of a building.

Before you start: comfort

Thermal comfort = feeling neither too hot nor too cold. It depends on air temperature, radiant (surface) temperature, humidity, air speed, clothing and activity. **Visual comfort** = enough light without glare.

Thermal comfort and its impact on design (2022 1b)

Thermal comfort (ASHRAE): "that state of mind which expresses satisfaction with the thermal environment". Six factors: four environmental (dry-bulb air temperature, mean radiant temperature, air speed, relative humidity) and two personal (metabolic rate, met; clothing insulation, clo: 1 clo = 0.155 m^2K/W).



Thermal comfort factors: physiological (activity, age, health), clothing, and environmental (air and surface temperature, humidity, air motion)

Impact of thermal control on design: achieving thermal comfort drives most building energy use, so the designer must

- Keep temperature uniform: zone by orientation (one thermostat per zone), mix air to avoid stratification

- Control radiant asymmetry: insulate walls and roof, avoid occupants next to hot/cold glass, use low-e glazing and shading
- Avoid drafts from large cold surfaces and diffusers; control humidity (high RH feels like higher temperature)
- Select HVAC (heating, ventilation and air-conditioning) controls for the five operation factors: air movement, air temperature, humidity, vertical gradient, radiant surface temperature

Design aspects for thermal, visual and acoustic comfort (2022 1c)

Comfort	Design aspects
Thermal	Orientation and shading, insulation and thermal mass, window type, natural ventilation, zoning, HVAC with good controls
Visual	Daylight through well-placed windows and skylights, glare control (shades, light shelves), user-controlled task lighting, uniform luminance, good colour rendering
Acoustic	Sound insulation of facade and partitions, absorptive surfaces, quiet HVAC, room shape and volume, separating noisy and quiet zones

Visual comfort and design considerations (2021 2a)

Visual comfort is "the subjective condition of visual well-being induced by the visual environment". A lighting system must give enough illumination for safety and movement, visual performance and colour perception, without glare.

- Electric lighting by user-controlled task lights wherever practical
- Localised controls with variable light levels accessible to users
- Variety of sources to allow user choice
- Design to light delivered at the task, not the general space; discourage over-illumination of non-task areas
- Maximum use of daylight with shading (light shelves, blinds) and daylight sensors; LEDs; caution with new sources until health effects are known

Wall and roof design for different climates (2022 2d)

Climate	Wall	Roof
Hot-humid (Bangladesh)	Light-weight, light-coloured, shaded, permeable for cross ventilation; small thermal mass	Insulated, reflective (white/cool roof), ventilated double roof, wide overhangs for rain and sun
Hot-dry	Heavy, thick walls with high thermal mass (time lag ~8-12 h), light colour, small openings	Thick, massive, reflective; roof pond / night radiative cooling
Cold	Heavy insulation, airtight, dark/absorptive on sunny side, double/triple glazing	Well insulated, airtight, dark if snow-free, sloped
Composite	Moderate mass + insulation, movable shading	Insulated, reflective, shaded

Effect of materials on visual comfort: light-coloured, matte interior finishes (high reflectance) spread daylight deeper and reduce contrast; glossy surfaces cause reflected glare; dark walls/roofs absorb light and need more electric lighting; translucent roof elements or clerestories bring diffuse light; external reflective surfaces may throw glare into neighbouring windows.

Remember: Thermal comfort: **4 environmental** (T, MRT, v, RH) + **2 personal** (met, clo).

A9 Urban heat island and urban wind island effects

Good to have · Possible · 40% · Medium

PAST-PAPER QUESTIONS

- **2024 Q2(b)** [10] Define urban heat island effect. With necessary diagram, briefly describe how urban heat island effect occurs.
- **2024 Q2(c)** [10] Wind is urban areas behaves differently compared to rural areas. Discuss the urban wind island effect and evaluate how urban geometry alters wind speed, turbulence, and flow direction.

Before you start: urban heat island

Cities are warmer than the countryside around them (by 2-8 °C at night) because concrete and asphalt store heat, there is less vegetation and evaporation, and cars and air-conditioners release heat. Tall buildings also change wind speed and direction.

Urban heat island (UHI) effect (2024 2b)

Definition: the urban heat island is the phenomenon in which a city's air and surface temperatures are noticeably higher (often 2-5 °C, more at night) than in the surrounding rural areas.

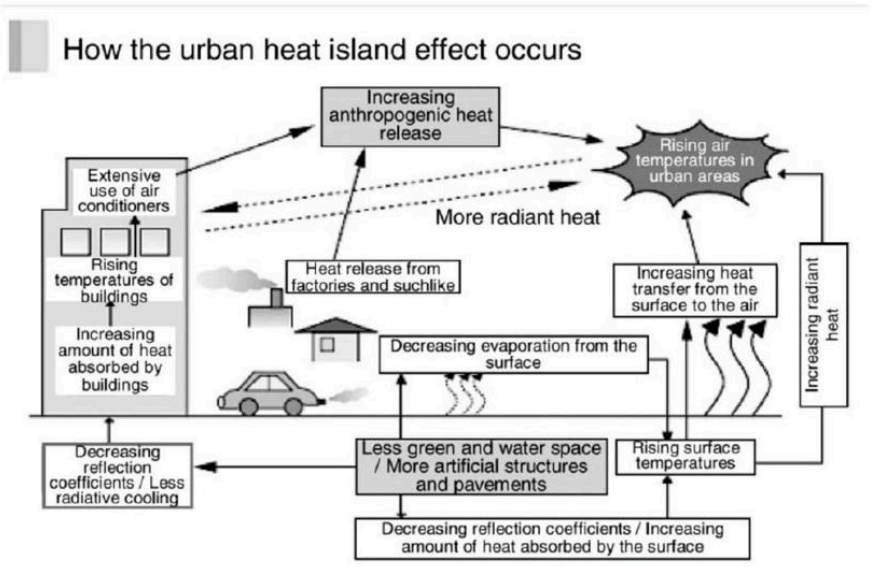


Fig. 1. How the Urban Heat Island occurs - image sourced from Yamamoto (2006).

http://dx.doi.org/10.1016/j.jenvman.2017.03.095

How the urban heat island effect occurs (slide diagram)

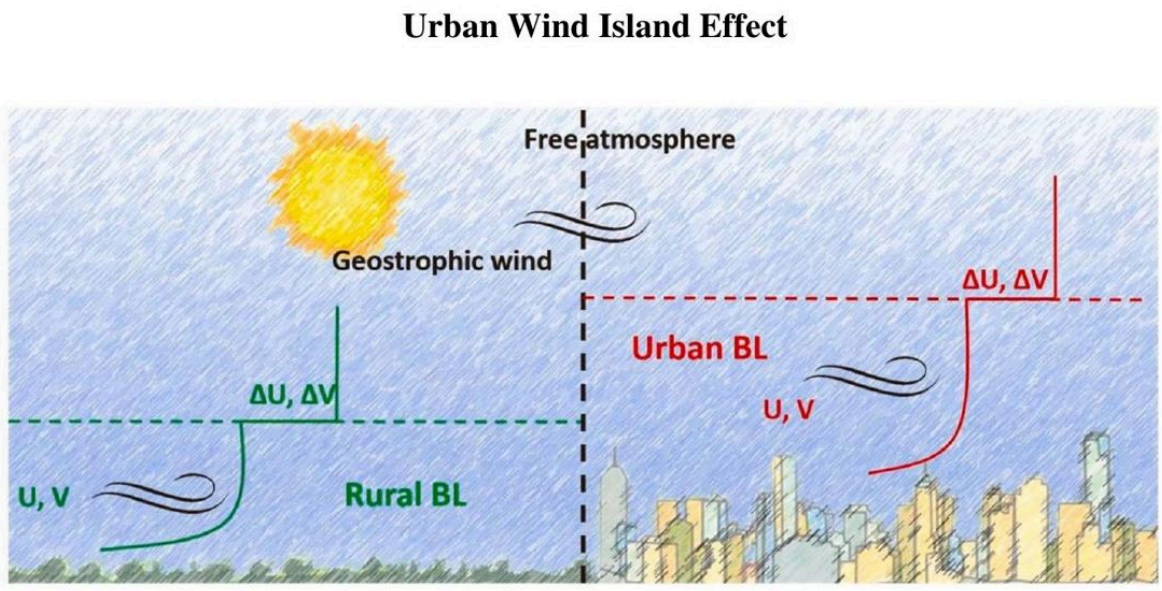
How it occurs (follow the diagram):

1. More anthropogenic heat: air conditioners, factories, vehicles release heat directly to the air
2. Less green and water space and more artificial surfaces: less evaporation and shading, so less evaporative cooling
3. Dark asphalt and concrete have low reflectance (albedo) and high absorptivity, so they absorb and store more solar heat during the day
4. Tall buildings and narrow streets (street canyons) trap radiation by multiple reflections and block the view of the cold sky, so night radiative cooling is reduced
5. Reduced wind speed in dense areas removes heat slowly

6. The stored heat is released at night; air temperature stays high, so more air conditioning is used, which releases even more heat (a vicious cycle)

Mitigation: cool/white roofs and pavements, green roofs and trees, water bodies, wider streets and ventilation corridors, less waste heat.

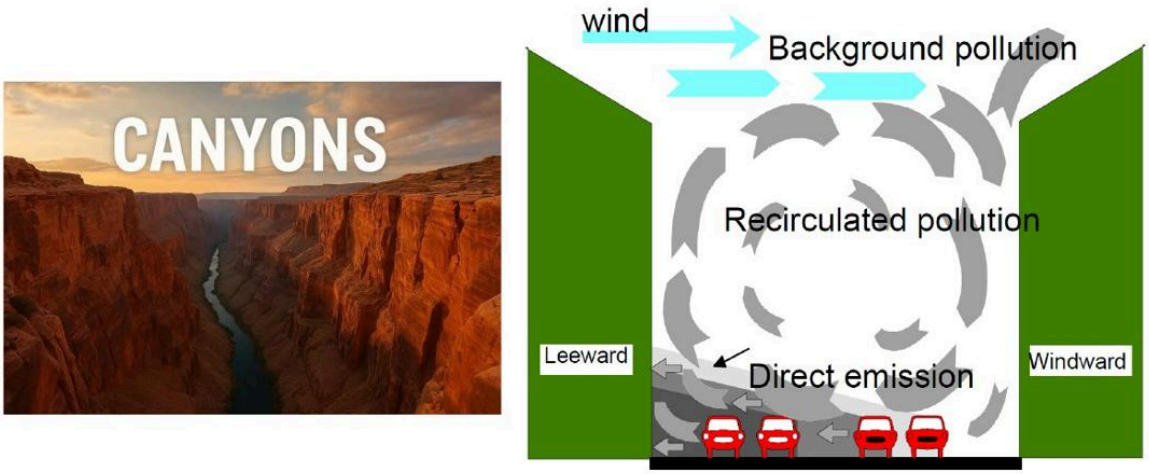
Urban wind island effect and urban geometry (2024 2c)



Urban wind island: the rural boundary layer is smooth; the urban boundary layer is deeper with changes of speed ΔU, ΔV

In rural areas wind meets few obstacles and flows smoothly with nearly constant speed and direction. In a city it is obstructed, deflected and sometimes amplified by buildings, so speed, direction and turbulence differ from the countryside. Cities can be windier than rural areas around sunrise and sunset, when temperatures are in transition; this is the **urban wind island effect**. It is hard to measure and model because of the complex and changing geometry.

Urban geometry	Effect on wind
Higher surface roughness (many buildings)	Lower mean wind speed near ground, deeper boundary layer, more turbulence
Tall buildings	Strong down-wash to street level, corner streams 1.2-2.2× ambient, large wake (wind shadow 3-4 times building size) with reverse flow
Gaps and narrow passages (Venturi)	Acceleration 1.2-1.6× ambient
Street canyons	Wind over the roofs forms a rotating vortex inside the canyon: air direction at street level reverses, pollution recirculates and is trapped
Street direction	Streets parallel to the wind channel and accelerate it; perpendicular streets shelter



Street canyon effect: recirculating vortex traps pollution at street level

Remember: UHI (urban heat island): **waste heat + dark surfaces + no green + canyons + low wind** → city hotter, especially at night.

Section B: Thermal performance, loads, annual energy, NZEB

B11 Net-zero energy and green buildings, LEED

Must study · Very likely · 100% · Easy

PAST-PAPER QUESTIONS

- **2024 Q8(b)** [7] List and describe key sustainability features of green buildings. Describe HVAC (heating, ventilation and air-conditioning) optimization strategies for net-zero buildings.
- **2022 Q8(a)** [12] Why net-zero energy building important? What energy efficiency measures should be taken in the building? How to design a green building?
- **2022 Q8(c)** [7] What are the functions of LEED (Leadership in Energy and Environmental Design (green building rating))? How to achieve a LEED certificate? Discuss.
- **2021 Q8(a)** [9] What is meant by nearly zero-energy building? Distinguish between net-zero site energy and net-zero source energy building.
- **2021 Q8(b)** [9] Discuss the principles for designing a net-zero energy building.
- **2021 Q8(c)** [8] Discuss about the challenges of technologies for low energy building.
- **2020 Q6(b)** [10] What is the basic difference between USA and European Union on the definition of zero-energy building? Why we need net-zero energy building?
- **2020 Q6(c)** [10] What are the principles of net-zero energy building design? Explain.

UNDERSTAND **Read this to understand; you do not need to write it**

Net-zero energy building: over one year the energy it makes on site (PV, solar thermal) equals the energy it uses. Step 1 is always **cut demand** (envelope, shading, daylight, efficient HVAC and lights); step 2 is **make the rest** with renewables.

Site energy counts kWh at the meter. **Source** energy counts the primary fuel burned at the power plant to deliver it (about 3 kWh of fuel per kWh of grid electricity), so a source-NZEB is harder to reach.

★ REMEMBER IT

- NZEB (net-zero energy building) recipe: **Reduce → Recover → Renew**. LEED = Leadership in Energy and Environmental Design: points for site, water, energy, materials, indoor quality; Certified < Silver < Gold < Platinum.

Before you start: net-zero energy building

A **net-zero energy building (NZEB)** makes as much energy (usually solar PV (photovoltaic (solar cells))) as it uses over a year. **LEED** (Leadership in Energy and Environmental Design) is a points-based green building rating (certified, silver, gold, platinum).

Definitions: USA vs EU (2020 6b, 2021 8a)

US Department of Energy: a zero-energy building is a building that produces enough renewable energy to meet its own annual energy consumption (net annual balance = 0).

European Union (EPBD Article 2): a **nearly zero-energy building (nZEB)** is a building with a very high energy performance whose very low energy need is covered to a very significant extent by renewable sources produced on-site or nearby.

Basic difference: the US definition demands an exact net-zero annual balance by renewables; the EU definition first demands very high efficiency (very low demand, set nationally in kWh/m²·yr) and then a large (not necessarily 100 %) renewable share, on-site or nearby.

Types of NZEB (2021 8a)

Type	Balance
Net-zero **site** energy	Produces as much energy as it consumes, measured at the site (meter)
Net-zero **source** energy	Produces as much energy as it uses over a year counted as primary energy at the source (includes generation and transmission losses; 1 kWh of grid electricity ≈ 3 kWh source)
Net-zero energy **cost**	Money earned from exported renewable energy ≥ money paid for grid energy over a year
Net-zero energy **emission**	Renewable generation offsets the emissions caused by the building's energy use

Site vs source: site counts kWh at the meter regardless of fuel; source weights each fuel by its primary-energy factor, so a building using gas on site and exporting electricity is judged differently. Source is fairer for the environment; site is easier to verify.

Why we need NZEB (2020 6b, 2022 8a)

- Buildings use about 40 % of energy and emit about 39 % of CO_2
- Reduce energy consumption and bills
- Reduce GHG emissions and global warming, ozone depletion and climate change
- Reduce dependence on fossil fuels and imports; energy security
- Protect the environment for future generations; better comfort and IEQ (indoor environmental quality)

Principles of NZEB design (2020 6c, 2021 8b)

A. Building envelope measures (reduce heat gain and HVAC loads): orientation to minimise HVAC demand; shading by overhangs, awnings, vegetation; avoid east/west glazing, use double glass or low-SHGC windows; more insulation and an envelope matched to local climate.

B. Energy efficiency measures: right-size systems (ASHRAE 90.1, simulation, part-load: VAV, VSDs, variable-capacity chillers/boilers/pumps); LED (light-emitting diode) lighting, dual-circuit task lighting, occupancy and daylight-dimming sensors; shift loads off-peak; heat-recovery chillers, underfloor air distribution, high-efficiency chillers, passive cooling, PCM thermal storage, CHP, natural ventilation; commissioning (5-10 % saving).

C. Renewable energy measures (only after A and B, as renewables are expensive): solar thermal and PV, wind (if steady 4.5-11.2 m/s), biomass, geothermal heat pumps (COP ≥ 3).

Energy-efficiency measures and HVAC optimisation (2022 8a, 2024 8b)

- Right-size HVAC; variable air volume, variable speed drives on fans and pumps, high-efficiency and heat-recovery chillers
- Demand-controlled ventilation (CO_2), energy recovery ventilators, economizer cycle
- Underfloor or displacement air distribution, radiant cooling, thermal storage with PCM or ice, night pre-cooling
- Dedicated outdoor air system with desiccant (B12), higher chilled-water temperature, BMS optimisation and commissioning

Green building: features and how to design (2022 8a, 2024 8b)

A green building (World Green Building Council) reduces or eliminates negative environmental impact during design, construction or operation and can create positive impacts on the climate and environment. **Key features:** renewable energy use; efficient energy use; water conservation (reuse/recycling, rainwater); pollution and waste reduction; healthy indoor environment (low-emission materials, good IAQ (indoor air quality), daylight); sustainable, non-toxic materials; adaptive design; focus on occupants and environment through the whole life cycle.

How to design one: sustainable site and orientation → passive design (shape, shading, daylight, natural ventilation) → high-performance envelope → efficient HVAC, lighting, water fixtures → renewables → green materials and waste plan → IEQ → commissioning, metering, certification (LEED).



Sustainable building: environmental, social and economic benefits

LEED: functions and certification (2022 8c)

LEED (Leadership in Energy and Environmental Design, US Green Building Council) is a globally recognised rating system that evaluates buildings on sustainability criteria. **Functions:** gives a common measurable standard for green buildings; rewards energy efficiency, water conservation, indoor air quality, material selection, sustainable sites, transport access, innovation and regional

priorities; gives third-party verification and market value. **To achieve it:** (1) register the project with USGBC/GBCI, (2) meet all prerequisites and earn credits in the categories (location and transport, sustainable sites, water efficiency, energy and atmosphere, materials and resources, IEQ, innovation, regional priority) through design and construction, (3) document and submit, (4) review by GBCI, (5) certification by points (out of 110): Certified 40-49, Silver 50-59, Gold 60-79, Platinum 80+. Many RMG factories in Bangladesh are LEED Platinum.

Challenges of low-energy building technologies (2021 8c)

- High initial cost and long payback (PV, high-performance glazing, heat recovery, BMS); limited financing
- Lack of skilled designers, contractors and operators; poor commissioning
- Performance gap between design and actual use; occupant behaviour
- Technology suited to climate: e.g. desiccant and evaporative cooling are hard in hot-humid climates; PV space on dense urban roofs
- Grid integration, net-metering policy, storage cost
- Lack of codes, incentives, local materials and testing; maintenance of complex systems

Remember: NZEB order: **Envelope → Efficiency → Renewables**. Types: **Site, Source, Cost, Emission**. US = exact zero; EU = nearly zero + very efficient.

B3 Smart windows (thermochromic, photochromic, electrochromic, LC)

Must study | Very likely · 100% | Easy

PAST-PAPER QUESTIONS

- **2024 Q5(d)** [7] Between photochromic and thermochromic window which one is more suitable for Bangladesh and why?
- **2022 Q5(c)** [0] (second half, marks counted in B1) Write short note on "smart window".
- **2021 Q6(c)** [9] What is smart window? Describe the various types of smart windows.
- **2020 Q4(d)** [8] Write a short note on "smart window".

★ REMEMBER IT

- Name tells the trigger: **thermo**chromic = heat, **photo**chromic = light, **electro**chromic = voltage (you switch it), liquid crystal = voltage (clear/frosted privacy). Bangladesh: thermochromic, because the problem is heat, and it darkens exactly when the glass is hot.

Before you start: smart windows

Glass that changes its tint by itself or on command: **thermochromic** (darkens with heat), **photochromic** (darkens with light), **electrochromic** (darkens when a small voltage is applied), **liquid crystal** (switches clear/opaque with voltage).

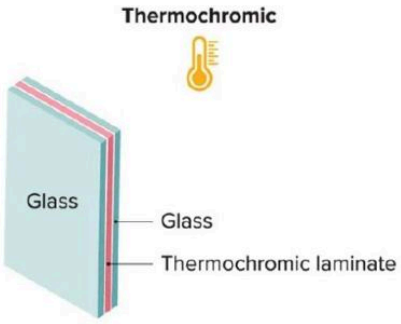
Smart windows (2020 4d, 2021 6c, 2022 5c)

Smart (dynamic or advanced) glazing is a new generation of glazing that changes its thermal/optical properties, such as shading coefficient and visible transmittance, in response to an electric signal or an environmental signal (temperature, light). It replaces trouble-prone motorised blinds and lets one window behave like clear glass in one condition and like tinted glass in another, saving cooling and lighting energy and controlling glare.

Type	Trigger	How it works
1. Thermochromic	Temperature (passive)	Gel or film between glass/plastic switches from clear when cold to a diffuse white reflective state when hot; visible transmittance falls as outdoor temperature rises
2. Photochromic	Incident light intensity (passive)	Glass darkens automatically in bright light and clears in low light (like photo-grey spectacles); drawback: also darkens under winter sun, increasing heating; large sizes not commercially available
3. Liquid crystal (LCD / PDLC)	Voltage (active)	Liquid crystal droplets in a film scatter light (opaque, privacy) with no voltage and align to make it clear with voltage; controls privacy and glare more than heat
4. Electrochromic	Small DC voltage (active, user/BMS controlled)	A thin multilayer coating (e.g. tungsten oxide) changes colour/tint when ions move under a low voltage and stays there with no power; the most controllable, can be linked to sensors

Thermochromic

1. Thermochromic materials whose transmissivity varies with temperature. As outdoor temperature increases, the visible light transmittance decreases, and vice versa. Gels sandwiched between glass and plastic switch from a clear state when cold to a more diffuse, white, reflective state when hot. In their switched-on state, less visible light is transmitted through the glazing.



Thermochromic glazing: a laminate that turns reflective when hot

Photochromic vs thermochromic for Bangladesh (2024 5d)

Thermochromic is more suitable. Bangladesh is hot and humid, the cooling load dominates almost all year, and there is little need for winter solar heating. A thermochromic window responds to temperature: when the glass and outdoor air get hot it turns reflective and cuts solar heat exactly when cooling is needed, while staying clear on cooler days and mornings. A photochromic window responds only to light: it would darken on any bright day (including cool winter days, causing gloom) and on hazy, hot, humid days with diffuse light it may not darken enough, so it does not track the cooling load. Photochromic glass also mainly reduces visible light rather than heat, and large panes are not commercially available.

Remember: Smart windows: **T-P-L-E** = Thermochromic (heat), Photochromic (light), Liquid crystal (voltage, privacy), Electrochromic (voltage, controllable). Bangladesh: **thermochromic**.

B10 Zoning, building shape and one-zone heat balance

Good to have

Very likely · 90% Easy

PAST-PAPER QUESTIONS

- **2024 Q6(c)** [7] A new residential building is being designed with two floor-plan options of the same total floor area: a compact square layout and a long narrow layout. Which option will have higher heat loss, and for what reason? Propose modifications to reduce this heat loss.
- **2022 Q6(d)** [8] What does mean by zoning of building? Why it is required? Explain with example.
- **2021 Q5(c)** [5] Why zoning of building is required?

★ REMEMBER IT

- Compact square plan = less wall area per floor area = less heat loss/gain. Long narrow plan = more envelope, but better daylight and cross ventilation (orient the long side north-south facing).

Before you start: zoning

A **zone** is a part of a building that can be controlled by one thermostat because it has similar loads (e.g. all south-facing offices). Different facades and uses need separate zones, otherwise some rooms overheat while others are too cold.

Zoning (2021 5c, 2022 6d)

A **zone** is a space or group of spaces whose temperature is controlled by one thermostat; the number of zones in a building equals the number of thermostats. **Zoning** is dividing the building into such zones and writing a heat balance for each.

Why it is required:

- Different orientations get sun at different times: east rooms peak in the morning, west rooms in the afternoon; one thermostat would over-cool one side and under-cool the other
- Different uses, internal gains and schedules (kitchen, conference room, server room, offices; occupied vs empty)
- Different required temperatures (a house with an attached sunspace, operating theatre vs ward)
- Interior vs perimeter spaces (interior needs cooling all year, perimeter may need heating)
- Energy saving: condition only occupied zones, set-back unused zones; comfort and control

Example: an office floor split into north, south, east, west perimeter zones (each about 4.5 m deep) and one interior core zone, each with its own VAV box and thermostat. Heat flow between two zones j and k: $\dot{Q}_{j\text{-}k} = U_{j\text{-}k}A_{j\text{-}k}(T_j - T_k)$.

One-zone heat balance (supporting)

$$K_{cond} = \Sigma U_k A_k = U_{glaz}A_{glaz} + U_{wall}A_{wall} + U_{roof}A_{roof}; \quad K_{tot} = K_{cond} + \rho c_p \dot{V}$$

$$Q_{sen} = K_{tot}(T_i - T_o) + Q_{floor} - Q_{gain} \pm Q_{stor}$$

$$\text{Unheated attached space: } T_{unheated} = [(UA)_{common}T_i + (UA)_{unheated}T_o] / [(UA)_{common} + (UA)_{unheated}]$$

Compact square vs long narrow plan (2024 6c)

For the same floor area the **long narrow layout has the higher heat loss**, because it has a larger perimeter, so more exposed wall area per unit floor area (higher surface-to-volume ratio). Since conduction loss = $\Sigma UA\Delta T$, more area means more loss (and more infiltration along longer joints).

> Example, 100 m², wall height 3 m: square 10 × 10 m → perimeter 40 m, wall area 120 m². Long 25 × 4 m → perimeter 58 m, wall area 174 m² (45 % more), so about 45 % more wall conduction loss.

Modifications to reduce the loss: add insulation to walls and roof (lower U), use double/low-e glazing and limit window area on the long walls, airtight construction and weather-stripping, orient the long side to the sun (south) for passive gain, thermal mass, buffer spaces (garage, store) on the cold side, earth berming or trees as windbreak, heat-recovery ventilation. (In hot-humid Bangladesh the narrow form is still often chosen for cross ventilation and daylight, with good shading.)

Remember: Zone = **one thermostat**. Same area: **longer perimeter = more loss**.

B5 Internal heat gains: people, lights, equipment; sensible vs latent Must study Very likely · 80% Easy

PAST-PAPER QUESTIONS

- **2024 Q6(b)** [8] Distinguish between sensible and latent internal heat gains and explain why they must be treated separately in load calculations.
- **2024 Q6(d)** [9] Describe the two methods used to estimate heat gains from lighting systems.
- **2022 Q5(b)** [9] What is the importance of internal heat gains in building heating and cooling load calculation? What are the main components of internal heat gains?
- **2022 Q5(d)** [7] How to calculate sensible heat gain and latent heat gain in a building?
- **2020 Q5(a)** [11] What are the major components of internal heat gains in a building?

UNDERSTAND **Read this to understand; you do not need to write it**

Sensible heat changes the air temperature (you feel it on a thermometer). **Latent** heat is moisture: people breathe and sweat water vapour, which the AC must condense out. They must be kept apart because the coil removes them differently: sensible by cooling, latent by dehumidifying (and latent heat never shows on a thermostat).

Lights: method 1 = installed watts × use factor × special allowance factor (ballast); method 2 = lighting power density (W/m²) × floor area. People: sensible + latent per person from the ASHRAE (American Society of Heating, Refrigerating and Air-Conditioning Engineers) table by activity.

CALCULATOR **Internal heat gain of a room**

Change any number; every step is recalculated and shown with the numbers put in.

1. People sensible: **20** × **75** = **1500** W
2. People latent: **20** × **55** = **1100** W
3. Lights: $W \times F_{ul} \times F_{sa}$ = **800** × **1** × **1.2** = **960** W
4. Total sensible: **1500** + **960** + **1500** = **3960** W
5. Total internal gain: **3960** + **1100** = **5060** W
6. Sensible heat ratio: **3960** / **5060** = **0.78**

Before you start: internal gains

Heat produced inside: people (about 100-130 W each when seated), lights (almost all their watts become heat), equipment. **Sensible** heat raises air temperature (handled by cooling); **latent** heat is moisture (sweat, breathing, cooking), which must be removed by dehumidifying.

Importance of internal heat gains (2022 5b)

Besides solar and conduction gains, heat released inside by people, lights and equipment is a major contributor to loads, especially in low-energy and commercial buildings. It adds to the summer

cooling load (must be removed) and partly offsets the winter heating demand (free heat, it lowers the balance-point temperature). Ignoring it oversizes heating and undersizes cooling equipment.

Components (2020 5a)

Component	Sensible / latent	How estimated
Occupants	Both: about 100 W (resting) to 500 W (heavy work) per person; latent at least 30 W (about 1 L of water a day)	Number of people × heat per person from ASHRAE tables for the activity
Lighting	Sensible only (part convective, part radiative)	$Q = P_L F_{UL} F_{SA}$ or LPD × floor area
Equipment and motors (computers, copiers, appliances, cooking)	Mostly sensible; cooking, kettles and washing add latent	$Q = (P_M/\eta_M) F_{UM} F_{LM}$ or tables of office equipment
Others: infiltration/ventilation air, process heat, hot water, plants	Sensible and latent	$\rho c_p \dot{V} \Delta T$ and $\rho h_{fg} \dot{V} \Delta W$

Heat Gains from Occupants

Heat gains from occupants depend on the level of physical activity characterized by the metabolic rate. The average amount of heat given off by a person depends on the level of activity, and can range from about 100W for a resting person to more than 500W for a physically very active person. Typical rates of heat dissipation by people are given in Table for various activities in various application areas.

		Total heat, W			
Degree of activity	Typical application	Adult male	Adjusted M/F/C[1]	Sensible heat, W	Latent heat, W
Seated at theater	Theater—matinee	115	95	65	30
Seated at theater, night	Theater—evening	115	105	70	35
Seated, very light work	Offices, hotels, apartments	130	115	70	45
Moderately active office work	Offices, hotels, apartments	140	130	75	55
Standing, light work; walking	Department or retail store	160	130	75	55
Walking, standing	Drug store, bank	160	145	75	70
Sedentary work	Restaurant[2]	145	160	80	80
Light bench work	Factory	235	220	80	80
Moderate dancing	Dance hall	265	250	90	90
Walking 4.8 km/h:					
light machine work	Factory	295	295	110	110
Bowling[3]	Bowling alley	440	425	170	255
Heavy work	Factory	440	425	170	255
Heavy machine work: lifting	Factory	470	470	185	285
Athletics	Gymnasium	585	525	210	315

Note: Tabulated values are based on a room temperature of 24°C. For a room temperature of 27°C, the total heat gain remains the same but the sensible heat values should be decreased by about 20 percent, and the latent heat values should be increased accordingly. All values are rounded to nearest 5 W. The fraction of sensible heat that is radiant ranges from 54 to 60 percent in calm air ($V < 0.2$ m/s) and from 19 to 38 percent in moving air ($0.2 < V < 4$ m/s).

[1]Adjusted heat gain is based on normal percentage of men, women, and children for the application listed, with the postulate that the gain from an adult female is 85 percent of that for an adult male and that the gain from a child is 75 percent of that for an adult male.

[2]Adjusted heat gain includes 18 W for food per individual (9 W sensible and 9 W latent).

[3]Figure one person per alley actually bowling, and all others are sitting (117 W) or standing or walking slowly (231 W).

Heat gain from occupants for various activities: total, sensible and latent (ASHRAE table in the slides)

Two methods for lighting heat gain (2024 6d)

1. Based on installed lighting systems:

$$\dot{Q}_{light} = P_L \cdot F_{UL} \cdot F_{SA}$$

P_L = total installed lamp wattage (sum of the ratings); F_{UL} = use (diversity) factor = wattage in use / installed (1 for shops, less for offices; varies with time); F_{SA} = special allowance factor = power drawn by the whole fixture (lamp + ballast) / rated lamp power (1 for incandescent, may be <1 for electronic ballasts, up to 2.2 for fluorescent; 1.2 recommended).

2. Based on space area (design stage): Q = LPD × floor area, where LPD is the maximum allowable lighting power density for the space type (ASHRAE 90.1): offices about 9.7-13 W/m² for fluorescent (about half for LED (light-emitting diode)), retail about 18 W/m².

Equipment: $\dot{Q}_{equip} = (P_M/\eta_M)F_{UM}F_{LM}$ (motor rating, efficiency, use factor, load factor).

Sensible vs latent and why separate (2024 6b, 2022 5d)

Sensible heat gain	Latent heat gain
Raises the dry-bulb temperature of the space air	Adds moisture (water vapour) to the air; changes humidity, not temperature
From conduction, solar, lights, equipment, sensible part of people and infiltration	From people (breathing, sweating), cooking, bathing, plants, infiltration/ventilation air
$Q_s = \rho c_p \dot{V}(T_o - T_i)$ for air; $= K_{tot}(T_i - T_o) + Q_{floor} - Q_{gain}$ for the space	$Q_l = \rho h_{fg} \dot{V}(W_o - W_i)$, with ρh_{fg} = 3010 W/(L/s) per unit W; plus occupants and equipment latent
Partly delayed: the radiative part is first absorbed by walls/furniture (thermal mass) and appears later as load	Instantaneous cooling load (no storage)
Removed by cooling the air (sensible coil capacity)	Removed only by condensing moisture on a coil below dew point or by desiccant

Why treat them separately: (1) latent load is instantaneous while sensible load is delayed and damped by thermal mass, so their time profiles differ; (2) the convective part goes directly to the air while the radiative part heats surfaces first; (3) the equipment must be selected for the right sensible heat ratio: the coil temperature/dew point, airflow and dehumidification capacity depend on the latent share; (4) in winter sensible gains offset heating, latent gains do not. Total load = sensible + latent.

Remember: Lighting: $\mathbf{P_L F_{UL} F_{SA}}$ or **LPD × area**. Latent = **instant**, sensible = **delayed** by mass.

B7 Infiltration, ventilation and the stack effect

Must study · Very likely · 100% · Medium

PAST-PAPER QUESTIONS

- **2024 Q7(c)** [9] Distinguish between ventilation and infiltration. What are the typical air-leakage site of a house?
- **2022 Q6(c)** [9] What factor influences the air infiltration through building envelope? Explain the tracer gas method to measure the infiltration rate.
- **2022 Q7(a)** [6] What is stack effect? Explain with schematic diagram.
- **2022 Q7(c)** [12] Distinguish between infiltration and natural ventilation. Explain air change method for calculating air infiltration rate.
- **2021 Q7(b)** [8] How to measure the air infiltration rate in a building? Distinguish between crack method and air-change method for prediction of air infiltration.
- **2020 Q5(b)** [12] Distinguish between infiltration and ventilation. How to prevent infiltration?

UNDERSTAND **Read this to understand; you do not need to write it**

Infiltration = air sneaking in through cracks (unplanned). Ventilation = air brought in on purpose (windows or fans). Stack effect = warm indoor air is lighter, rises and leaks out at the top, pulling cold air in at the bottom, like a chimney.

Air-change method: $\dot{V}$ = ACH (air changes per hour) × room volume / 3600 (m³/s). Tracer gas: $C(t) = C_0 e^{-Nt}$, so $N = \ln(C_0/C) / t$.

CALCULATOR **Infiltration load (air-change method) and tracer-gas ACH**

Change any number; every step is recalculated and shown with the numbers put in.

1. Infiltration flow: $\dot{V}$ = ACH × V / 3600 = **0.5** × **300** / 3600 = **0.0417** m³/s
2. Sensible infiltration load: $Q = \rho c_p \dot{V} \Delta T$ = 1.2 × 1005 × **0.04167** × (**22** - **2**) = **1005** W
3. Air change rate from tracer gas: $N = \ln(C_0/C)/t$ = ln(**100** / **60**) / **1** = **0.511** 1/h

Before you start: air leakage

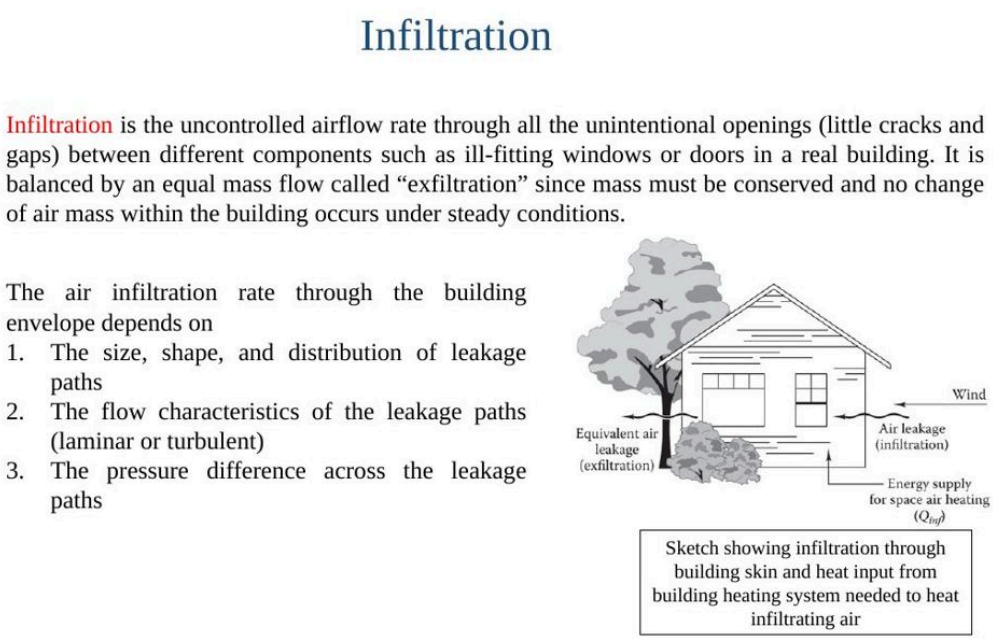
Infiltration = outdoor air leaking in through cracks (unwanted). **Ventilation** = fresh air supplied on purpose. **ACH** (air changes per hour) = volume of air entering per hour / room volume. **Stack effect**: warm indoor air is lighter, rises and leaks out at the top, pulling cold air in at the bottom (like a chimney).

Infiltration vs ventilation (2020 5b, 2022 7c, 2024 7c)

Infiltration	Ventilation
Uncontrolled flow of outdoor air through unintentional openings: cracks, gaps around windows and doors, joints	Intentional supply of outdoor air
Balanced by equal exfiltration; varies with wind and temperature (much higher in winter)	**Natural:** deliberate opening of windows/doors, variable, cannot be controlled well. **Mechanical:** fans, controlled to meet IAQ (indoor air quality)
Not a reliable fresh-air source; wastes energy (30-40 % of winter heat loss of a leaky house)	Provides required fresh air (0.35 ACH, 7.5 L/s per person), dilutes CO_2 below 1000 ppm

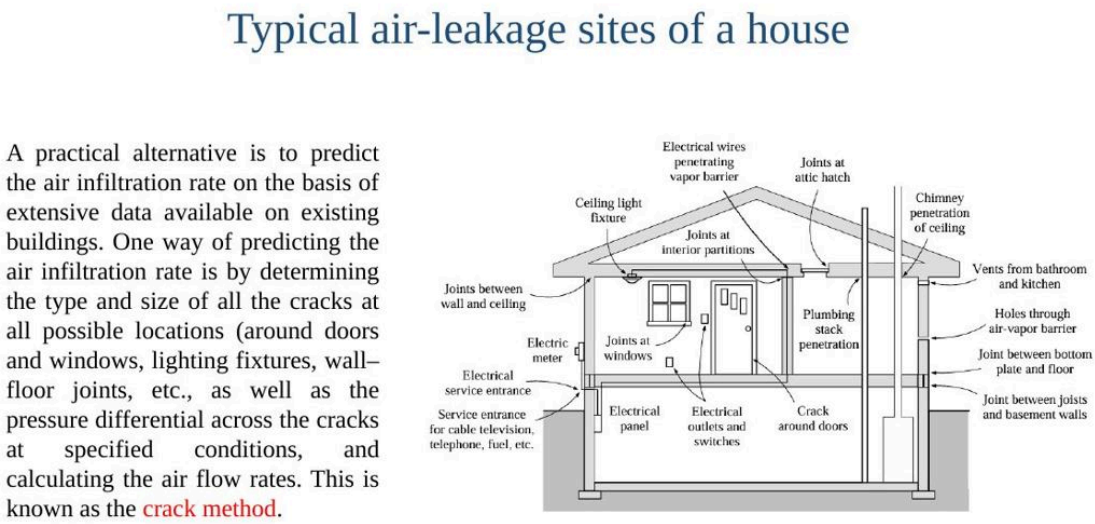
Factors that influence infiltration (2022 6c)

1. Size, shape and distribution of the leakage paths (tightness of construction, workmanship, age)
2. Flow characteristics of the paths (laminar in fine cracks, turbulent in large openings)
3. Pressure difference across the paths, created by wind (speed, direction, exposure), stack effect (indoor-outdoor temperature difference, building height) and mechanical ventilation imbalance



Infiltration through the building skin; the heating system must heat the infiltrating air

Typical air-leakage sites of a house (2024 7c)



Typical air-leakage sites of a house

Joints between wall and ceiling and wall and floor; around windows and doors and their frames; electrical outlets and switches; recessed ceiling lights; plumbing and duct penetrations; chimney and flue penetrations; attic hatch; vents from bathrooms and kitchen (fans and dampers); dryer vents; cracks in walls and around the sill plate.

Measuring infiltration (2021 7b, 2022 6c)

Tracer gas method: a harmless, easily measured gas (SF_6, CO_2, N_2O) is injected and mixed in the building; then its concentration is recorded as it decays. For a well-mixed space $C(t) = C_0e^{-Nt}$, so the air change rate N (ACH) = $\ln(C_0/C)/t$, and $\dot{V} = N \times V$. (Constant-injection and constant-concentration versions also exist.)

Pressurisation (blower door): a large fan in a door or window pressurises the building to 10-75 Pa; the airflow needed to hold each pressure is measured, giving the leakage curve $\dot{V} = C\Delta p^n$ and the effective leakage area.

Air-change method vs crack method (2021 7b, 2022 7c)

```
ACH = flow rate of indoor air into the building per hour / internal volume = V̇(m³/h) / V(m³)
V̇_inf = ACH × V;  Q_sensible = ρc_pV̇(T_i - T_o)
```

Air-change method: estimate from experience with similar buildings how many times per hour the whole indoor air is replaced (e.g. 0.35 ACH minimum for houses; tight new house 0.3-0.5, old leaky house 1-2 ACH). Simple, used for design. **Example:** house 300 m^3, 0.5 ACH → $\dot{V}$ = 150 m^3/h = 0.0417 m^3/s; with ΔT = 20 K, Q = 1.2×1005×0.0417×20 ≈ 1.0 kW.

Crack method: identify the type and length/area of every crack (doors, windows, fixtures, joints) and the pressure difference across each, then compute the flow of each crack and add them:

Turbulent (orifice, ≥ 3 mm): $\dot{V} = C_D A_{op}(2\Delta p/\rho)^{1/2}$; laminar hairline: Couette flow $\propto r^4\Delta p/(\mu L)$

General crack equation: $\dot{V} = C\Delta p^n$ (n = 0.5 turbulent ... 1 laminar, about 0.65 for buildings)

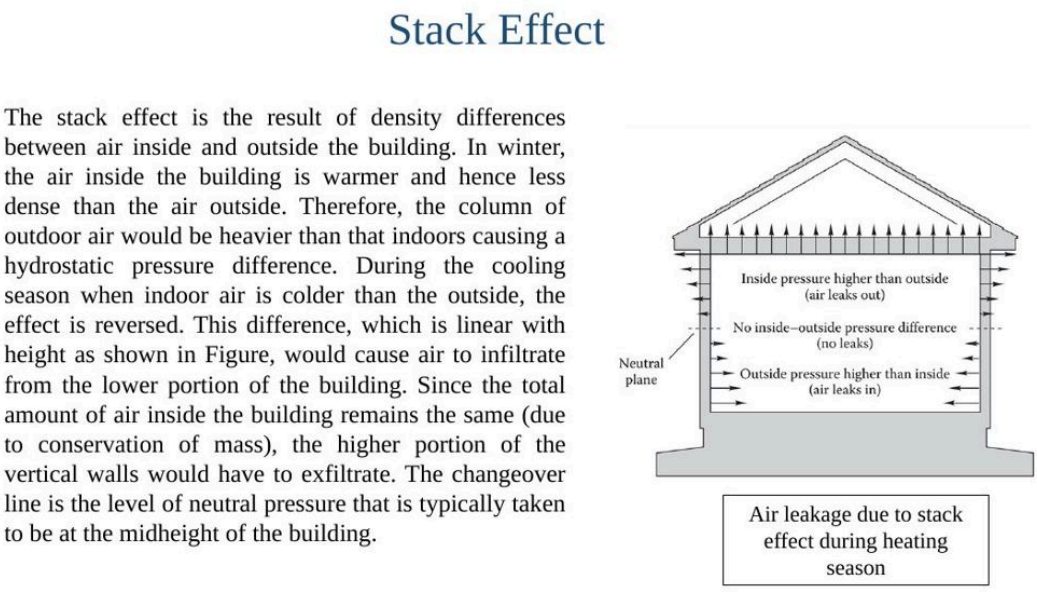
Air-change method	Crack method
Based on experience/whole-building average	Based on actual cracks and pressure differences
Quick, needs only volume	Detailed, needs crack data and wind/stack pressures
Less accurate; does not show where leaks are	More accurate, shows leakage sites, needed for tall/commercial buildings

LBL model: $\dot{V} = A_{leak}\sqrt{(a_s\Delta T + a_w v^2)}$ combines stack and wind in quadrature.

Driving pressures and the stack effect (2022 7a)

$\Delta p = \Delta p_{wind} + \Delta p_{stack} + \Delta p_{vent}$; $\Delta p_{wind} = C_p \rho v^2/2$

Stack effect: in winter the inside air is warmer and lighter than outside, so the outdoor air column is heavier. The hydrostatic pressure difference, linear with height, makes outside air infiltrate through the lower part of the building and inside air exfiltrate through the upper part. The changeover height is the **neutral pressure level**, usually near mid-height. In summer with AC the effect reverses.



Air leakage due to stack effect during the heating season: in at the bottom, out at the top, neutral plane at mid-height

$\Delta p_{stack} = -(\rho_o - \rho_i)g\Delta h = -\rho_i g\Delta h(T_i - T_o)/T_o$; real buildings: × C_d (draft coefficient)

Preventing infiltration (2020 5b)

- **Caulking** with a caulking gun where two stationary surfaces meet (wall-window frame, joints, penetrations)
- **Weather-stripping** (metal, vinyl, rubber, felt or foam strips) on moving joints of doors and windows; door sweeps

- Air barriers / house wrap, sealing outlets, recessed lights, attic hatches, duct and pipe penetrations
- Vestibules/revolving doors, dampers on exhaust fans and chimneys; slight over-pressurisation of commercial buildings by mechanical ventilation
- Then supply controlled fresh air mechanically (with heat recovery) if the building becomes tighter than 0.35 ACH

Remember: ACH = $\dot{V}/V$. Crack: $\dot{V} = C\Delta p^n$. Stack: **in at bottom, out at top** in winter, neutral plane mid-height. Seal: **caulk the fixed, weather-strip the moving**.

B1 Heat transfer through windows: U-factor, double pane, low-e, glazing selection

Must study · Very likely · 100% · Medium

PAST-PAPER QUESTIONS

- **2024 Q5(a)** [8] Explain how glazing selection affects overall occupant comfort and building performance beyond just heat gain and heat loss.
- **2024 Q5(c)** [9] A homeowner in a cold northern climate complains that their living room loses heat rapidly at night through large double-pane windows. What emissivity level should be chosen for the interior-facing glass surface, and why would this be more effective than increasing the glass thickness or reducing the air gap?
- **2022 Q5(c)** [10] How color and number of glazing affect the heat transfer through window? Write short note on "smart window".
- **2022 Q7(d)** [4] What are the desirable characteristics of window glasses in cold location?
- **2021 Q6(a)** [9] Explain the typical characteristics of window glazing in cold and hot location with schematic diagram.
- **2021 Q6(b)** [9] How does color affect the heat transfer through building walls? Explain. Explain the ways to minimize the air space between the glasses of double-pane window.
- **2020 Q4(a)** [5] Distinguish between 'low-e coating' and 'selective low-e'.

UNDERSTAND **Read this to understand; you do not need to write it**

A window loses or gains heat in three ways: **conduction** through the glass and frame, **convection** of the air in the gap, and **long-wave radiation** between the panes. Each fix attacks one: double glazing adds an air gap (conduction), a gas fill like argon slows convection, and a **low-e coating** (low emissivity, a thin metal oxide film) stops radiation.

Cold climate: put the low-e coating on the inner pane, surface facing the gap (surface 3), and use high-SHGC glass so winter sun still heats the room. Hot climate: low-e on surface 2 (outer pane, gap side) and low SHGC (solar heat gain coefficient) to keep solar heat out.

U-factor (W/m²K) = heat flow per m² per degree difference. Single glass ≈ 5.9, double ≈ 2.8, double low-e argon ≈ 1.5–1.8.

CALCULATOR **Window heat loss: single vs double vs low-e**

Change any number; every step is recalculated and shown with the numbers put in.

1. Heat loss: $Q = U\,A\,(T_i - T_o)$ = **2.8** × **4** × (**22** - **(-5)**) = **302** W
2. Same window with single glass (U = 5.9): 5.9 × **4** × (**22** - **(-5)**) = **637** W
3. Same window, double low-e argon (U = 1.6): 1.6 × **4** × (**22** - **(-5)**) = **173** W
4. Saving of low-e over your glazing: (**302.4** - **172.8**) / **302.4** × 100 = **43** %

This is why the 2024 5(c) homeowner should fit low-e (emissivity about 0.05–0.1 instead of 0.84 for plain glass) on surface 3 of the double pane.

Before you start: U-factor

U-factor (W/m^2·K) = how much heat passes through 1 m^2 of window for every 1 °C difference between inside and outside. Lower U = better insulation. Q = U A ΔT.

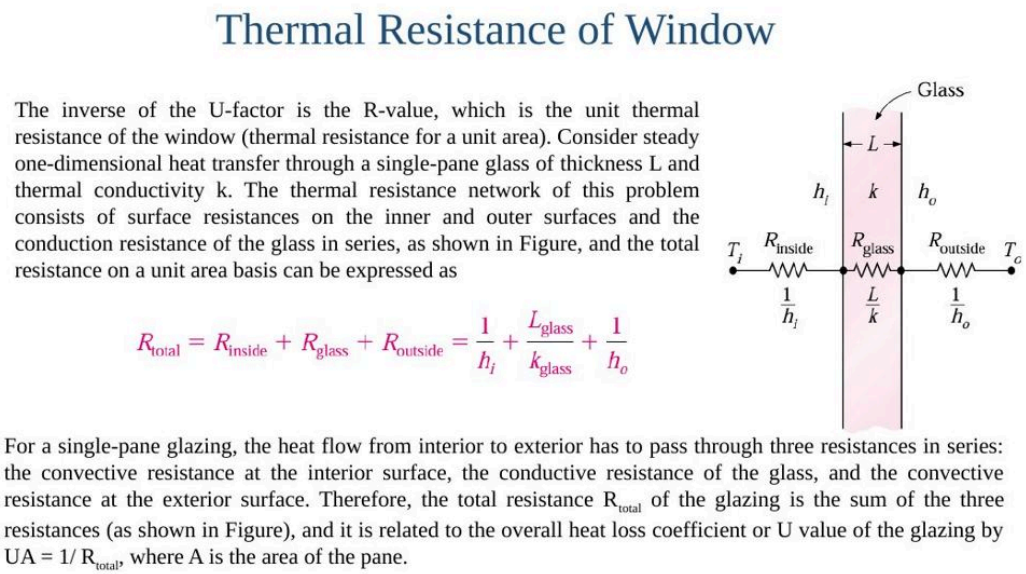
Single glass U ≈ 6; double glazing traps an air/gas layer, U ≈ 2.5-3; a **low-e** (low emissivity) coating reflects long-wave heat radiation and lowers U further.

Heat transfer through a window

Windows offer the least resistance in the envelope: about one third of winter heat loss of a house goes through them, most infiltration occurs at their edges, and their solar gain is a major part of the summer cooling load. Heat flows through three regions: centre-of-glass (1-D), edge-of-glass (2-D) and frame.

$$Q_{window} = U_{window}A_{window}(T_{in} - T_{out})$$
$$U_{window} = (U_{cg}A_{cg} + U_{eg}A_{eg} + U_{f}A_{f})/A_{window}$$

Single pane (resistances in series): $R_{total} = 1/h_i + L/k + 1/h_o = 1/8.29 + 0.003/0.92 + 1/34 = 0.121 + 0.003 + 0.029 = 0.153$ m²°C/W, so U ≈ 6.5 W/m²°C. The glass itself is only 0.003/0.153 ≈ **2 %** of the resistance, so a thicker glass does almost nothing.



Thermal resistance network of a single-pane window

Double pane: trapped still air (k = 0.025 W/m°C, 1/30 of glass) gives the resistance: 1 cm air ≈ 30 cm glass.

$$1/U_{double} = 1/h_i + 1/h_{space} + 1/h_o, \quad h_{space} = h_{rad,space} + h_{conv,space}$$

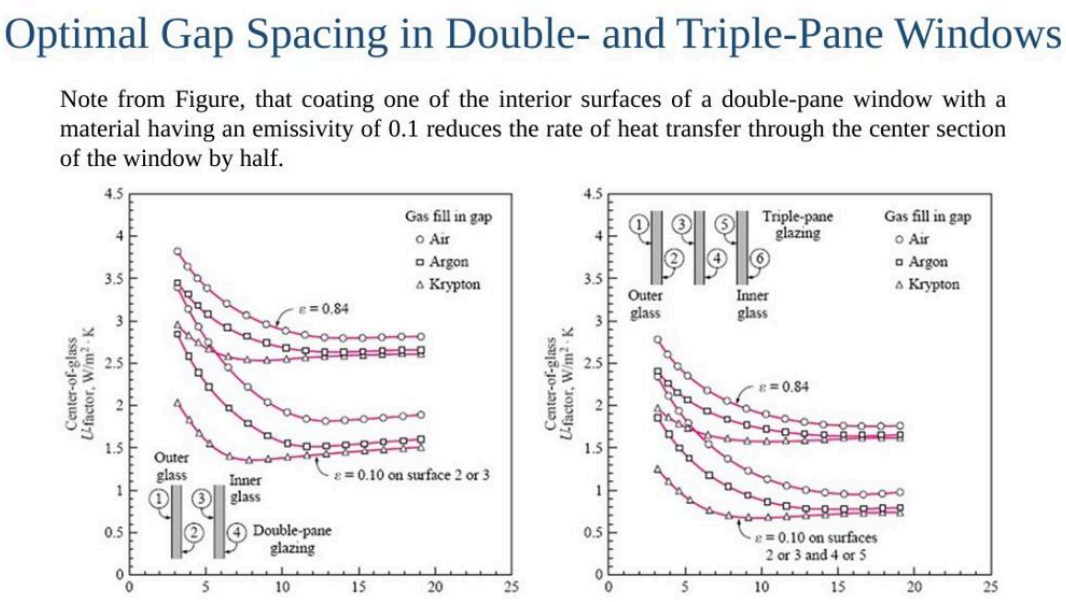
Roughly half the heat crossing the gap is radiation and half conduction/convection.

Ways to minimise heat transfer across the air space (2021 6b)

1. **Cut radiation** with a low-emissivity (low-e) coating on a surface facing the gap: $\varepsilon_{eff} = 1/(1/\varepsilon_1 + 1/\varepsilon_2 - 1)$. Coating one surface with ε = 0.1 makes ε_{eff} ≈ 0.1 and halves the centre-of-glass heat

transfer; coating one surface is enough and cheaper.

2. **Cut conduction** by widening the gap up to the optimum (about 13 mm for air): beyond it convection currents start and h_{space} rises again; wider gaps also need thicker, costlier frames.
3. Fill the gap with a heavier, low-conductivity gas (argon, krypton) or evacuate it; add a third pane (triple glazing).
4. Use warm-edge spacers and insulated frames for the edge/frame regions.



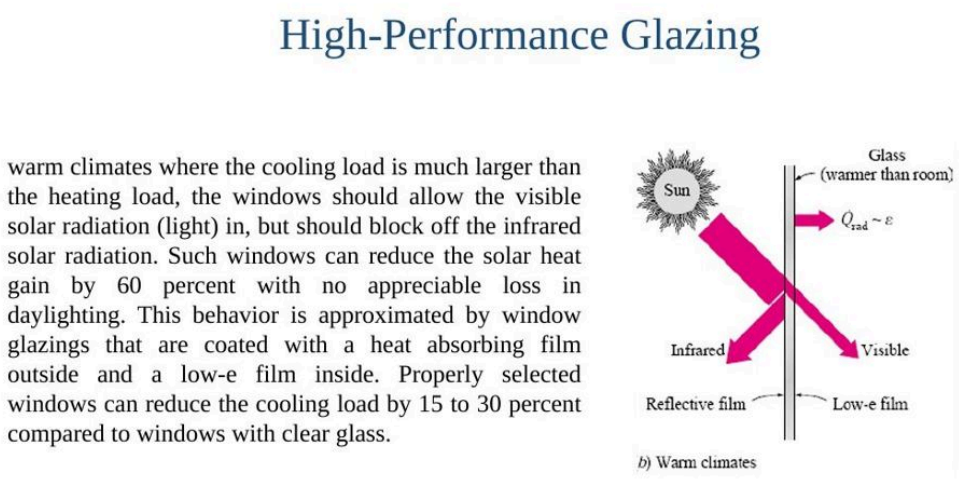
Centre-of-glass U-factor vs gap width for double and triple glazing with air, argon, krypton and low-e (ε = 0.10)

Cold-climate homeowner losing heat at night (2024 5c)

Choose a **low-emissivity coating, ε ≈ 0.05-0.10**, on the interior-facing surface of the gap (surface 3, the inner pane's face toward the air space; a hard-coat low-e on the room-facing surface 4 also works). At night there is no sun, so all heat loss is long-wave: the warm inner pane radiates across the gap to the cold outer pane. Low-e cuts ε_{eff} from about 0.72 to about 0.1, removing most of the radiative half of h_{space}, and it reflects room-generated infrared back indoors; the window's U-value drops by about 30-50 %.

Why this beats the alternatives: the glass is only about 2 % of the total resistance, so doubling its thickness changes U by about 2 %. Reducing the air gap actually *increases* conduction ($h_{conv} \approx k/d$ grows as d falls); widening it beyond about 13 mm triggers convection. Only the low-e coating attacks the largest term (radiation). The inner glass also stays warmer, so less radiant discomfort and condensation.

Glazing for cold and hot locations (2021 6a, 2022 7d)



Hot climate glazing: heat-absorbing film outside, low-e film inside; visible light in, solar infrared out

Cold location (heating dominated)	Hot location (cooling dominated)
High transmissivity for the whole solar spectrum (let solar heat in)	High visible transmittance but block solar infrared (spectrally selective)
High reflectivity (low emissivity) for far-infrared from room surfaces: low-e inside	Heat-absorbing/reflective film outside + low-e film inside: cuts solar gain about 60 % with little loss of daylight
Low U-value: double/triple glazing, argon, insulated frames, airtight	Low SHGC; with shading; can reduce cooling load 15-30 % vs clear glass
Large south (equator-facing) windows act as net energy gainers	Small east/west glazing

Desirable characteristics in cold locations (2022 7d, 4 marks): low U-value (multi-pane, gas fill, low-e), high SHGC on the sunny side, high visible transmittance, airtight edges and frames, warm inner surface to avoid condensation and drafts.

Colour and number of glazing (2022 5c)

Colour/tint: tinted (bronze, grey, green) glass absorbs more solar radiation (higher α_s), lowering transmitted solar heat (tinted double: 39 % vs clear 83 %) but also daylight (29 %); the absorbed heat partly flows inward, and the glass gets hot. Reflective coatings reflect more, reducing both gain and light. **Number of panes:** each extra pane plus gap adds a resistance $1/h_{space}$: U drops from about 6 (single) to about 2.8 (double) and 1.8 (triple) W/m²K, and the solar heat gain coefficient falls slightly (each pane reflects and absorbs some).

How colour affects heat transfer through walls (2021 6b, 2020 4c): see B4.

Low-e vs spectrally selective low-e (2020 4a)

Low-e coating	Spectrally selective low-e
Thin metal/metal-oxide layer with low emissivity for long-wave (far IR) radiation	Low-e coating designed also to reflect the near-infrared part of sunlight
Reduces radiative heat transfer across the gap and reflects room heat back (lower U)	Lower U and also low SHGC while keeping high visible transmittance
Lets most solar radiation through: good for cold climates (passive heating)	Lets visible light in, keeps solar heat out: ideal for hot climates

How glazing selection affects comfort and performance beyond heat gain/loss (2024 5a)

- **Daylighting** (visible transmittance) and the electric lighting saved; **view** and connection to outside
- **Glare** and visual comfort; **privacy**
- **Thermal comfort near windows:** radiant asymmetry and **cold down-drafts** from cold inner glass; warm glass in summer
- **Condensation control** (warm-edge, low U) and mould
- **Ultraviolet control** (fading of furnishings, skin), **acoustic control** (laminated, double glazing), **colour effects** of tinted glass
- **Air infiltration and water seepage** at frames; workmanship and durability
- **HVAC** (heating, ventilation and air-conditioning) **sizing and peak demand**, and occupant productivity

Remember: Glass = **2 %** of resistance. Gap: half **radiation** (kill with low-e) + half **conduction** (optimum ~13 mm). Cold: **let sun in, keep IR in**. Hot: **light in, IR out**.

B12 Evaporative cooling and solar-assisted desiccant dehumidification

Must study | Very likely · 90% | Medium

PAST-PAPER QUESTIONS

- **2024 Q8(c)** [11] For a Net-zero Energy Building (NZEB) in a hot humid climate. Propose an appropriate cooling strategy that addresses both latent and sensible loads while minimizing energy consumption, and describe the key process steps involves.
- **2024 Q8(d)** [11] Discuss about solar assisted desiccant dehumidification with air conditioning.
- **2022 Q8(b)** [8] Explain the evaporative cooling system with solar-assisted desiccant dehumidification system.
- **2021 Q8(d)** [9] Describe the evaporative cooling system with desiccant dehumidification.

LEARN IT FROM ZERO

Desiccant cooling, from zero

Bangladesh air is hot **and** wet. A normal AC removes moisture by over-cooling the air below its dew point and then often reheating it: very wasteful. A **desiccant** (silica gel, like the packet in a shoe box) grabs water vapour directly, so the air comes out dry (and a bit warmer).

Once the air is dry, simple **evaporative cooling** (blowing it over wet pads) can cool it cheaply, because dry air can absorb water and drop in temperature. The desiccant wheel slowly turns: one half dries the supply air, the other half is dried out again (**regenerated**) by hot air at 60–90 °C, which can come from **solar collectors**. So the sun pays for the dehumidifying.

Order to remember on the supply side: **dry (desiccant wheel) → cool (heat-recovery wheel) → cool more (evaporative cooler) → room**. Return side: evaporative cool → heat wheel → solar heater → regenerate desiccant → out.

Before you start: cooling by evaporation and desiccants

Evaporating water absorbs a lot of heat, so blowing air over wet pads cools it (works best in dry air). A **desiccant** (silica gel, lithium chloride) absorbs moisture from humid air; it is then dried again (regenerated) with heat, which can come from the sun.

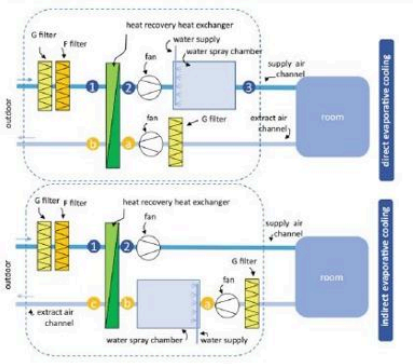
Evaporative cooling

Evaporating water absorbs its latent heat from the air or surfaces, cooling them (as sweating cools the body). **Direct evaporative cooling:** air is blown through a wetted pad or water spray; sensible heat removed equals latent heat gained (adiabatic cooling at constant wet-bulb): the air gets cooler but more humid. Best in hot, dry climates. **Indirect:** the exhaust (or secondary) air is evaporatively cooled and then cools the supply air through a heat exchanger at constant moisture content, so no humidity is added; for hot climates with moderate humidity.

Evaporative Cooling Methods

Evaporative cooling can be either **direct** or **indirect**.
In **direct evaporative cooling**, water droplets are sprayed directly into the flow of supply air. This results in air with a lower temperature but higher humidity entering the living spaces, which can potentially reduce indoor comfort. Direct evaporative cooling is most effective in climates with **high summer temperatures and low air humidity**.

To avoid increasing the humidity of the supply air, an **indirect evaporative cooling** process can be used. In this method, extracted air is cooled through evaporation **before entering the heat recovery exchanger** in the air handling unit (AHU). The supply air is then cooled in the heat exchanger at **constant absolute humidity**, meaning the amount of water vapor in the supply air does not increase. This approach is commonly applied in climates with **high summer temperatures and moderate air humidity**.



Direct and indirect evaporative cooling arrangements in an AHU

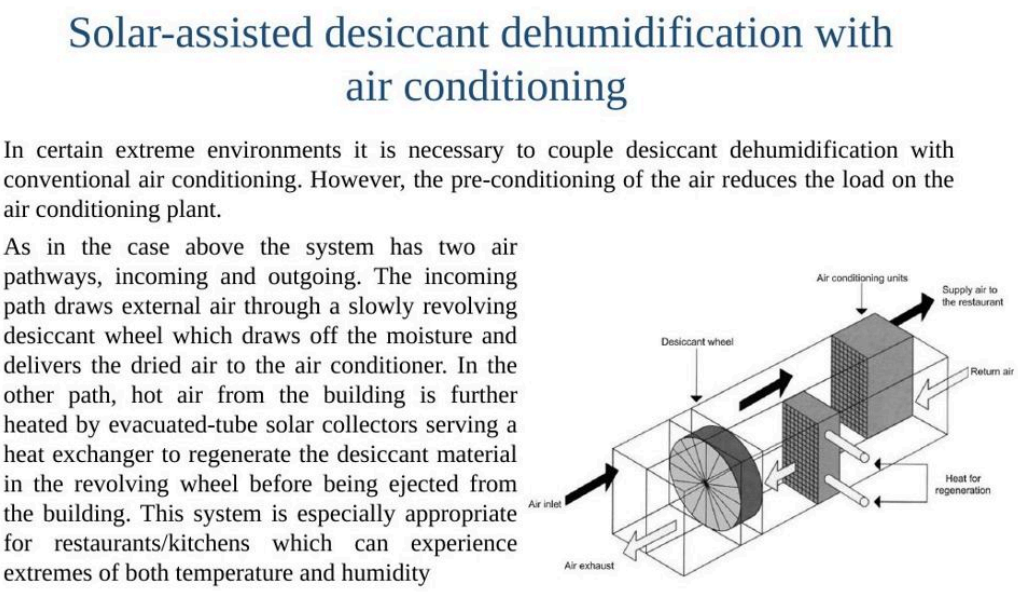
Desiccant dehumidification with evaporative cooling (2021 8d, 2022 8b)

A desiccant is a hygroscopic material (silica gel, activated alumina, lithium salts, triethylene glycol) that takes moisture from air: solid desiccants by adsorption in pores, liquids by absorption. It must be regenerated (dried) by heating to 60-90 °C, which can come from evacuated-tube solar collectors with gas backup.

1. Outdoor (supply) air passes through a slowly rotating **desiccant wheel**: it is dried (its temperature rises from the adsorption heat)
2. It then passes through a **thermal (heat recovery) wheel** that cools it
3. If needed, a **direct evaporative cooler** cools it further (now it has room for moisture) before it is supplied to the building
4. Return/exhaust air from the room is first evaporatively cooled, then passes the thermal wheel and cools it (so the wheel can cool the supply)
5. The exhaust air, now warmed, is further heated by the **solar collector heat exchanger** (60-90 °C) and passes through the desiccant wheel to drive the moisture out (regeneration), then is expelled

This "desiccant evaporative cooling" (DEC) can fully replace a conventional chiller in moderate climates, using mainly low-grade solar heat and fans.

Solar-assisted desiccant dehumidification with air conditioning (2024 8d)



Solar-assisted desiccant dehumidification with air conditioning: desiccant wheel, solar heat exchanger for regeneration, AC unit

In extreme (very humid and hot) climates the desiccant is coupled with conventional air conditioning. Two air paths: **incoming** outdoor air is drawn through the slowly rotating desiccant wheel, which removes the moisture (latent load) and delivers dry air to the **air conditioner**, which only has to remove the sensible load (it can run with a higher evaporator temperature and better COP (coefficient of performance), with no reheat). **Outgoing** hot air from the building is further heated by **evacuated-tube solar collectors** through a heat exchanger and passed through the wheel to regenerate the desiccant before being expelled. Pre-conditioning the air reduces the load on the AC plant considerably. It suits restaurants and kitchens with extremes of temperature and humidity, hospitals, and humid tropical buildings.

NZEB (net-zero energy building) cooling strategy for a hot-humid climate (2024 8c)

Proposed strategy: decouple latent and sensible loads: a solar-assisted desiccant **dedicated outdoor air system (DOAS)** handles all the latent load, and a high-efficiency, high-temperature sensible system (radiant chilled ceilings/slabs or chilled beams, or a high-SEER inverter/VRF unit fed by rooftop PV (photovoltaic (solar cells))) handles the sensible load.

1. **Reduce loads first:** shading, low-SHGC glazing, reflective insulated roof, airtight envelope, efficient LED (light-emitting diode) and equipment, ceiling fans (raise set-point to 26-27 °C)
2. **Energy-recovery wheel:** outdoor air first exchanges heat and moisture with the exhaust air (pre-cools and pre-dries)
3. **Solar desiccant wheel:** dries the outdoor air to the required dew point (latent load removed); regenerated at 60-90 °C by evacuated-tube solar collectors (gas or waste-heat backup)
4. **Post-cooling:** the warm dry air is sensibly cooled by a heat-recovery wheel / indirect evaporative cooler / cooling coil with 15-18 °C water
5. **Sensible cooling in the rooms:** radiant panels or chilled beams with 16-18 °C chilled water (no condensation because the air is dry), produced by a high-COP chiller or heat pump at a high evaporating temperature
6. **Supply** the dry outdoor air at the ventilation rate only (demand-controlled by CO_2)

7. **Renewables:** PV supplies the fans, pumps and chiller; thermal storage and night pre-cooling shift the peak
8. **Controls and commissioning:** BMS keeps dew point and temperature, monitoring confirms net-zero

Why it saves energy: a conventional coil must overcool air to about 10-12 °C to remove moisture and then often reheat; here moisture is removed by solar heat and sensible cooling happens at a high temperature, so chiller COP rises and electricity use falls by roughly 30-50 %.

Related passive options: night-time ventilation cooling (works with night temperatures below 20 °C; offsets 20-30 W/m² naturally, 50 W/m² mechanically) and ammonia absorption cooling driven by solar heat.

Remember: Desiccant = **latent**, cooler = **sensible**. Regenerate at **60-90 °C** with **evacuated-tube solar**. Wheels: **desiccant → thermal → evaporative**.

B8 Annual energy: balance point, degree-day and bin methods

Must study · Very likely · 100% · Hard

PAST-PAPER QUESTIONS

- **2024 Q7(d)** [8] Explain heating degree-day method with necessary equations.
- **2024 Q8(a)** [6] Define annual energy consumption and explain why it differs from peak load calculation.
- **2022 Q7(b)** [13] What is balance point temperature? Explain the degree-day method to estimate the heating and cooling load of building. Differentiate between degree-day method and bin method.
- **2021 Q7(c)** [8] What are the elements should be considered while estimate the annual energy consumption of a building in design stage. Explain the Bin method.
- **2021 Q7(d)** [8] What are meant by heating degree-day and cooling degree-day? How to determine the maximum threshold temperature to switching between air-conditioning and ventilation only?
- **2020 Q6(a)** [10] Explain the degree-day method to calculate the annual energy consumption of building. What are its advantages over bin method?

LEARN IT FROM ZERO

Balance point and degree-days, from zero

A building gets free heat from people, lights and sun. On a mild day that free heat alone keeps it warm. The outdoor temperature at which the free heat exactly balances the losses is the **balance-point temperature**: $T_{bal} = T_i - Q_{gain}/K_{tot}$. Heating is needed only when it is colder than T_{bal}.

A **heating degree-day** counts "how cold and for how long": each day, take (T_{bal} − daily mean outdoor temperature) if positive, and add them for the whole season. A day at 5 °C with T_{bal} = 17 °C gives 12 degree-days. Annual heating energy is then just **$Q = K_{tot} \times HDD$ (heating degree-days) $\times 24 / \eta$**.

The **bin method** does the same but by temperature bins (e.g. 2 °C wide) and hours in each bin, so it can include efficiency that changes with outdoor temperature (heat pumps) and day/night schedules. Degree-day is quicker; bin is more accurate.

Symbols used here

Symbol	Name and meaning	Unit
K_{tot}	Total heat loss coefficient = $\Sigma UA + \rho c_p \dot{V}$ (W per K of difference)	W/K
T_{bal}	Balance-point temperature	°C
Q_{gain}	Free (internal + solar) heat gains	W
HDD	Heating degree-days, $\Sigma(T_{bal} - \bar{T}_o)^+$	K·day
η	Furnace / boiler efficiency (or COP (coefficient of performance) for a heat pump)	–

CALCULATOR **Balance point and annual heating energy**

Change any number; every step is recalculated and shown with the numbers put in.

1. Balance-point temperature: $T_{bal} = T_i - Q_{gain}/K_{tot}$ = **20** - **900** / **300** = **17.0** °C
2. Annual heating energy: $Q = K_{tot} \times HDD \times 24 / \eta$ = **300** × **2000** × 24 / **0.9** / 1000 = **16000** kWh
3. Same in GJ: **16000** × 0.0036 = **57.60** GJ

Before you start: annual energy in one picture

Peak load sizes the machine; annual energy sets the electricity bill. **Balance-point temperature** = outdoor temperature at which free heat (sun, people, lights) exactly balances heat loss. Heating is needed only below it.

Degree-day: for each day, how many degrees the average outdoor temperature is below the balance point; add over the year. Annual heating energy = K_{tot} × degree-days × 24 h / efficiency. The **bin method** does the same with hour counts in temperature bins.

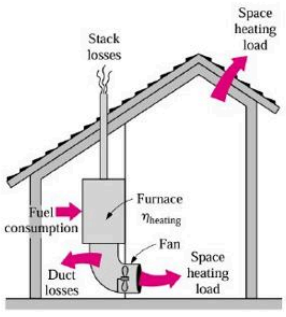
Annual energy consumption vs peak load (2024 8a)

Annual energy consumption is the total energy (fuel or electricity) used for heating/cooling over a year: $Q_{yr} = \int Q_h(t)/\eta_h(t)\, dt \approx \Sigma\, Q_h(h)/\eta_h(h)$ over 8760 h. It sets the operating cost and life-cycle cost.

Why it differs from peak load: the peak (design) load sizes the equipment for the most demanding, worst weather (design temperatures, peak solar, full occupancy). Annual consumption uses average usage under average weather over the whole season, includes part-load efficiency and system/distribution losses. So the method is different, and a big peak load does not mean a big annual use.

Annual Energy Consumption

For buildings that are at the design or construction stage, the evaluation of annual energy consumption involves the determination of (1) the space load for heating or cooling due to heat transfer through the building envelope and infiltration, (2) the efficiency of the furnace where the fuel is burned or the COP of cooling or heat pump systems, and (3) the parasitic energy consumed by the distribution system (pumps or fans) and the energy lost or gained from the pipes or ducts. The determination of the space load is similar to the determination of the peak load, except the average conditions are used for the weather instead of design conditions.



The various quantities involved in the evaluation of the annual energy consumption of a building.

Quantities in the annual energy estimate: space load, furnace/COP efficiency, distribution (duct, pump, fan) losses

Elements to consider at the design stage (2021 7c): (1) the space heating/cooling load from envelope heat transfer and infiltration under average weather, (2) the efficiency of the furnace/boiler or the COP of the chiller/heat pump (at part load), (3) parasitic energy of fans and pumps and losses or gains from ducts and pipes; plus internal and solar gains, schedules and set-points. For existing buildings, analyse a year of utility bills.

Balance-point temperature (2022 7b)

The **balance-point temperature** T_{bal} is the outdoor temperature at which the building's total heat loss equals the free heat gains (sun, people, lights, equipment), for a fixed indoor temperature T_i. Below it heating is needed, above it cooling.

$$K_{tot}(T_i - T_{bal}) = Q_{gain} \quad \Rightarrow \quad \mathbf{T_{bal} = T_i - Q_{gain}/K_{tot}}$$

$$K_{tot} = \Sigma UA + \rho c_p \dot{V} \quad \text{or} \quad K_{tot} = Q_{design}/(T_{i,design} - T_{o,design}) \quad \text{(winter design only)}$$

Heating degree-day method (2024 7d, 2020 6a)

$$\dot{Q}_h = (K_{tot}/n_h)[T_{bal} - T_o(t)]^+ \quad \text{(only when } T_o < T_{bal}\text{)}$$

$$Q_{h,yr} = (K_{tot}/n_h) \int [T_{bal} - T_o(t)]^+ dt$$

$$HDD(T_{bal}) = (1 \text{ day}) \Sigma_{days}(T_{bal} - \bar{T}_o)^+ \quad [\text{K·day}]; \quad \text{degree-hours: } (1 \text{ day}/24 \text{ h}) \Sigma_{hours}(T_{bal} - T_o)^+$$

$$\mathbf{Q_{h,yr} = K_{tot} \cdot HDD(T_{bal}) / n_h} \quad (\times 24 \text{ h/day to get W·h})$$

> **Example:** K_{tot} = 300 W/K, T_i = 20 °C, Q_{gain} = 900 W → T_{bal} = 20 − 900/300 = 17 °C. If HDD(17 °C) = 2000 K·day and η = 0.85: Q_{yr} = 300 × 2000 × 24 / 0.85 = 16.9 MWh.

Cooling degree-days and the threshold T_{max} (2021 7d)

$$CDD(T_{bal}) = (1 \text{ day}) \Sigma(\bar{T}_o - T_{bal})^+; \quad Q_{c,yr} = K_{tot} \cdot CDD(T_{bal})/n_c$$

$$HDD(T_{bal}) - CDD(T_{bal}) = 365 \text{ days} \times (T_{bal} - \bar{T}_{o,yr})$$

In summer you can open windows (or use economizer ventilation), which raises the loss coefficient to K_{max}. The air conditioner is needed only when T_o exceeds the **maximum threshold temperature**:

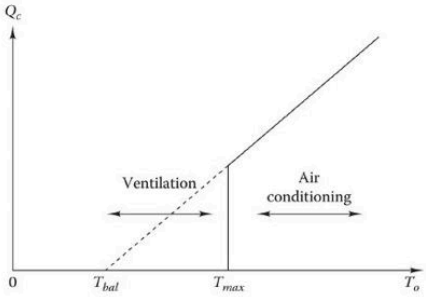
$$\mathbf{T_{max} = T_i - Q_{gain}/K_{max}} \quad (K_{max} \text{ = loss coefficient with windows open / full ventilation})$$

$$Q_{c,yr} = K_{tot}[CDD(T_{max}) + (T_{max} - T_{bal})N_{max}]/n_c \quad (N_{max} \text{ = days with } T_o > T_{max})$$

the dashed line is the load if K_{tot} is kept constant. The annual cooling load for this mode is the sum of the cooling load during the days when $T_o > T_{max}$ with windows closed plus that needed to cool down the building from T_{max} to T_{bal} during these same days:

$$Q_{c,yr} = K_{tot}[\mathrm{CDD}(T_{max}) + (T_{max} - T_{bal}) \times N_{max}]$$

where $\mathrm{CDD}(T_{max})$ is the cooling degree-days for base T_{max} and N_{max} is the number of days during the season when T_o rises above T_{max}. This model of air conditioning is very schematic, of course. In practice, heat gains and ventilation rates are variable, and, more importantly, the erratic occupant behavior in utilizing the windows and air conditioner has an important effect on cooling energy



Cooling load as a function of outdoor temperature T_o. Below T_{max}, cooling can be avoided by using ventilation.

Cooling load vs outdoor temperature: below T_{max} cooling is avoided by ventilation

Bin method (2021 7c, 2022 7b)

Outdoor temperatures are grouped into bins (e.g. 2.8 K / 5 °F wide) with the number of hours N_{bin} in each (often split into time-of-day shifts). The load and the equipment efficiency are evaluated at the mid-temperature of each bin and summed:

$$Q_{bin} = N_{bin}(K_{tot}/n)(T_{bal} - T_o)^+; \quad Q_{yr} = \Sigma_{bins}\, Q_{bin}$$

Use it when K_{tot}, the equipment efficiency (heat pumps, chillers, boilers at part load) or T_{bal} vary with outdoor temperature or time of day, or when solar loads and occupancy schedules matter.

Degree-day method	Bin method
One balance temperature, constant K_{tot} and efficiency	Each temperature bin (and time shift) calculated separately
Needs only HDD/CDD data; very quick (advantage, 2020 6a)	Needs hourly bin data; more work
Cannot handle efficiency varying with T_o, solar bias, schedules	Handles part-load efficiency, heat pumps, schedules, economizers
Good for residences and heating-dominated, simple buildings	Good for commercial buildings and heat pumps

Advantages of degree-day over bin (2020 6a): simple, uses widely published climatic data (degree-days for any base), quick hand calculation, gives good insight into how climate and T_{bal} affect energy, adequate for residential heating with steady efficiency.

Remember: $T_{bal} = T_i - Q_{gain}/K_{tot}$; $Q_{yr} = K_{tot}\cdot DD/\eta$; $T_{max} = T_i - Q_{gain}/K_{max}$; Bin = **$\Sigma\, N_{bin}K(T_{bal} - T_o)/\eta$.**

Same answer, step by step with reasons

Step (write this)	Why / what it means
$T_{bal} = T_i - Q_{gain}/K_{tot}$	Free heat gains lower the temperature at which heating must start.
Daily: $DD = (T_{bal} - \bar{T}_o)$ if positive, else 0	Only cold days count.
HDD = sum of DD over the year (K·day)	
$Q_{yr} = K_{tot} \times HDD \times 24\ h / n_h$	Heat loss coefficient × degree-hours, divided by furnace efficiency.

Memory trick: "K times Degree-days times 24".

★ REMEMBER IT

- Peak load sizes the **machine** (worst day). Annual energy sizes the **bill** (all days, average weather, part-load efficiency). Different questions, different methods.

B6 Transient heat flow: diurnal wall response, CTF, finite difference

Must study · Very likely · 100% · Hard

PAST-PAPER QUESTIONS

- **2024 Q7(a)** [7] How does diurnal outdoor temperature variation affect heat transfer through different wall constructions? Compare the thermal responses of lightweight and heavyweight walls under same sol-air temperature profile.
- **2024 Q7(b)** [11] Explain in detail how boundary nodes are treated in the finite difference method when exposed to combined convection and radiation. Derive the final rearranged equation using the biot number.
- **2022 Q2(c)** [8] Graphically show and discuss the variation of temperature and heat flow of a wall for a typical day from mid-night to mid-night.
- **2022 Q6(a)** [9] With the help of schematic diagram, mathematically explain the thermal transient heat transfer through wall.
- **2021 Q6(d)** [8] Why building thermal load should be determined on transient condition? Explain the CTF (conduction transfer function) model.
- **2020 Q5(c)** [7] Why should building heat load be determined under transient condition? Explain.

LEARN IT FROM ZERO

Transient heat flow, from zero

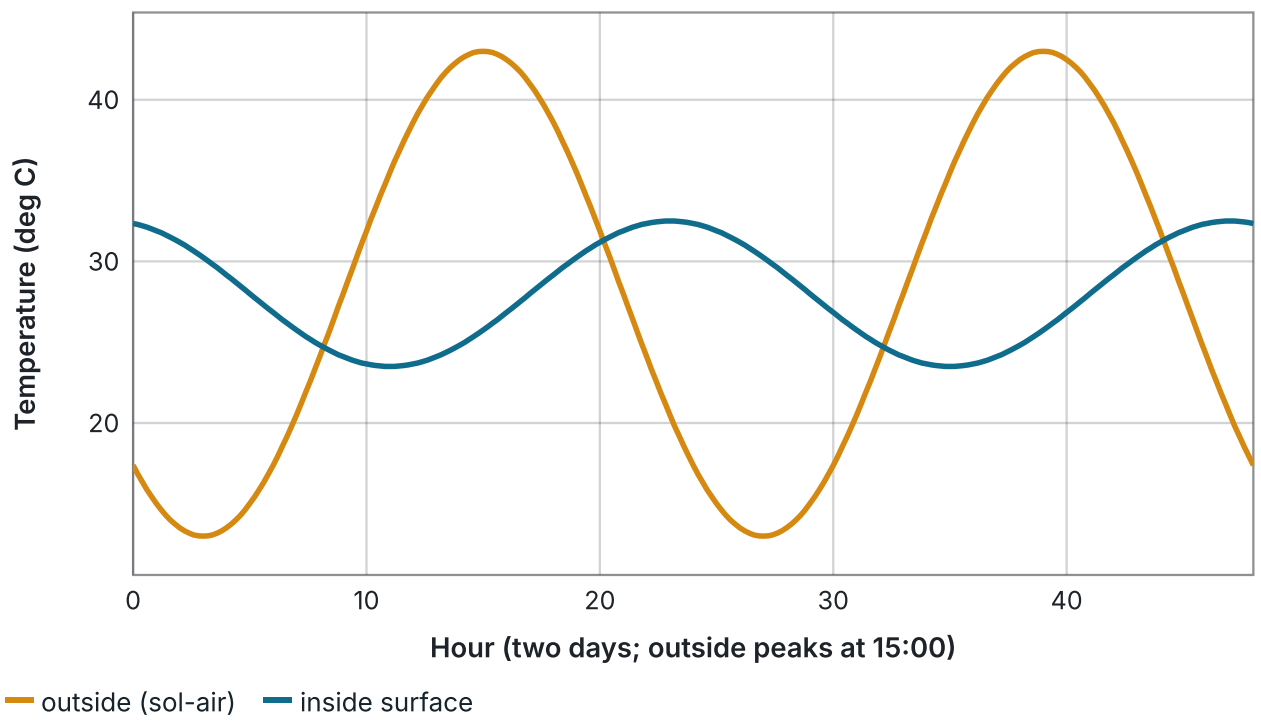
Steady state ($Q = UA\Delta T$) assumes the wall instantly passes on every change. Real walls are **heat sponges**. In the morning sun the outside face warms, but that heat has to soak through 20–30 cm of brick or concrete before it reaches the room. By the time it arrives (6–10 hours later for heavy walls) it is also **smaller**, because part was stored and part leaked back out.

Two words describe this: **time lag** (how many hours the inside peak comes after the outside peak) and **decrement factor** (inside swing / outside swing, less than 1). A light steel wall: lag ≈ 1 h, decrement ≈ 0.9. A 300 mm concrete wall: lag ≈ 8–10 h, decrement ≈ 0.1–0.2. That is why heavy walls keep rooms cool in the afternoon and release heat at night.

CTF (conduction transfer function) is the computer way to do this: today's heat flow inside = a weighted sum of the present and past few hours' outside temperatures, inside temperatures and past heat flows. **Finite difference** chops the wall into thin slices (nodes) and does an energy balance on each one every time step.

INTERACTIVE **Wall response to a daily temperature swing**

Move the sliders and watch the curve change.



Inside peak at: **23.0** h · Inside swing: **4.5** ±°C

Light steel siding: f ≈ 0.9, lag ≈ 1 h. 300 mm concrete: f ≈ 0.15, lag ≈ 9 h.

Before you start: transient heat flow

Steady state means temperatures do not change with time (Q = UAΔT). Real walls heat up and cool down over the day, storing heat. Heavy walls delay the peak (**time lag**) and shrink it (**decrement**). CTF (conduction transfer functions) and finite-difference methods calculate this hour by hour.

Why loads must be found under transient conditions (2020 5c, 2021 6d)

1. Thermal mass of the envelope responds slowly to the daily swing of outdoor temperature and solar radiation
2. Transmitted solar radiation is first absorbed by internal mass (slabs, walls, furniture) and released later
3. Internal mass interacts with schedules of lights, equipment and occupants (daily fluctuation of internal loads)
4. Internal mass interacts with control strategies (thermostat set-up/set-back, varying air volume)

A steady-state calculation (Q = UAΔT) ignores this storage, so it gets the peak load size and its timing wrong; heavy buildings show a time lag and a reduced (decremented) peak. Correct transient loads give correctly sized and cheaper equipment.

1-D transient conduction (2022 6a)

Transient Heat Flow

Building heat loads have to be determined under transient conditions, which arise, for example, due to

1. Thermal mass of the building envelope subject to changes in outdoor ambient air conditions and in solar radiation over the day
2. Transmitted solar radiation, which is first absorbed in the internal mass of the various elements within a room/space (such as slabs, internal walls, and furniture) and released later in time
3. Internal mass interacting with operational schedules causing diurnal fluctuations in internal load (lights, equipment, and occupants)
4. Internal mass interacting with control strategies of the indoor environment (diurnal thermostat setup or setback or variations in supply air volume)

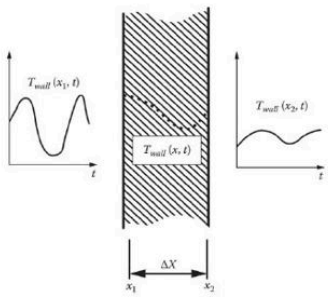


Illustration of boundary condition and temperature distribution across a homogenous wall

Transient heat flow through a homogeneous wall: boundary conditions and temperature profile

Fourier: $q = -k\,\partial T/\partial x$; energy: $\partial q/\partial x = -\rho c_p \partial T/\partial t$

$\Rightarrow \partial T/\partial t = \alpha\,\partial^2 T/\partial x^2$, $\alpha = k/(\rho c_p)$ (thermal diffusivity)

With convective boundary conditions $h_o(T_{sol\text{-}air} - T_0)$ outside and $h_i(T_L - T_i)$ inside. It is solved by finite differences, thermal networks, or response factors/CTF.

Daily temperature and heat flow of a wall (2022 2c)

Temperature Distribution

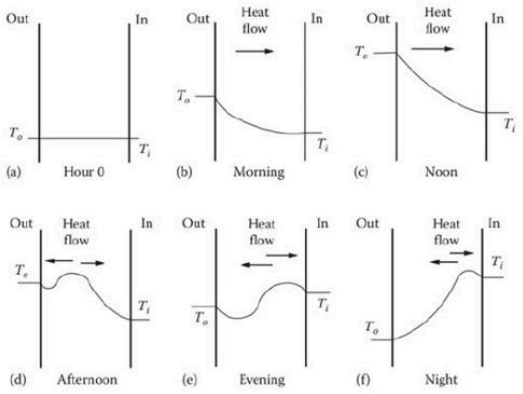


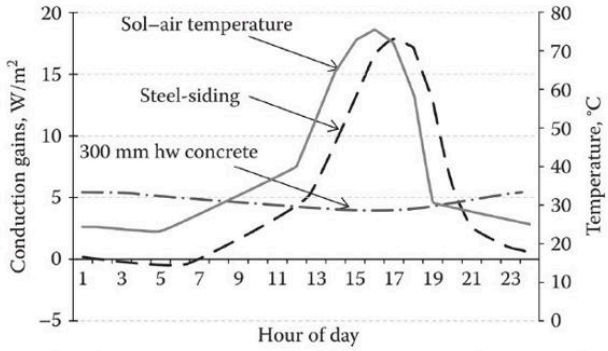
Figure : Diurnal temperature distribution and heat flows through a building element

Diurnal temperature distribution and heat flow through a building element: (a) midnight, (b) morning, (c) noon, (d) afternoon, (e) evening, (f) night

At midnight and early morning the outside is cold, heat flows outward; in the morning the outer surface warms and a heat wave starts into the wall while the inner part still loses heat; by noon/afternoon the outer layers are hot and heat flows inward; in the evening the outer surface cools and heat flows both ways from a warm core; at night stored heat leaves both to outside and to

the room. The inside surface peak arrives hours after the outside peak (time lag) and is smaller (decrement).

Lightweight vs heavyweight wall (2024 7a)



Plots illustrating the large difference in diurnal heat gain distributions from two different wall constructions when subject to the same boundary sol–air temperature profile.

Same sol-air profile, very different heat gains: steel siding (light) vs 300 mm heavyweight concrete

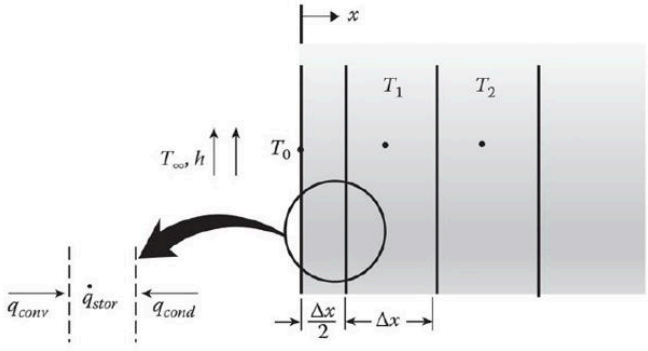
Lightweight wall (steel siding, thin panel)	Heavyweight wall (300 mm concrete, thick brick)
Small heat capacity: inside heat flow follows the sol-air temperature almost immediately	Large heat capacity stores heat: peak delayed 6-12 h
High, sharp peak around 15-17 h (coincides with the cooling peak)	Low, flat curve (large decrement): peak reduced, heat released at night
Bigger cooling equipment needed	Smaller peak load; good for hot-dry climates with large daily swing and night ventilation

Finite difference method (2024 7b)

Numerical Methods: Finite Differences

A very flexible and widely used numerical method for solving differential equations with any type of boundary condition is the finite difference approach. It involves discretizing the derivatives of the temperature profile both spatially and temporally. The solid, or in our specific case, the building wall assumed to be a homogeneous element with constant conductivity, is divided into a number of finite layers, as shown in Figure.

Each of the layers or control volume elements is assumed to be at a uniform temperature lumped at its center, that is, all the mass of the wall element is located at the node. The boundary or surface nodes are assumed to be half the control volume of the interior nodes as shown. An energy balance on node m (assuming unit cross-sectional area) over time increment Δt yields



Wall divided into layers; nodes at layer centres; surface nodes hold half a control volume

Interior node m (unit area, layer Δx): heat in from m−1 minus heat out to m+1 = storage:

$$k(T_{m-1} - T_m)/\Delta x - k(T_m - T_{m+1})/\Delta x = \rho c_p \Delta x\,(T_m^{t+\Delta t} - T_m^t)/\Delta t$$

$$\text{Explicit: } T_m^{t+\Delta t} = \varphi_{Fo}(T_{m-1}^t + T_{m+1}^t) + (1 - 2\varphi_{Fo})T_m^t, \quad \varphi_{Fo} = \alpha\Delta t/\Delta x^2 \quad (\text{stable if } \varphi_{Fo} \le 1/2)$$

Boundary node 0 with combined convection and radiation: the surface node holds only half a control volume (Δx/2). Heat arriving by convection from air at T_∞ and by radiation from surroundings (sky) at T_{surr}, plus conduction from node 1, equals storage in Δx/2:

$$h_c(T_\infty - T_0) + \varepsilon\sigma(T_{surr}^4 - T_0^4) + k(T_1 - T_0)/\Delta x = (\rho c_p \Delta x/2)(T_0^{t+\Delta t} - T_0^t)/\Delta t$$

$$\text{Linearise radiation: } \varepsilon\sigma(T_{surr}^4 - T_0^4) = h_r(T_{surr} - T_0), \quad h_r = \varepsilon\sigma(T_0 + T_{surr})(T_0^2 + T_{surr}^2)$$

$$\text{If } T_{surr} \approx T_\infty\text{: } h_o = h_c + h_r \text{ (combined)}$$

$$\text{Multiply by } 2\Delta t/(\rho c_p \Delta x)\text{:}$$

$$T_0^{t+\Delta t} = 2h_o\Delta t/(\rho c_p \Delta x)\cdot(T_\infty - T_0) + 2k\Delta t/(\rho c_p \Delta x^2)\cdot(T_1 - T_0) + T_0$$

$$\text{Using } \varphi_{Fo} = \alpha\Delta t/\Delta x^2 \text{ and the finite-difference Biot number } \boldsymbol{\varphi_{Bi} = h_o\Delta x/k}\text{:}$$

$$\boldsymbol{T_0^{t+\Delta t} = 2\varphi_{Fo}(T_1^t + \varphi_{Bi}T_\infty) + (1 - 2\varphi_{Fo} - 2\varphi_{Bi}\varphi_{Fo})T_0^t}$$

$$\text{Stability: } 1 - 2\varphi_{Fo} - 2\varphi_{Bi}\varphi_{Fo} \ge 0 \Rightarrow \varphi_{Fo}(1 + \varphi_{Bi}) \le 1/2$$

Finite Difference

Finally, on rearranging,

$$T_0^{t+1} = \frac{2h_o\Delta t}{\rho \cdot c_p \cdot \Delta x}\left(T_\infty^t - T_0^t\right) + \frac{2k\Delta t}{\rho \cdot c_p \cdot \Delta x^2}\left(T_1^t - T_0^t\right) + T_0^t$$

$$= 2\phi_{Bi}\phi_{Fo}\left(T_\infty^t - T_0^t\right) + 2\phi_{Fo}\left(T_1^t - T_0^t\right) + T_0^t$$

$$= 2\phi_{Fo}\left(T_1^t + \phi_{Bi}T_\infty^t\right) + (1 - 2\phi_{Fo} - 2\phi_{Bi}\phi_{Fo})T_0^t \qquad (8)$$

Where the finite difference form of the biot number,

$$\phi_{Bi} = \frac{h_o\Delta x}{k} \qquad (9)$$

Slide: rearranged boundary-node equation with the finite-difference Biot number

With separate radiation temperatures, keep the two terms: $T_0^{t+\Delta t} = 2\varphi_{Fo}[T_1 + \varphi_{Bi,c}T_\infty + \varphi_{Bi,r}T_{surr}] + [1 - 2\varphi_{Fo}(1 + \varphi_{Bi,c} + \varphi_{Bi,r})]T_0$; a solar term $\alpha q_{solar}\Delta x/k$ can be added the same way. The implicit (backward) scheme is unconditionally stable.

CTF model (2021 6d)

$$\dot{Q}_{cond,t} = -\Sigma_{n\ge1}\, d_n \dot{Q}_{cond,t-n\Delta t} + A[\ \Sigma_{n\ge0}\, b_n T_{os,t-n\Delta t} - T_{rc}\Sigma_{n\ge0}\, c_n\ ]$$

The conduction transfer function gives the heat gain through a roof or wall at hour t from the present and past sol-air (outside surface) temperatures T_{os}, the past heat gains, and the constant room temperature T_{rc}; A = area, Δt = 1 h, b_n, c_n, d_n = CTF coefficients tabulated for each wall type. The past heat gains are not known at the start, so one assumes zero and repeats the daily cycle

until the result becomes periodic. It captures the thermal mass (time lag and damping) with simple arithmetic and is used in ASHRAE (American Society of Heating, Refrigerating and Air-Conditioning Engineers)'s transfer function method.

Remember: $\partial T/\partial t = \alpha \partial^2 T/\partial x^2$. Interior node: $\boldsymbol{\varphi_{Fo}}$; surface node: half volume + $\boldsymbol{\varphi_{Bi} = h\Delta x/k}$. Heavy wall = **lag + decrement**.

B2 Solar heat gain coefficient (SHGC) Must study Likely · 70% Medium

PAST-PAPER QUESTIONS

- **2024 Q5(b)** [11] Define the solar heat gain coefficient (SHGc). Discuss why the statement "Lower SHGc is always preferable" is not appropriate for all climate or orientations.
- **2022 Q6(b)** [9] What is meant by solar heat gain coefficient (SHGC)? How to calculate SHGC (solar heat gain coefficient) of a single pane glass window?

LEARN IT FROM ZERO

SHGC, from zero

Sunlight falls on a window. Part goes straight through (transmitted, τ), part bounces off (reflected, ρ) and part is soaked up by the glass (absorbed, α): τ + ρ + α = 1. The absorbed part warms the glass, and some of that heat then flows **into** the room.

The **solar heat gain coefficient** asks one simple question: *of every 100 W of sun hitting the window, how many watts end up inside?* SHGC = (transmitted + inward share of absorbed) / incident. Clear glass about 0.86, tinted or reflective 0.25–0.5.

The inward share of the absorbed heat is α × U / h_o (the glass loses heat to both sides; the inside gets the fraction U/h_o). So **SHGC = τ + αU/h_o**.

CALCULATOR **SHGC of a single pane**

Change any number; every step is recalculated and shown with the numbers put in.

1. U-factor of the glass: U = $h_i h_o/(h_i+h_o)$ = **8.29** × **34** / (**8.29**+**34**) = **6.66** W/m²K
2. Solar heat gain coefficient: SHGC = τ + αU/h_o = **0.84** + **0.08** × **6.665** / **34** = **0.856**
3. Solar heat entering per m²: q = SHGC × I = **0.8557** × **600** = **513** W/m²

Before you start: SHGC

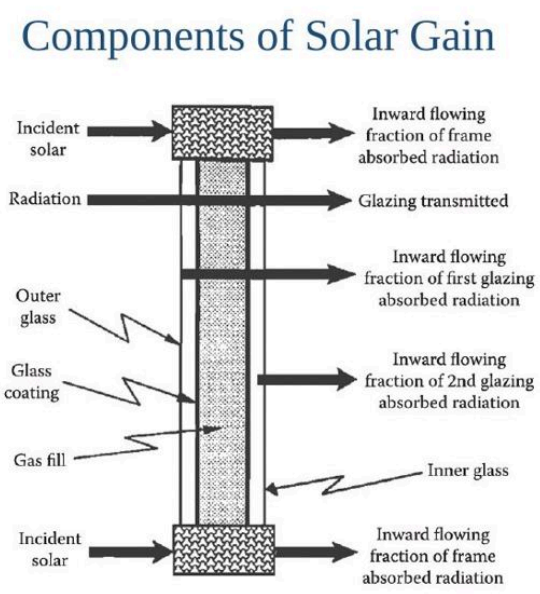
Solar heat gain coefficient = fraction of the sunlight striking a window that ends up as heat inside (transmitted directly + absorbed by the glass and released inward). 0.8 for clear single glass, 0.25-0.4 for good low-e glass. Lower SHGC = less cooling load.

Definition (2022 6b, 2024 5b)

When sunlight hits glass, $\tau_s + \rho_s + \alpha_s = 1$ (about 8 % reflected for clear glass, 5-50 % absorbed). The total solar heat gain is the transmitted part plus the inward-flowing fraction of the absorbed part. The **solar heat gain coefficient** is

SHGC = solar heat gain through the window / solar radiation incident on the window = $\dot{q}_{solar,gain}$ / $\dot{q}_{solar,incident}$

$\dot{Q}_{solar}/A = SHGC_{dir} \cdot I_{T,dir} + SHGC_{dif} \cdot I_{T,dif}$, $I_{T,dir} = I_{beam}\cos\theta_i$



Components of solar gain through a double-glazed window

Derivation for single-pane glass

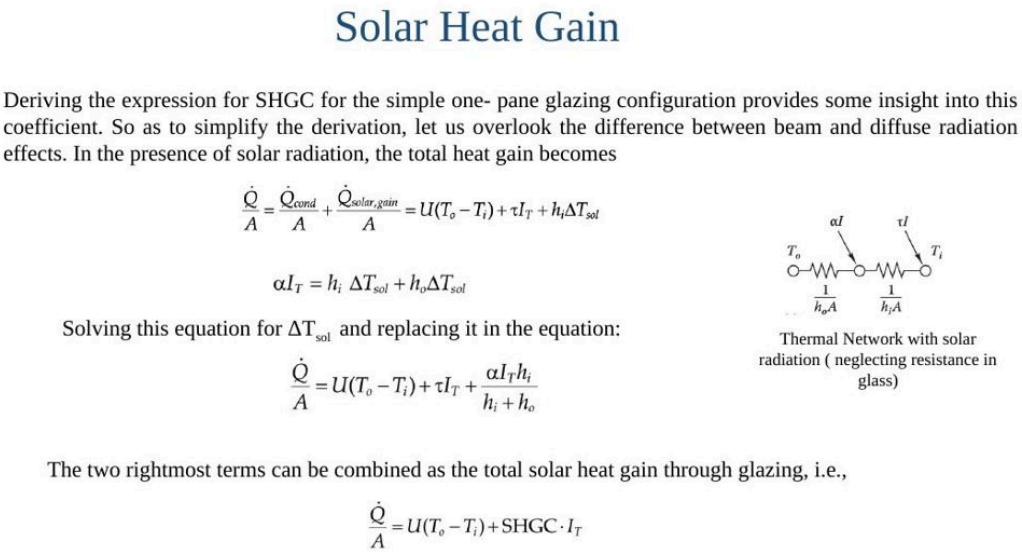
Neglect the glass resistance. The glass absorbs αI_T, which leaves by convection/radiation to inside (h_i) and outside (h_o). Heat balance on the glass node:

$$Q/A = U(T_i - T_o) + \tau I_T + h_i \Delta T_{sol}, \quad \text{with } \alpha I_T = (h_i + h_o)\Delta T_{sol}$$

$$\Rightarrow Q/A = U(T_i - T_o) + [\tau + \alpha h_i/(h_i + h_o)]I_T = U(T_i - T_o) + SHGC \cdot I_T$$

$$\mathbf{SHGC = \tau + \alpha U/h_o}, \quad U = h_i h_o/(h_i + h_o)$$

$$\text{Double glazing: } SHGC = \tau + \alpha_o U/h_o + \alpha_i U(1/h_o + 1/h_s)$$



Slide derivation of SHGC for a single pane

> **Worked example:** clear 3 mm glass, $\tau = 0.84$, $\alpha = 0.08$, $h_i = 8.29$, $h_o = 34$ W/m²K.
> $U = 8.29 \times 34/(8.29+34) = 6.66$ W/m²K; $SHGC = 0.84 + 0.08 \times 6.66/34 = 0.84 + 0.016 = \mathbf{0.86}$

Why "lower SHGC is always preferable" is wrong (2024 5b)

- **Climate:** in cold, heating-dominated climates a high SHGC on equator-facing windows gives free passive solar heating; a low SHGC raises the heating bill. Low SHGC pays only where cooling dominates.
- **Orientation:** on the south (equator-facing) side a properly sized overhang already blocks summer sun, so a higher SHGC glass can admit winter sun; on east/west a low SHGC is valuable; on the north side SHGC hardly matters.
- **Daylight:** many low-SHGC glasses (tinted, reflective) also cut visible light, increasing electric lighting and its internal heat. The right metric is a high light-to-solar-gain ratio (VT/SHGC), not the lowest SHGC.
- **Season and shading:** with external shading or movable devices the glazing SHGC can stay moderate; very low SHGC can make spaces dull and need more heating in winter.
- Low SHGC glass absorbs more and gets hot, giving radiant discomfort; it also costs more.

Remember: SHGC = $\tau + \alpha U/h_o$. Pick SHGC by **climate and orientation**: high on sunny side in cold climates, low on E/W in hot climates.

★ **REMEMBER IT**

- "Lower SHGC is always better" is false: in cold climates or on south glass with an overhang you **want** winter sun. Judge glass by VT (visible transmittance)/SHGC (light in per heat in), not SHGC alone.

B4 Sol-air temperature, wall colour, and the sun-lit wall numerical Must study Likely · 70% Hard

PAST-PAPER QUESTIONS

- **2024 Q6(a)** [11] What is sol-air temperature? Develop the equation of sol-air temperature considering a sun-exposed wall.
- **2021 Q5(b)** [7] What is sol-air temperature? Why sol-air temperature differs from ambient temperature?
- **2021 Q5(d)** [13] The south masonry wall of a house is made of 10-cm-thick red face brick, 10-cm-thick common brick, 19-mm-thick air space, and 13-mm-thick gypsum board, and its overall heat transfer coefficient is 1.8 W/m²·°C, which include the effect of convection on both sides. The cooling system is to be sized on the basis of the heat gain at 3 PM solar time and sol-air temperature in the location at that time is 48 °C. The interior of the house is to be maintained at 22 °C and the area of the wall is 20 m². If the design ambient air temperature at that time is 35 °C, determine: (i) the design heat gain through the wall, (ii) the fraction of this heat gain due to solar heating.
- **2020 Q4(b)** [8] What is sol-air temperature? Explain the effect of absorptivity of the surface on sol-air temperature.
- **2020 Q4(c)** [9] How does color affect heat transfer through walls?

LEARN IT FROM ZERO

Sol-air temperature, from zero

Stand barefoot on a black road at noon. The air may be 35 °C but the road feels like 60 °C, because it soaks up sunlight. A sunlit wall is the same. To calculate heat entering the wall we would need to handle the hot air **and** the sunshine separately.

Sol-air temperature is a trick that folds the sunshine into one imaginary air temperature: "how hot would the outside air need to be to push the same heat into this wall with no sun at all?" Then the heat gain is just the familiar $\mathbf{Q = U\,A\,(T_{sol\text{-}air} - T_{inside})}$.

Derivation in one line: heat into the surface = convection + absorbed sun − sky radiation loss = $h_o(T_a - T_s) + \alpha I - \varepsilon \Delta R$. Take h_o outside the bracket and you get $h_o(T_{sol\text{-}air} - T_s)$ with $\mathbf{T_{sol\text{-}air} = T_a + \alpha I/h_o - \varepsilon \Delta R/h_o}$. The last term is about 0 for walls and 4 °C for roofs.

Colour enters through α: dark brick α ≈ 0.9 doubles the solar rise compared with white paint α ≈ 0.45.

Symbols used here

Symbol	Name and meaning	Unit
T_a	Ambient (outdoor air) temperature	°C
$T_{sol\text{-}air}$	Sol-air temperature: equivalent outdoor temperature including sunshine	°C
T_i	Indoor temperature	°C
α_s	Solar absorptivity of the outer surface (colour)	0 to 1
I ($\dot{q}_{solar}$)	Solar radiation falling on the wall	W/m²
h_o	Outside combined convection + radiation coefficient	W/m²K (about 17)
$\varepsilon\Delta R/h_o$	Long-wave sky radiation correction	°C (0 wall, 4 roof)
U	Overall heat transfer coefficient of the wall	W/m²K
A	Wall area	m²

INTERACTIVE **Sol-air temperature vs surface colour**

Move the sliders and watch the curve change.



Light wall: **50.9** °C · Dark wall: **66.8** °C

CALCULATOR **Wall heat gain and solar fraction (2021 5d type)**

Change any number; every step is recalculated and shown with the numbers put in.

1. Heat gain through the wall: $Q = UA(T_{sol\text{-}air} - T_i) = \mathbf{1.8} \times \mathbf{20} \times (\mathbf{48} - \mathbf{22}) = \mathbf{936}$ W
2. Gain if there were no sun: $UA(T_a - T_i) = \mathbf{1.8} \times \mathbf{20} \times (\mathbf{35} - \mathbf{22}) = \mathbf{468}$ W
3. Solar fraction of the gain: $(T_{sa} - T_a)/(T_{sa} - T_i) = (\mathbf{48} - \mathbf{35}) / (\mathbf{48} - \mathbf{22}) \times 100 = \mathbf{50.0}$ %

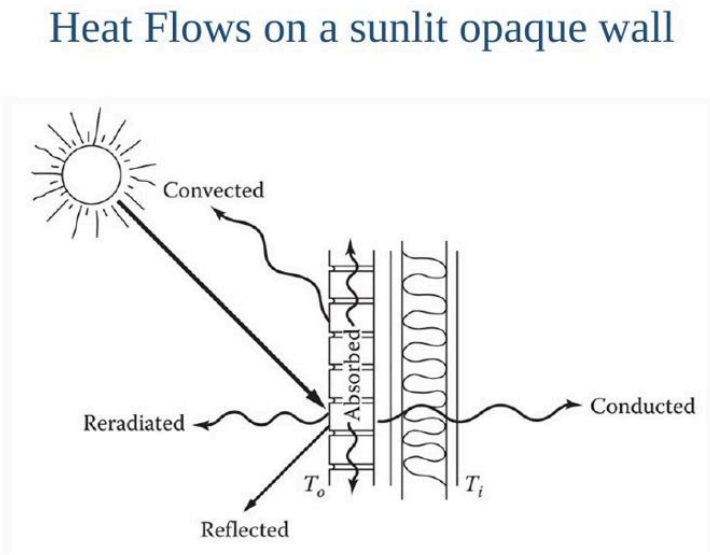
Before you start: sol-air temperature

A wall in the sun gets heat from the hot air **and** from absorbed sunshine. Sol-air temperature rolls both into one imaginary outdoor air temperature: $T_{sol\text{-}air} = T_{air} + \alpha q_{solar}/h_o$ (– a small sky-radiation correction for roofs).

α = solar absorptivity (dark surface high, light low), q_{solar} = sunshine on the wall (W/m^2), h_o = outside heat transfer coefficient (about 17 W/m^{2}K). Then wall heat gain $Q = U\,A\,(T_{sol\text{-}air} - T_{inside})$.

Sol-air temperature (2020 4b, 2021 5b, 2024 6a)

Sol-air temperature is the equivalent outdoor air temperature that would give the same rate of heat transfer to an exterior surface as the actual combination of incident solar radiation, convection with ambient air, and long-wave radiation exchange with the sky and surroundings.



Heat flows on a sunlit opaque wall: absorbed and reflected solar, convection, re-radiation, conduction inward

Derivation (2024 6a)

Energy balance at the outside surface (temperature $T_{surface}$) of a wall of area A_s:

$$\dot{Q}_{surface} = \dot{Q}_{conv+rad} + \dot{Q}_{solar} - \dot{Q}_{radiation\ correction}$$
$$= h_o A_s (T_{ambient} - T_{surface}) + \alpha_s A_s \dot{q}_{solar} - \varepsilon A_s \sigma (T_{ambient}^4 - T_{surr}^4)$$
$$= h_o A_s [\, T_{ambient} + \alpha_s \dot{q}_{solar}/h_o - \varepsilon\sigma (T_{ambient}^4 - T_{surr}^4)/h_o - T_{surface} \,]$$
$$= h_o A_s (T_{sol\text{-}air} - T_{surface})$$
$$\mathbf{T_{sol\text{-}air} = T_{ambient} + \alpha_s \dot{q}_{solar}/h_o - \varepsilon\sigma (T_{ambient}^4 - T_{surr}^4)/h_o}$$

h_o = combined convection-radiation coefficient (about 17 W/m²K in summer). The last (radiation correction) term is about 0 for vertical walls and about 4 °C for roofs facing the cold sky. Then the wall heat gain and the solar part are:

$\dot{Q}_{wall} = UA_s(T_{sol\text{-}air} - T_{inside})$; $\Delta T_{solar} = \alpha_s \dot{q}_{solar}/h_o$
Solar fraction of the heat gain = $(T_{sol\text{-}air} - T_{amb})/(T_{sol\text{-}air} - T_{in})$
Fraction of incident solar transferred inside = $U\alpha_s/h_o$

Why it differs from ambient temperature (2021 5b)

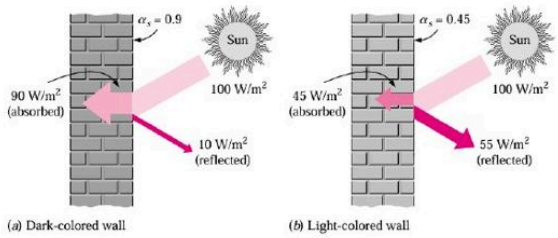
A sunlit surface absorbs solar radiation ($\alpha_s q_{solar}$), which raises its temperature well above the air, so the wall behaves as if the outside air were hotter by $\alpha_s q/h_o$ (often 15-35 °C for dark walls). At night or on roofs it is slightly lower than ambient because the surface radiates to the cold sky. It also depends on surface colour, orientation and time of day, while ambient temperature does not.

Effect of absorptivity / colour (2020 4b, 4c, 2021 6b)

Being conservative and taking h_o 17W/m² .℃,the summer design values of the ratio α_s/h_o for light- and dark-colored surfaces are determined to be

$$\left(\frac{\alpha_s}{h_o}\right)_{light} = \frac{0.45}{17\ \text{W/m}^2 \cdot °\text{C}} = 0.026\ \text{m}^2 \cdot °\text{C/W}$$

$$\left(\frac{\alpha_s}{h_o}\right)_{dark} = \frac{0.90}{17\ \text{W/m}^2 \cdot °\text{C}} = 0.052\ \text{m}^2 \cdot °\text{C/W}$$



Dark-colored buildings absorb most of the incident solar radiation whereas light colored ones reflect most of it.

Dark surfaces absorb most of the incident solar radiation; light surfaces reflect most of it

The solar rise $\Delta T_{solar} = \alpha_s q/h_o$ is directly proportional to absorptivity. With h_o = 17 W/m²K: light colours $\alpha_s \approx 0.45$, α/h_o = 0.026 m²K/W; dark colours $\alpha_s \approx 0.90$, α/h_o = 0.052 m²K/W. For q = 600 W/m²: light wall +16 °C, dark wall +32 °C. So a dark wall or roof roughly doubles the solar part of the heat gain. Light/white (or "cool") colours reduce cooling load in hot climates; dark colours help in cold climates. Emissivity matters too: a high-emissivity surface loses more at night.

Numerical (2021 5d)

Given: U = 1.8 W/m²°C (includes both surface films), A = 20 m², $T_{sol\text{-}air}$ = 48 °C, T_{in} = 22 °C, T_{amb} = 35 °C

(i) $\dot{Q}_{wall} = UA(T_{sol\text{-}air} - T_{in}) = 1.8 \times 20 \times (48 - 22) = 1.8 \times 20 \times 26 =$ **936 W**

(ii) $\Delta T_{solar} = T_{sol\text{-}air} - T_{amb} = 48 - 35 = 13$ °C; overall ΔT = 26 °C

Solar fraction = 13/26 = **0.50 = 50 %** of the heat gain is due to solar heating

(Extra, if asked) Red brick is dark: $\alpha_s/h_o = 0.052$, fraction of incident solar entering = $U\alpha_s/h_o = 1.8 \times 0.052 = 0.094 \approx 9$ %

Same structure as Cengel Example 16-3 in the slides (west wall, U = 1.65, 19.5 m², $T_{sol\text{-}air}$ = 63.7 °C, 24 and 32 °C → Q = 1277 W, solar fraction 79 %).

Remember: $T_{sol\text{-}air} = T_{amb} + \alpha q/h_o - \varepsilon\Delta R/h_o$. $Q = UA(T_{sol\text{-}air} - T_i)$. Solar fraction = $(T_{sa} - T_{amb})/(T_{sa} - T_i)$. Dark α/h = **0.052**, light **0.026**.

Same answer, step by step with reasons

Step (write this)	Why / what it means
`U = 1.8 W/m²°C, A = 20 m², T_sol-air = 48, T_in = 22, T_amb = 35 °C`	List the data.
`Q = U A (T_sol-air - T_in) = 1.8 × 20 × 26 = 936 W`	Same as Q = UAΔT, but the outside temperature is replaced by the sol-air temperature to include sunshine.
`Temperature rise due to sun = 48 - 35 = 13 °C`	Sol-air minus real air temperature = the solar part.
`Solar fraction = 13 / 26 = 50 %`	Solar part / total driving temperature difference.

Memory trick: "Sun adds αq/h" to the air temperature.

WRITE IN EXAM **Write this in the exam (this much is enough for full marks)**

2021 5(d) Given: U = 1.8 W/m²°C (with both films), A = 20 m², $T_{sol\text{-}air}$ = 48 °C, T_i = 22 °C, T_a = 35 °C

(i) Heat gain: $Q = UA(T_{sol\text{-}air} - T_i)$

$= 1.8 \times 20 \times (48 - 22) = 1.8 \times 20 \times 26$

$=$ **936 W**

(ii) Solar fraction: $(T_{sol\text{-}air} - T_a) / (T_{sol\text{-}air} - T_i)$

$= (48 - 35) / (48 - 22) = 13 / 26$

$=$ **0.50 = 50 %**

So half of the heat entering the wall is caused by the sun; the other half by the hot air.

If the paper gives the layers instead of U: $U = 1 / (1/h_o + \Sigma R_{layers} + 1/h_i)$ first, then the same two lines.

B13 Energy audit and building energy management economics Good to have

Possible · 50% Easy

PAST-PAPER QUESTIONS

- **2022 Q8(d)** [8] How to overcome the economic challenges of building by building energy management?
- **2021 Q7(a)** [11] What are the major components of energy auditing of a building? Discuss about the economic goal of energy management in building?

Before you start: energy audit

An energy audit is a check-up: measure where energy is used, find waste, propose measures and calculate payback (cost / yearly saving). Walk-through audit = quick; detailed audit = measured and modelled.

Components of a building energy audit (2021 7a)

1. **Preliminary (walk-through) audit:** collect 1-3 years of utility bills, building drawings, occupancy and operating hours; quick inspection to spot obvious waste
2. **Energy use breakdown:** benchmark (kWh/m²·yr) against similar buildings; split use into HVAC (heating, ventilation and air-conditioning), lighting, hot water, equipment, plug loads
3. **Detailed survey and measurement:** envelope (insulation, glazing, air leakage via blower door, thermography), HVAC efficiency, lighting levels and power density, motors, controls, metering and logging
4. **Analysis:** identify energy conservation measures (ECMs), estimate savings and costs; simulation if needed
5. **Economic evaluation:** simple payback, NPV, life-cycle cost, IRR; rank the measures
6. **Report and implementation plan,** then **monitoring and verification** of savings; energy targeting for the future

Economic goal of energy management (2021 7a)

The goal is not to minimise energy alone but to **minimise the total life-cycle cost** (first cost + operating energy cost + maintenance) while keeping comfort and productivity. A more efficient but more expensive system is justified only if its energy savings over its life exceed the extra first cost (positive net present value, acceptable payback). Energy management also aims to reduce peak demand charges, protect against price rises and add asset value.

Overcoming economic challenges by building energy management (2022 8d)

- Start with no-cost/low-cost measures (good housekeeping: switch-off policy, set-points, schedules, maintenance) that pay back immediately and fund later steps
- Energy policy, an energy champion, staff involvement, targets, notices, walk-rounds and meter readings (slides' good housekeeping programme)
- Sub-metering and a BMS to find and fix waste; continuous commissioning
- Prioritise measures by payback; bundle quick-payback with long-payback measures
- Use financing tools: energy service companies (ESCO) and performance contracts, green loans, government incentives, net-metering for PV (photovoltaic (solar cells))
- Plan upgrades at replacement time (lower marginal cost), use life-cycle costing in purchases, and demand-side management to cut peak charges

Remember: Audit: **Data → Benchmark → Survey → Analyse → Economics → Report → Verify**. Goal = **minimum life-cycle cost**.

B9 Below-grade heat loss: basement walls, floors, slab on grade Good to have Possible · 50% Medium

PAST-PAPER QUESTIONS

- **2022 Q5(a)** [9] How to calculate heat losses from the vertical basement walls?
- **2021 Q5(a)** [10] Explain the ASHRAE (American Society of Heating, Refrigerating and Air-Conditioning Engineers) method of calculating heat losses from walls and floor of basement. Write the equation to calculate transient temperature of soil.

Before you start: ground heat loss

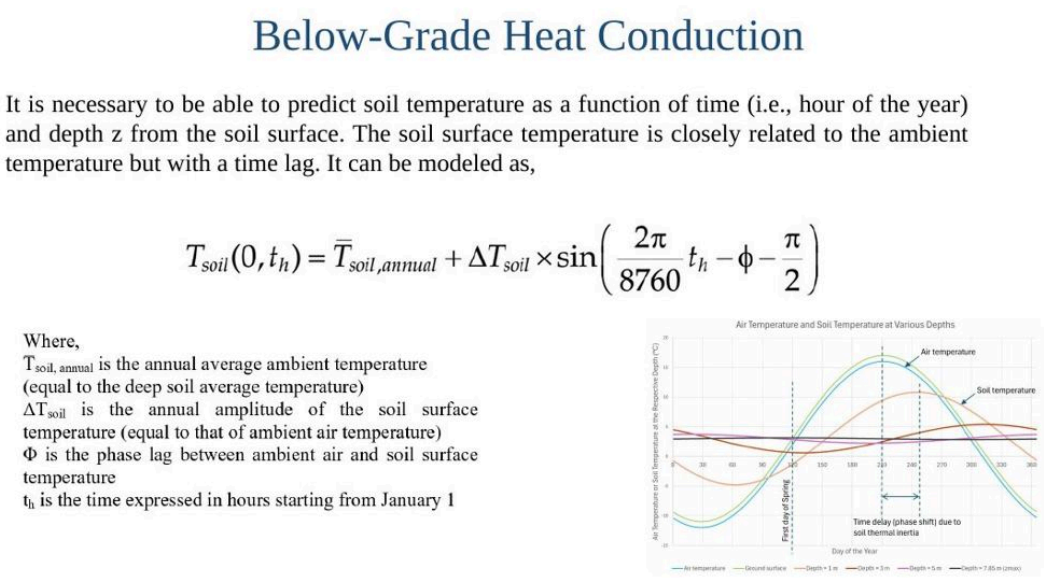
Heat leaks from basements and floors into the ground. The ground is a large heat store with fairly steady temperature, so heat loss is calculated with special path-length methods, and is largest near the edges (the slab perimeter).

Ground coupling and soil temperature (2021 5a)

Heat transfer between a basement or slab and the earth is called **ground coupling**, a conduction problem that depends on the indoor-soil temperature difference, the resistance of the wall/floor and the soil conductivity (soil type, density, porosity, depth, moisture). The soil surface temperature follows the ambient with a time lag:

$$T_{soil}(0,\ t_h) = \bar{T}_{soil,annual} + \Delta T_{soil}\ \sin(\ 2\pi t_h/8760 - \varphi - \pi/2\ )$$

$\bar{T}_{soil,annual}$ = annual average ambient (= deep soil) temperature, ΔT_{soil} = annual amplitude of soil surface temperature, φ = phase lag between ambient and soil temperature, t_h = hours from 1 January. Amplitude decays and lag increases with depth.



Below-Grade Heat Conduction

It is necessary to be able to predict soil temperature as a function of time (i.e., hour of the year) and depth z from the soil surface. The soil surface temperature is closely related to the ambient temperature but with a time lag. It can be modeled as,

$$T_{soil}(0,t_h) = \bar{T}_{soil,annual} + \Delta T_{soil} \times \sin\left(\frac{2\pi}{8760} t_h - \phi - \frac{\pi}{2}\right)$$

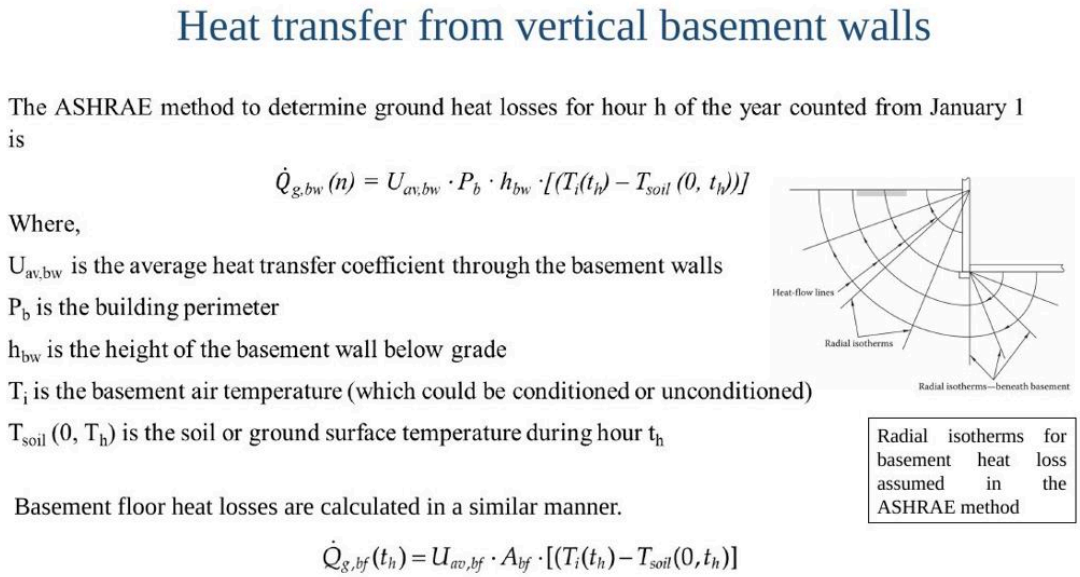
Where,
$T_{soil,\ annual}$ is the annual average ambient temperature (equal to the deep soil average temperature)
ΔT_{soil} is the annual amplitude of the soil surface temperature (equal to that of ambient air temperature)
Φ is the phase lag between ambient air and soil surface temperature
t_h is the time expressed in hours starting from January 1

Transient soil temperature model (slide)

ASHRAE method for vertical basement walls (2022 5a)

$$\dot{Q}_{g,bw}(t_h) = U_{avg,bw} \cdot P_b \cdot h_{bw} \cdot [\, T_i(t_h) - T_{soil}(0,\, t_h) \,]$$

$U_{avg,bw}$ = average heat transfer coefficient of the basement wall (from tables, depends on wall insulation and depth; it already includes the soil path), P_b = basement perimeter, h_{bw} = height of wall below grade, T_i = basement air temperature (conditioned or not), $T_{soil}(0,t_h)$ = ground surface temperature. The method assumes heat flows along **radial (circular) paths** from the wall to the ground surface, so the path is longer and U lower deeper down.



Heat transfer from vertical basement walls

The ASHRAE method to determine ground heat losses for hour h of the year counted from January 1 is

$$\dot{Q}_{g,bw}(n) = U_{av,bw} \cdot P_b \cdot h_{bw} \cdot [(T_i(t_h) - T_{soil}(0,\, t_h))]$$

Where,

$U_{av,bw}$ is the average heat transfer coefficient through the basement walls

P_b is the building perimeter

h_{bw} is the height of the basement wall below grade

T_i is the basement air temperature (which could be conditioned or unconditioned)

T_{soil} $(0, T_h)$ is the soil or ground surface temperature during hour t_h

Basement floor heat losses are calculated in a similar manner.

$$\dot{Q}_{g,bf}(t_h) = U_{av,bf} \cdot A_{bf} \cdot [(T_i(t_h) - T_{soil}(0, t_h)]$$

Radial isotherms for basement heat loss assumed in the ASHRAE method

ASHRAE basement wall method with radial isotherms

Basement floor: same form, $\dot{Q}_{g,bf} = U_{avg,bf} \cdot A_{bf} \cdot [T_i - T_{soil}(0, t_h)]$, with $U_{avg,bf}$ depending on floor width and depth.

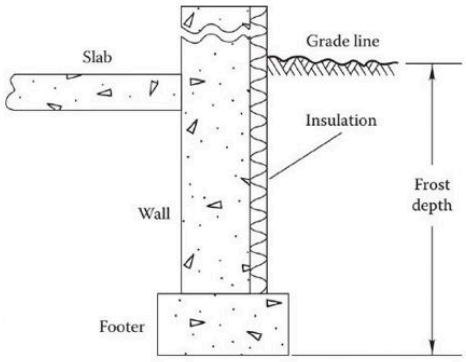
Slab on grade: loss is proportional to the perimeter, not the area: $\dot{Q}_{g,slab} = F_p P_{slab}(T_i - T_o)$, F_p = perimeter heat loss coefficient (W/m·K, table 7.5), with outdoor air temperature.

ASHRAE Method for Slab on Grade Heat Losses

ASHRAE assumes that heat losses from *slabs on grade* are proportional to the perimeter of the slab, not the area of the slab, as for below-grade slabs. The governing equation based on work in the 1940s at the National Bureau of Standards is as follows:

$$\dot{Q}_{g,slab} = F_p \cdot P_{slab} \cdot (T_i - T_o)$$

where the heat loss coefficient F_p, is given in Table 7.5, P_{slab} is the perimeter of the slab, and the temperature difference is between the building interior air temperature and exterior air temperature.



Slab-on-grade heat loss through the edge (perimeter) with edge insulation

Remember: Basement wall: $\mathbf{Q = U_{avg} P\, h\, (T_i - T_{soil})}$, radial paths. Slab: $\mathbf{Q = F_p P (T_i - T_o)}$, perimeter not area.